

Chance Constrained Optimization with Kernel Density Estimation

Master-Thesis

for the degree of

Master of Science (M.Sc.)

Data Science

in the Technischen Fakultät at
Friedrich-Alexander-Universität Erlangen-Nürnberg

vorgelegt am Datum

from **Syed Ansab Waqar Gillani**

Betreuer/in: Prof. Dr. Michael Schuster



Friedrich-Alexander-Universität
Naturwissenschaftliche Fakultät

Contents

1	The Problem of Optimization Under Uncertainty	1
1.1	Where Deterministic Optimization Falls Short	1
1.1.1	The mean-value approach and the flaw of averages	1
1.1.2	Four domains where uncertainty bites	2
1.1.3	A worked example: the cost of trusting the mean	3
1.1.4	A failure rate that recurs across problems	4
1.2	Two Extremes and the Gap Between Them	5
1.2.1	Ignoring uncertainty and worst-case robustness	5
1.2.2	When each extreme is the right tool	6
1.2.3	Chance constraints as the unifying middle ground	7
1.3	What Chance Constraints Are and Why They Are Hard	8
1.3.1	Individual and joint chance constraints	9
1.3.2	Three structural sources of difficulty	10
1.4	The Case for Nonparametric Estimation	13
1.4.1	The parametric assumption and its routine failure	13
1.4.2	What a wrong distribution does to the guarantee	13
1.4.3	Kernel density estimation and its fit to optimization	14
1.4.4	The price of going nonparametric	15
1.5	The Residual-Space Insight	16
1.5.1	From uncertainty space to residual space	16
1.5.2	Three dividends of the change of variables	17
1.5.3	Joint constraints, bandwidth, and the safety guarantee	19
1.6	Thesis Scope and Positioning	19
1.6.1	Disciplinary positioning and anchor literature	20
1.6.2	Research questions and roadmap	21
2	Stochastic Optimization and Chance Constraints	22

2.1	A Taxonomy of Optimization Under Uncertainty	22
2.1.1	Four paradigms	23
2.1.2	A conceptual map and where this thesis sits	24
2.2	Classical Chance-Constrained Programming	25
2.2.1	Origins and canonical formulations	25
2.2.2	Convexity islands: the log-concavity theory	26
2.2.3	The general case	28
2.3	Standard Approximation Methods and Their Trade-offs	28
2.3.1	Parametric and Gaussian reformulations	29
2.3.2	The scenario approach	30
2.3.3	Sample average approximation	30
2.3.4	CVaR and convex restrictions	31
2.3.5	Quantile smoothing	32
2.3.6	The blackbox approach	33
2.3.7	Trade-offs in one view	36
2.4	Kernel Density Estimation	38
2.4.1	The Parzen-Rosenblatt estimator	39
2.4.2	Kernel choices	39
2.4.3	Bandwidth and the bias-variance trade-off	41
2.4.4	Multivariate KDE and the curse of dimensionality	42
2.5	KDE in the Chance-Constrained Literature: A Literature Map	43
2.5.1	First constructions: process systems and aerospace	44
2.5.2	The safety line: biased kernel density estimation	45
2.5.3	Networks, breadth, and the arrival of theory	45
3	Smooth Approximation with Standard (Unbiased) KDE	47
3.1	The Residual-Space Transformation	47
3.2	Residual Samples	48

3.3	KDE of the Residual Distribution	49
3.4	The Integrated Kernel Reformulation	49
3.5	Gradient Computation	50
3.6	Stochastic Optimization Reformulated via Unbiased KDE	51
3.7	Properties of the Unbiased KDE Approximation	52
4	The Biased KDE Approach and Conservative Reformulation	54
4.1	Why Smoothing Alone Is Not Safe	54
4.2	The Biasing Construction	55
4.3	Specific Biased Kernels	56
4.4	Stochastic Optimization Reformulated via Biased KDE	57
4.5	Joint Chance Constraints: Three Approaches	58
4.6	Log-Sum-Exp Smoothing of the Scalar Violation Statistic	59
4.7	Bandwidth Continuation and Kernel Switching	61
5	Convergence Theory	63
5.1	The Gap the Theory Fills	63
5.2	Setup and Assumptions	65
5.3	The Passive Constraint Case	65
5.4	Limits of Approximate Solutions Are Optimal	66
5.5	Existence of a Convergent Solution Sequence	68
5.6	What the Theory Does and Does Not Cover	68
5.7	The Gap Between the Two Anchor Papers	70
6	Optimal Control Formulation	72
6.1	Stochastic optimal-control problem	72
6.2	Direct transcription used in the experiments	73
6.3	Local lunar landing model	74
6.4	A conservation audit of the local objective	75

6.5	Stage logic and numerical interpretation	76
6.6	Scope of the formulation	76
7	Implementation and Validation	78
8	Experimental Study	79
8.1	Design, estimands, and provenance	79
8.2	Static nonlinear benchmark	80
8.3	Energy dispatch and exact distributional check	82
8.4	Bandwidth, sample size, and joint statistics	83
8.5	Lunar continuation record	84
8.6	Evidence hierarchy	85
8.7	Reproducibility and reporting protocol	85
9	Discussion and Practical Guidance	87
9.1	What the static evidence says	87
9.2	Scenario optimization and KDE answer different questions	88
9.3	Bandwidth and data requirements	88
9.4	Joint constraints require a real allocation experiment	89
9.5	The lunar result is a numerical audit	89
9.6	Threats to validity	90
9.7	Practical decision rule	91
	Literatur	98
	Lebenslauf	100

1 The Problem of Optimization Under Uncertainty

Every optimization model is a promise. It promises that if the decision it recommends is implemented, the constraints of the system will hold and the objective will be as good as the mathematics claims. That promise is honest only if the numbers inside the model are honest, and in almost every real deployment they are not. The demand a grid operator schedules against is a forecast, not a fact. The thrust a lander commands is corrupted by disturbances that sensors resolve only imperfectly. The returns a portfolio is built on are realizations of a process that is not available in closed form. This chapter establishes precisely what goes wrong when uncertainty is ignored, why the two classical remedies, ignoring it and armoring against the worst case, each have important limitations in the settings this thesis targets, and why the chance constraint, a constraint required to hold with a prescribed probability, offers a principled middle ground between them. It then exposes the three structural reasons chance constraints are computationally hard, builds the case for estimating the relevant probability distributions from data rather than assuming them, and introduces the single conceptual shift on which the entire thesis rests: moving the estimation problem from the space of the uncertainty to the space of the constraint residual. The chapter closes by fixing the scope, the anchor literature, and the research questions of the thesis.

1.1 Where Deterministic Optimization Falls Short

This section establishes why the deterministic template struggles under uncertainty. It first identifies the mechanism behind the failure, then illustrates it across four application domains, works through a numerical example in which the failure can be quantified exactly, and closes by showing that the observed failure rate is largely independent of the application.

1.1.1 The mean-value approach and the flaw of averages

The standard nonlinear program

$$\underset{x \in X}{\text{minimize}} \ f(x) \quad \text{subject to} \quad g(x, \xi) \leq 0, \tag{1}$$

treats the parameter vector ξ as a known constant. Decades of solver development have made problems of the form Eq. (1) routinely solvable at scale, and this success

has had a quiet side effect: the deterministic template is applied by default, even when ξ is plainly a random quantity. The usual workaround is to replace ξ with a point estimate, most often its mean or a nominal forecast, and solve the resulting deterministic problem. This is the mean-value approach, and its shortcomings are systematic rather than occasional.

The failure mechanism is worth stating carefully, because it is the mechanism behind the examples in this section. An optimizer pushes its solution to the boundary of the feasible region; that is what optimality means whenever a constraint is binding. If the constraint $g(x, \bar{\xi}) \leq 0$ is active at the mean-value optimum x_{mv}^* , then the residual $g(x_{mv}^*, \xi)$ sits exactly at zero when ξ equals its mean. The moment ξ is allowed to fluctuate around that mean, roughly half of its realizations push the residual positive and the constraint is violated. The optimizer does not merely tolerate risk; it tends to select the configuration in which the risk of violation is largest, because that configuration is where the objective is best. In this sense optimization and uncertainty often work against each other, a phenomenon known in the planning literature as the flaw of averages (Savage, 2009): a plan built on average inputs need not behave like an average plan, and can be considerably more fragile.

1.1.2 Four domains where uncertainty bites

Four application domains make this concrete. They are chosen deliberately, because each reappears later in the thesis as a benchmark or a case study, and because together they span the distributional pathologies that motivate the nonparametric approach of Section 1.4.

Autonomous lunar landing. A powered-descent guidance problem asks for a thrust profile that brings a lander from orbit to a designated site while respecting bounds on velocity, glide slope, and fuel. The dynamics are integrated forward under thrust perturbations and imperfectly known gravitational and terrain parameters. Keil u. a. (2021) show that disturbance distributions in this setting are naturally non-Gaussian; a mixture arises, for example, when a thruster operates in one of two regimes. A trajectory optimized for nominal thrust grazes the velocity constraint at touchdown, so under perturbation the lander exceeds its safe impact speed on a large fraction of descents. A lunar landing offers no opportunity to repeat a failed attempt. Chance-constrained formulations of exactly this problem are the application anchor of the thesis (Keil u. a., 2020, 2021; Blackmore u. a., 2010).

Energy dispatch under renewable generation. A system operator commits generation a day ahead against uncertain demand and uncertain wind and solar output. Forecast errors for renewable infeed are well documented to be skewed and heavy-tailed rather than Gaussian (Hodge u. Milligan, 2011; Bienstock u. a., 2014). A dispatch that balances the forecast exactly is short of energy whenever net demand lands above its forecast, which by construction happens with probability near one half, and reserve activations, load shedding, or price spikes follow. This domain has produced some of the strongest data-driven chance-constraint literature (Ciftci u. a., 2019; Wu u. a., 2021; Wu u. Kargarian, 2023).

Chemical process control. A reactor recipe is optimized for yield subject to temperature and purity limits, but the composition of the feed material varies from batch to batch. Running the process at the deterministic optimum places the temperature constraint on the boundary, and ordinary feedstock variation then produces off-specification batches at a steady rate. Process systems engineering was, in fact, the field where data-driven kernel-smoothed chance constraints were first formulated (Calfa u. a., 2015).

Portfolio allocation. An investor maximizes expected return subject to a cap on the probability of a large loss. Asset returns exhibit heavy tails and volatility clustering; these are among the most robust stylized facts in empirical finance (Cont, 2001). A portfolio optimized as if returns were fixed at their historical means concentrates in the highest-mean assets and breaches its loss cap far more often than any Gaussian calculation predicts (Rockafellar u. Uryasev, 2000; Liu u. a., 2022).

1.1.3 A worked example: the cost of trusting the mean

The mechanism is easiest to see in a problem small enough to solve by inspection. Consider a single-bus dispatch model. A controllable resource of capacity x megawatts must cover an uncertain net demand ξ , and capacity is costly, so the planner solves

$$\underset{x \geq 0}{\text{minimize}} \quad c x \quad \text{subject to} \quad \xi - x \leq 0. \quad (2)$$

Let net demand follow a moderately skewed lognormal law with mean $\bar{\xi} \approx 102$ MW, a realistic shape for net load once renewable infeed is subtracted (Hodge u. Milligan, 2011). The mean-value method replaces ξ by $\bar{\xi}$ and returns $x_{\text{mv}}^* = \bar{\xi}$: build exactly

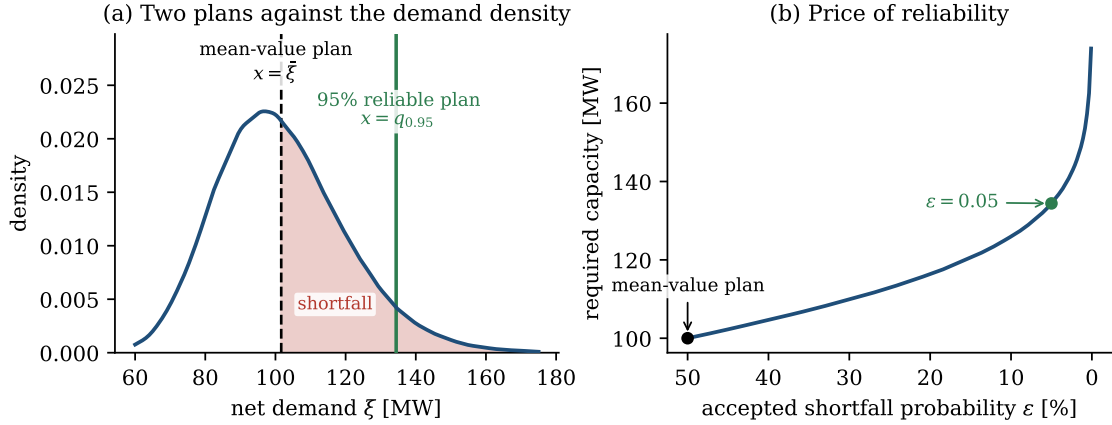


Figure 1: The mean-value fallacy in a one-dimensional dispatch problem with lognormal net demand (Monte Carlo with 2×10^5 samples). (a) The mean-value plan $x = \bar{\xi} \approx 102$ MW leaves the shaded shortfall region uncovered, so the constraint $\xi \leq x$ fails on 46% of days; covering demand with 95% reliability requires $x = q_{0.95} \approx 134$ MW. (b) Required capacity as a function of the accepted shortfall probability ε . Reliability is bought at an accelerating price as $\varepsilon \rightarrow 0$; the chance constraint makes this price explicit and lets the planner choose a point on the curve deliberately.

average capacity. Figure 1(a) shows what this decision looks like against the actual demand density. The constraint is violated on every day that demand exceeds its mean, and for this distribution that is 46% of all days. A plan that appears exact on paper fails almost every other day in operation.

The repair is equally easy to read off the figure. If the planner is willing to accept a shortfall on at most 5% of days, the appropriate decision is the 95% quantile of demand, $x = q_{0.95} \approx 134$ MW, about 32% more capacity than the mean-value plan. Figure 1(b) generalizes this observation into the central trade-off of the thesis: required capacity, and hence cost, rises steeply as the accepted violation probability ε shrinks, and rises without bound as $\varepsilon \rightarrow 0$ because the lognormal has unbounded support. Reliability is a commodity with a price, and the role of a properly formulated stochastic optimization problem is to make that price visible and chooseable rather than implicit and accidental.

1.1.4 A failure rate that recurs across problems

One might hope that the 46% figure is an artifact of the simple model. It is not. Figure 2 repeats the experiment for stylized versions of three of the domains above: dispatch under skewed demand, a landing-corridor constraint under bimodal thrust

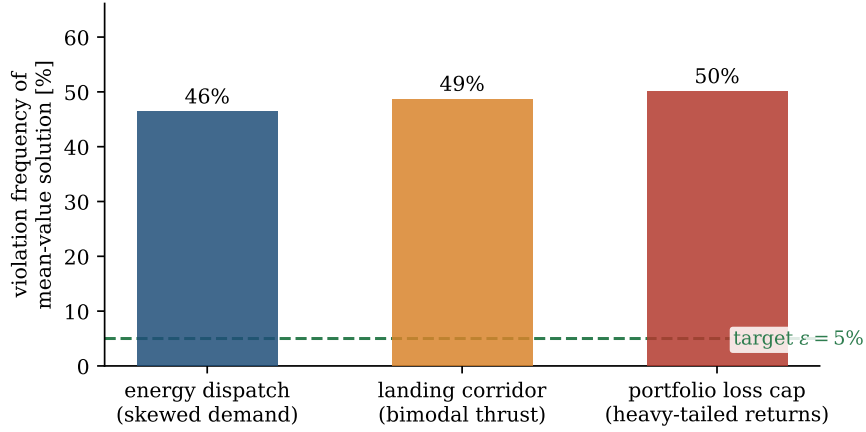


Figure 2: Empirical violation frequency of the mean-value solution in three stylized problems drawn from the application domains of this thesis (Monte Carlo estimates). A typical engineering target of $\varepsilon = 5\%$ is shown for scale. The recurring value close to 50% reflects the boundary mechanism: optimizers drive binding constraints to the edge of feasibility, where roughly half of all perturbations cause failure.

disturbance, and a portfolio loss cap under heavy-tailed (Student- t) returns. In every case the mean-value solution violates its constraint between 46% and 50% of the time, well beyond a typical design target. The violation frequency stays near one half largely because of the boundary mechanism described above: where the constraint is binding at the deterministic optimum, the median of the residual sits at or near zero, and this holds across the applications considered. The value therefore reflects a property of the optimization itself rather than of any particular application domain.

1.2 Two Extremes and the Gap Between Them

This section examines the two classical responses to the failure just described, the deterministic status quo and worst-case robustness, identifies the settings in which each is appropriate, and introduces the chance constraint as a formulation that connects the two through a tunable reliability level.

1.2.1 Ignoring uncertainty and worst-case robustness

Once the limitations of the mean-value approach are recognized, the question becomes what should replace it. Two answers dominate, and they sit at opposite ends of a spectrum.

The first answer is to ignore uncertainty entirely, which is the mean-value approach al-

ready examined in Section 1.1. Its defining property is worth restating in the language used for the rest of the thesis: the deterministic solution carries essentially no reliability statement. The formulation Eq. (1) does not bound the probability of failure, and Fig. 2 shows that this probability is typically near one half. Whatever reliability a deployed mean-value solution exhibits usually comes from safety margins added by hand, outside the optimization. Such hand-tuned margins can be read as an indication that the optimization, as posed, did not capture the intended requirement.

The second answer is worst-case robust optimization (Soyster, 1973; Ben-Tal u. a., 2009). The uncertain parameter is confined to an uncertainty set Ξ , and the constraint is enforced for every element of that set:

$$g(x, \xi) \leq 0 \quad \text{for all } \xi \in \Xi. \quad (3)$$

Robust optimization is a mature and elegant theory; for many set geometries, Eq. (3) admits exact tractable reformulations, and when a failure is not acceptable under any circumstance it is arguably the model of choice. Its weakness is the obverse of its strength: protection is purchased against every point of Ξ , including points that are vanishingly unlikely or physically implausible, and the objective pays for all of them. In the dispatch example, the demand distribution has unbounded support, so a literal worst-case formulation has no feasible solution at all; the set Ξ must be truncated by fiat, and the answer then depends sensitively on a truncation the data does not specify. Even with a defensible truncation, the protection level is largely a fixed property of the set rather than a dial the decision maker controls. Empirically, robust solutions in power systems and aerospace routinely sacrifice a large fraction of attainable objective value to guard against scenarios whose combined probability is far below any operational concern (Ben-Tal u. a., 2009; Keil u. a., 2021).

1.2.2 When each extreme is the right tool

Fairness demands an acknowledgment before the middle ground is claimed: each extreme has a legitimate home. The mean-value approach is correct when uncertainty is genuinely negligible relative to the constraint margins, or when violations carry only a linear cost that the objective already prices in, as in inventory models where a short-fall is simply an expense. Worst-case robustness is correct when the uncertainty set is small, well characterized, and physically bounded, and when a single failure would be severe and difficult to reverse: structural safety factors are a familiar example. The

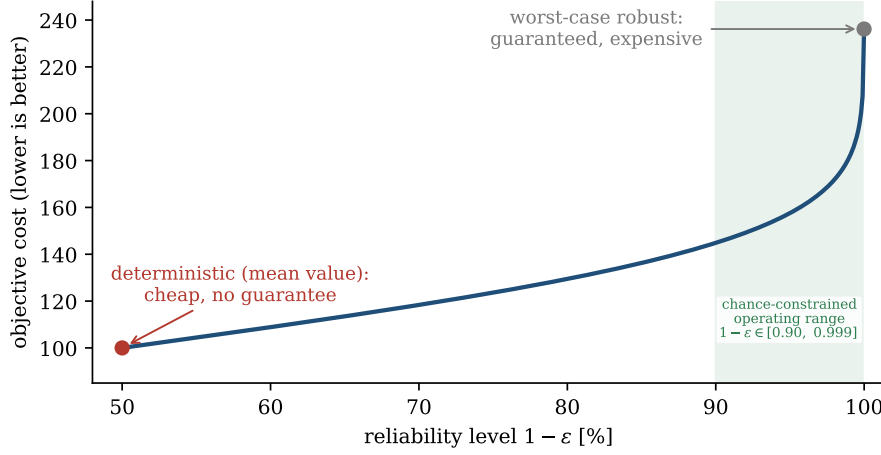


Figure 3: The reliability spectrum for the dispatch example of Fig. 1. The deterministic mean-value plan is cheap and offers no guarantee; the worst-case robust plan is guaranteed and expensive, and is not even well defined under unbounded support. Chance constraints parameterize the entire curve between the extremes: the decision maker selects a reliability level $1 - \varepsilon$, and the optimization returns the cheapest decision that honors it.

problems this thesis targets sit in neither home. Their uncertainties are large, data-described, and unbounded or effectively so, and their failures are costly but tolerable at a small, explicit rate. A lander program accepts a quantified risk per descent; a grid code tolerates a bounded loss-of-load expectation per year. For this regime, neither extreme expresses what the decision maker actually wants.

1.2.3 Chance constraints as the unifying middle ground

Between a method with no guarantee and a method that guarantees too much lies the formulation this thesis is about. A chance constraint, introduced by Charnes u. Cooper (1959) and developed into a full branch of stochastic programming by Prékopa (1995), requires the constraint to hold with a prescribed probability:

$$\mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon. \quad (4)$$

The risk tolerance $\varepsilon \in (0, 1)$ is small, typically between 0.001 and 0.1 depending on the stakes, and $1 - \varepsilon$ is the reliability level. Parts of the literature, including the convergence theory of Schuster (2025), write the required satisfaction probability directly as $\alpha = 1 - \varepsilon$; the two conventions are interchangeable, and this thesis works with ε throughout. Figure 3 places the three formulations on a single axis for the dispatch example. Rather than a compromise between the extremes, the chance constraint can

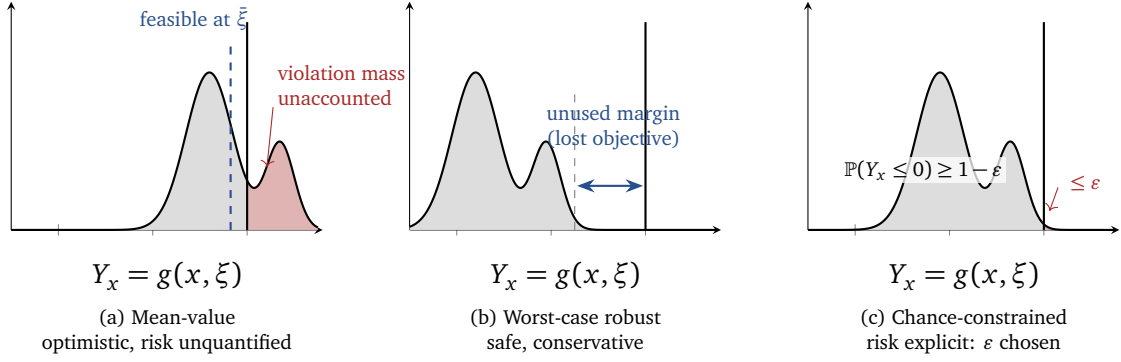


Figure 4: Three responses to constraint uncertainty, shown on the density of the residual $Y_x = g(x, \xi)$; the constraint is satisfied for $Y_x \leq 0$. **(a)** The mean-value solution satisfies the constraint at $\bar{\xi}$ but leaves an unquantified violation mass (red). **(b)** The worst-case robust solution eliminates all violations by pushing the distribution well clear of zero, at the cost of large unused margin. **(c)** The chance-constrained solution places exactly the permitted fraction ε of probability mass beyond the boundary, making the accepted risk explicit and controllable.

be viewed as generalizing both of them. As $\varepsilon \rightarrow 0$ it approaches worst-case protection over the support of ξ , and at $\varepsilon = 0.5$ it roughly reproduces the mean-value behavior for symmetric residuals. Every intermediate point is available, and each carries an explicit, auditable meaning: the system fails at most a fraction ε of the time. This closely mirrors the language in which engineering requirements are commonly written. Grid codes prescribe loss-of-load expectations, flight rules prescribe acceptable failure probabilities per descent, and risk mandates prescribe loss probabilities per quarter. The chance constraint is the optimization-native translation of a reliability requirement, which is why it is a natural framing for many engineering and operational problems, and why the difficulty of solving it, the subject of Section 1.3, has remained an active research frontier for over six decades. Figure 4 makes the three positions concrete. Each panel shows the same bimodal residual density $Y_x = g(x, \xi)$, with the constraint boundary at zero; the two modes reflect the non-Gaussian disturbance shapes that motivate the nonparametric approach developed from subsection 1.4 onward

1.3 What Chance Constraints Are and Why They Are Hard

This section fixes the notation used throughout the thesis, formalizes individual and joint chance constraints, and isolates the three structural properties that make probabilistic constraints computationally demanding. These three difficulties also organize the survey of solution methods in Chapter 2.

1.3.1 Individual and joint chance constraints

This section fixes notation used throughout the thesis and isolates the three structural sources of computational difficulty. Let $x \in X \subseteq \mathbb{R}^n$ be the decision vector, let ξ be a random vector on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ taking values in $\mathbb{R}^d$, and let $g : \mathbb{R}^n \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a constraint function. The *individual chance constraint* is

$$\mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon. \quad (5)$$

When the system carries several uncertain requirements $g_1, \dots, g_R$ that must hold simultaneously, the appropriate model is the *joint chance constraint*

$$\mathbb{P}(g_r(x, \xi) \leq 0, r = 1, \dots, R) \geq 1 - \varepsilon, \quad (6)$$

which bounds the probability that *all* requirements hold at once. The distinction matters: enforcing each g_r individually at level $1 - \varepsilon$ does not bound the joint failure probability, while the classical repair via Boole's inequality, splitting the budget as $\sum_r \varepsilon_r \leq \varepsilon$, is sound but conservative (Prékopa, 1995; Keil u. a., 2021). The full chance-constrained program studied in this thesis is

$$\underset{x \in X}{\text{minimize}} \ f(x) \quad \text{subject to} \quad \mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon. \quad (7)$$

A word on the role of ε is warranted, because it is the one quantity in Eq. (7) that mathematics cannot supply. The risk tolerance is a design parameter set by the consequences of failure and by external requirements: a 5% daily shortfall tolerance may be acceptable for a microgrid with backup supply (Ciftci u. a., 2019), while a descent constraint may demand $\varepsilon \leq 10^{-2}$ or far less (Blackmore u. a., 2010). Two facts about this parameter shape everything downstream. First, the difficulty of the problem grows as ε shrinks, because the constraint then probes ever deeper into the tail of a distribution that must be known or estimated ever more accurately there. Second, ε is a budget, and for joint constraints Eq. (6) the budget must cover all R requirements simultaneously, which is what makes the joint form strictly harder than the individual one. Throughout the thesis, ε is treated as given and the question is how to honor it efficiently and honestly.

1.3.2 Three structural sources of difficulty

Writing Eq. (7) is easy; solving it is not, and the obstruction is threefold. Each difficulty below is independent of the others, and any viable solution method must answer all three. The pattern repeats through the entire methodological literature, so these three points are the lens through which every method in this thesis, including the KDE reformulation itself, should be judged.

Difficulty 1: the feasible set is generally non-convex. Convexity of g in x does not transfer to the probabilistic feasible set $\{x : \mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon\}$. A minimal example makes the geometry vivid. Take a linear constraint $\xi^\top x \leq 1$ where ξ takes one of two values $\xi^{(1)}, \xi^{(2)}$ with probability one half each, and set $\varepsilon = 0.5$. The chance constraint then requires $\xi^\top x \leq 1$ for *at least one* of the two scenarios, so the feasible set is the union of two half-spaces. Figure 5(a) shows the result: two points, each satisfying a different scenario, are individually feasible, while their midpoint satisfies neither and is infeasible. A union of convex sets is not convex, and this toy construction scales up: with general discrete or continuous distributions the feasible set can shatter into combinatorially many pieces. Convexity survives only under special structure, most famously when the distribution of ξ is log-concave and g has suitable form, the well-known theory of Prékopa (1995), or when ξ is Gaussian and the constraint is linear in ξ , in which case Eq. (5) collapses to a second-order cone constraint (Charnes u. Cooper, 1963; Shapiro u. a., 2014). Outside these islands, non-convexity is the rule, and global optimality claims must be abandoned in favor of high-quality local solutions, which is precisely the regime in which nonlinear programming solvers operate.

Difficulty 2: the probability has no closed form. Define the parameterized failure region $M(x) = \{\xi : g(x, \xi) > 0\}$. Evaluating the constraint in Eq. (7) means computing

$$\mathbb{P}(g(x, \xi) \leq 0) = 1 - \int_{M(x)} \rho_\xi(\xi) d\xi, \quad (8)$$

an integral of the density ρ_ξ over a region whose shape changes with every candidate decision x . For nonlinear g the region $M(x)$ is curved and possibly disconnected, the integral is multidimensional, and a closed form exists only in textbook cases. Worse, in the data-driven settings that motivate this thesis the density ρ_ξ itself is unknown; all that exists is a finite collection of samples from simulation, historical records, or exper-

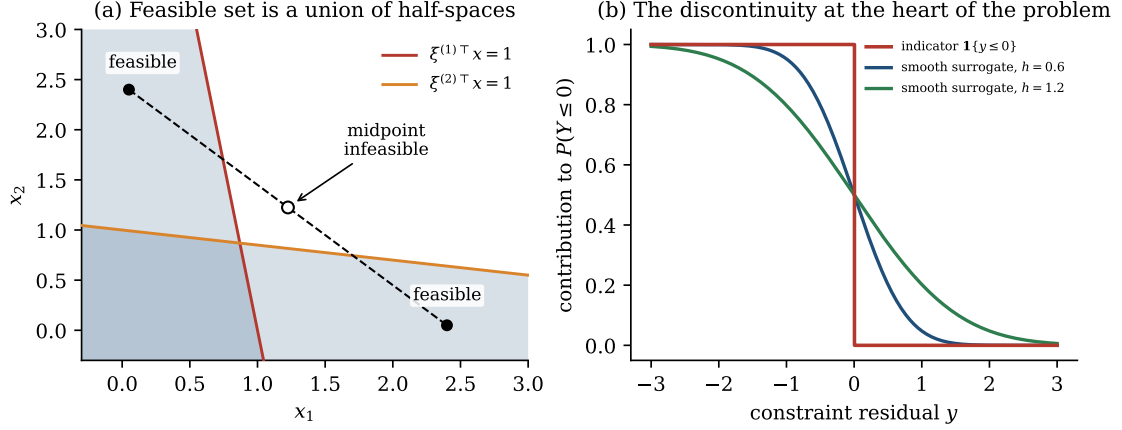


Figure 5: The two geometric faces of chance-constraint hardness. (a) For a two-point distribution of ξ and $\varepsilon = 0.5$, the feasible set of $\mathbb{P}(\xi^\top x \leq 1) \geq 0.5$ is the union of two half-spaces (shaded). The marked endpoints are feasible while their midpoint is not, certifying non-convexity. (b) The contribution of a single sample to the probability is the discontinuous indicator $\mathbf{1}\{y \leq 0\}$ of the residual; its derivative is zero almost everywhere and undefined at the jump. Smooth surrogates obtained by integrating a kernel, shown for two bandwidths h , restore the differentiability that gradient-based solvers require, at the price of a bias controlled by h . This single picture contains the central tension of the thesis.

iments. The constraint function that the optimizer must evaluate, thousands of times, at thousands of points x , is therefore an unknown integral of an unknown density over an intractable region. Any practical method replaces Eq. (8) with an approximation, and the quality of that approximation governs everything downstream.

Difficulty 3: the probability is built from a discontinuous function. Rewriting the probability as an expectation,

$$\mathbb{P}(g(x, \xi) \leq 0) = \mathbb{E}[\mathbf{1}\{g(x, \xi) \leq 0\}], \quad (9)$$

exposes the third obstruction: the indicator $\mathbf{1}\{\cdot\}$ jumps from one to zero at the constraint boundary, as plotted in Fig. 5(b). The natural sample-based estimate of Eq. (9), count the fraction of samples satisfying the constraint, inherits this discontinuity. As a function of x it is piecewise constant: its gradient is zero almost everywhere and undefined on a measure-zero set, which is poorly suited to the sequential quadratic programming and interior-point methods that are widely used in nonlinear optimization (Shapiro u. a., 2014). The established escape routes each pay a recognizable price. Mixed-integer reformulations of the counting constraint are exact but combinatorial and scale poorly (Luedtke u. Ahmed, 2008). The scenario approach enforces the

constraint on all N sampled scenarios, regaining convexity and even finite-sample feasibility certificates for convex problems (Calafiore u. Campi, 2006; Campi u. Garatti, 2008), but it cannot exploit the slack that ε permits and grows conservative as N increases. Conservative convex surrogates such as conditional value-at-risk restrict the feasible set from inside (Rockafellar u. Uryasev, 2000; Nemirovski u. Shapiro, 2006), guaranteeing feasibility at a cost in objective value. Smoothing methods, the family to which this thesis belongs, replace the indicator with a differentiable approximation (Peña-Ordieres u. a., 2020), restoring the gradients that nonlinear solvers need while introducing a controlled bias, visible in Fig. 5(b) as the gap between the step and the smooth curves.

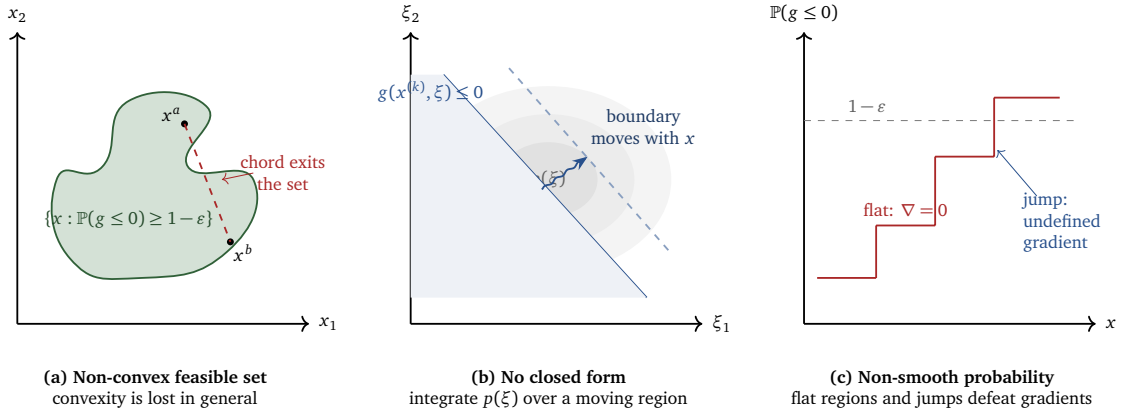


Figure 6: The three structural sources of difficulty in chance-constrained programming: (a) the feasible set is non-convex, (b) the constraint probability has no closed form because the safe region moves with x , and (c) the probability function is flat almost everywhere and non-differentiable at the boundary.

The combination of the three difficulties, and not any one of them alone, is what makes Eq. (7) substantially harder than a deterministic NLP. Non-convexity generally precludes global guarantees, the missing closed form precludes exact evaluation, and non-smoothness makes naive sample averaging unsuitable inside derivative-based solvers. Every approximation method discussed in this thesis is an attempt to buy back tractability by conceding something: exactness, generality, or a margin of conservatism. The contribution of the KDE-based reformulation is a specific and unusually favorable bargain on all three fronts, and the remainder of this chapter explains why.

1.4 The Case for Nonparametric Estimation

This section develops the argument for estimating distributions from data. It reviews the parametric alternative and the evidence against its default assumptions, quantifies the consequences of a misspecified distribution, introduces kernel density estimation, and closes with an account of the costs that the nonparametric route incurs.

1.4.1 The parametric assumption and its routine failure

Among the classical escape routes from Section 1.3, the oldest and still most widely used is parametric: assume the law of ξ belongs to a named family, almost always the Gaussian, and exploit the structure that follows. Under Gaussian ξ and a constraint linear in ξ , the chance constraint Eq. (5) reduces exactly to a deterministic second-order cone constraint involving the mean, the covariance, and a quantile of the standard normal (Charnes u. Cooper, 1963; Shapiro u. a., 2014). The reformulation is convex, smooth, and supported by every modern conic solver. When the Gaussian assumption holds, this reformulation is difficult to improve upon, and the experimental chapters use it as the reference baseline for that reason.

The assumption, however, is routinely false in the very domains where chance constraints matter most, and the evidence is not anecdotal. Wind-power forecast errors are skewed and heavy-tailed, with documented deviations from normality that grow in the tails, which is precisely where a chance constraint lives (Hodge u. Milligan, 2011; Wan u. a., 2020). Equity and commodity returns exhibit heavy tails and aggregational Gaussianity only at long horizons; at decision-relevant horizons their kurtosis is far above three (Cont, 2001). Thrust disturbances and navigation errors in powered descent arise from mixtures of operating regimes and are naturally multimodal (Keil u. a., 2021). Process feedstocks vary by supplier and batch in ways no single unimodal family captures (Calfa u. a., 2015). The Gaussian is often chosen more for analytical convenience than for its fit to the observed behavior of the system.

1.4.2 What a wrong distribution does to the guarantee

The consequences of the mismatch are asymmetric and quantifiable. Figure 7 dissects both failure modes. Panel (a) shows a bimodal disturbance law next to its moment-matched Gaussian fit: the fit invents probability where there is none and erases the two modes that actually govern system behavior. Panel (b) shows the more dangerous

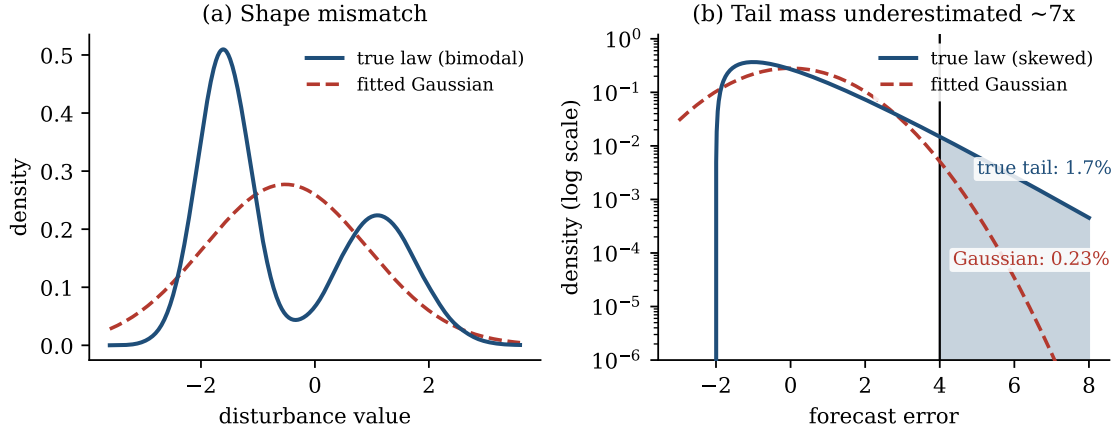


Figure 7: What a wrong distributional assumption does to a chance constraint. (a) A bimodal disturbance law, of the kind produced by two-regime actuator behavior, against its best Gaussian fit: the fit places probability mass between the modes where the true law has almost none, and misplaces both shoulders. (b) A skewed forecast-error law (centered Gamma) on a logarithmic density scale. Beyond the marked threshold the Gaussian fit reports 0.23% tail mass where the true law carries 1.7%, an underestimate by a factor of about seven. A planner who certifies a 0.5% risk level with the Gaussian model deploys a system whose true failure rate is more than three times the certificate.

case on a logarithmic scale. For a skewed forecast-error law, the Gaussian fit underestimates the mass beyond a critical threshold by a factor of seven: it reports 0.23% where the truth is 1.7%. A decision certified to a 0.5% failure probability under the Gaussian model in fact fails more than three times as often as certified. The error can also point the other way: when the true law is lighter-tailed than the fit, the parametric constraint is needlessly conservative and pays an invisible objective penalty. Whether the result is optimistic or conservative, the direction is governed by distributional geometry that the parametric fit does not capture, and the certificate that the chance constraint was intended to provide, $\mathbb{P}(\text{failure}) \leq \varepsilon$, may no longer hold. A reliability guarantee that rests on an unverified distributional assumption therefore provides considerably weaker assurance than its form suggests.

1.4.3 Kernel density estimation and its fit to optimization

The alternative pursued in this thesis is to estimate the distribution from the data instead of assuming it. Kernel density estimation, the canonical nonparametric estimator introduced by Rosenblatt (1956) and Parzen (1962), reconstructs a density

from samples $y_1, \dots, y_N$ as

$$\hat{\rho}_h(y) = \frac{1}{Nh} \sum_{j=1}^N K\left(\frac{y - y_j}{h}\right), \quad (10)$$

where the kernel K is a smooth density and the bandwidth $h > 0$ sets the degree of smoothing. The estimator makes no shape commitment: given enough data it recovers skewness, multimodality, and heavy tails alike, with well-understood consistency theory and practical bandwidth rules (Silverman, 1986; Devroye u. Györfi, 1985). Three properties make KDE specifically suited to the optimization context, rather than merely to statistics. First, $\hat{\rho}_h$ is as smooth as its kernel, so quantities built from it are differentiable, which addresses Difficulty 3 head-on. Second, its integral is available in closed form whenever the kernel's cumulative distribution function is, which addresses the evaluation half of Difficulty 2. Third, it is distribution-free, which removes the implicit distributional assumption discussed above.

This is not a speculative combination. A coherent body of work already demonstrates KDE-based chance constraints across the application domains of Section 1.1: kernel-smoothed individual and joint chance constraints in process planning (Calfa u. a., 2015), KDE-reformulated optimal control for launcher ascent and the Goddard problem (Caillau u. a., 2018), biased-KDE optimal control for soft lunar landing (Keil u. a., 2020, 2021; Keil u. Rao, 2021), adaptive-KDE microgrid energy management (Ciftci, 2017; Ciftci u. a., 2019; Babić u. a., 2023), multivariate-KDE joint chance constraints for economic dispatch and power-system scheduling (Wu u. a., 2021; Wu u. Kargarian, 2023), KDE-driven distributionally robust load shedding (Qi u. a., 2026), and KDE-based tail-risk portfolio optimization (Liu u. a., 2022). Adjacent nonparametric formulations reinforce the same direction in robotics and forecasting (Gopalakrishnan u. a., 2018; Wan u. a., 2020; Wang u. a., 2021). What this literature has lacked, until recently, is a unifying account connecting the engineering constructions to convergence theory; supplying and stress-testing that connection is the work of this thesis.

1.4.4 The price of going nonparametric

A nonparametric estimator does not come free, and the costs are named now because they structure the experimental program of the application chapter. KDE suffers the curse of dimensionality: its convergence rate degrades rapidly as the dimension of the estimated variable grows (Silverman, 1986). The bandwidth h is a genuine hyperpa-

parameter whose role inside an optimization problem, where it shapes the feasible region and the gradients the solver sees, is more delicate than its role in density estimation. A plain KDE smooths in both directions, so it can overestimate the safe probability and thereby weaken the very bound it is meant to enforce. A fourth cost concerns the tails: a chance constraint with a small violation budget asks about a rare event, and a kernel estimate is least accurate in sparsely sampled tail regions. The severity of this effect is problem dependent rather than universal, since at moderate risk levels the binding region lies in the bulk of the distribution, where the estimator behaves well, while stringent levels push it into the tails. Each of these costs has a specific answer in this thesis: the first is mitigated by the residual-space construction of Section 1.5, the second is treated as a first-class experimental question, the third is addressed by the biased-KDE construction of Keil u. a. (2021), and the fourth is handled by reporting results across a range of risk levels.

1.5 The Residual-Space Insight

This section introduces the change of variables on which the methodological part of the thesis is built. It defines the constraint residual, explains why estimating its distribution is preferable to estimating the distribution of the uncertainty itself, and previews the treatment of joint constraints together with the role of the bandwidth.

1.5.1 From uncertainty space to residual space

Everything assembled so far points toward a single conceptual move, simple to state and central to the approach developed in this thesis. The naive way to use KDE in Eq. (7) is to estimate the density of the uncertainty ξ itself and then integrate the estimate over the safe region $\mathbb{R}^d \setminus M(x)$. This inherits both halves of Difficulty 2: the estimation runs in d dimensions, exactly where the curse of dimensionality bites, and the integration region still deforms with every candidate x . A more effective object to estimate is not the uncertainty itself but the *constraint residual*

$$Y(x) = g(x, \xi), \tag{11}$$

the scalar random variable obtained by pushing the uncertainty through the constraint function at the current decision. The residual is the quantity the constraint actually speaks about: $Y(x) \leq 0$ is constraint satisfaction, by definition, and the chance con-

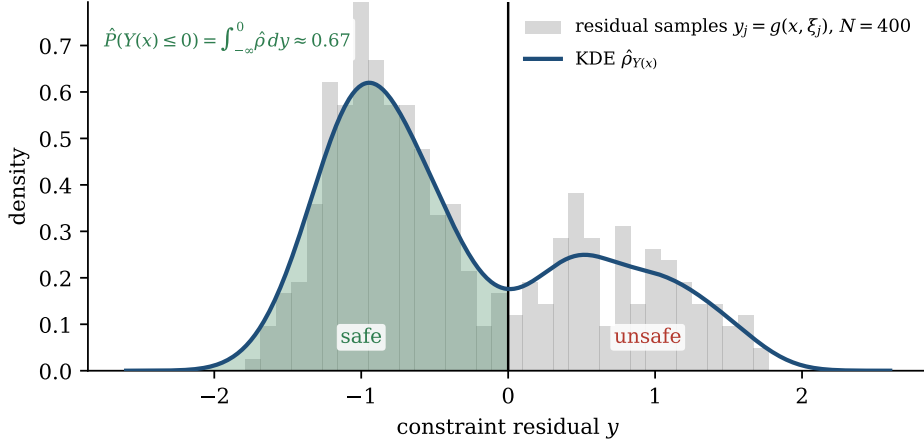


Figure 8: The residual-space construction at a fixed candidate decision x . Samples ξ_j of the uncertainty are pushed through the constraint to give residual samples $y_j = g(x, \xi_j)$ (histogram, $N = 400$, bimodal by construction); a KDE $\hat{\rho}_{Y(x)}$ smooths them, and its integral over the fixed safe region $(-\infty, 0]$ (shaded) estimates the safe probability, here $\hat{P} \approx 0.67$. The estimation problem is one-dimensional regardless of the dimension of ξ , and the resulting constraint $\hat{F}_{Y(x)}(0) \geq 1 - \varepsilon$ is a smooth, differentiable function of x .

straint Eq. (5) is nothing but a statement about the cumulative distribution function of $Y(x)$ at zero,

$$\mathbb{P}(Y(x) \leq 0) = F_{Y(x)}(0) \geq 1 - \varepsilon. \quad (12)$$

Schuster (2025) makes this change of variables the foundation of the convergence theory: the probability is rewritten as an integral of the density of the *transformed* variable $g(x, \xi)$ over the fixed set $(-\infty, 0]$, rather than an integral of the density of ξ over the moving set $\mathbb{R}^d \setminus M(x)$.

1.5.2 Three dividends of the change of variables

The benefits of this shift are worth enumerating, because each one addresses a specific difficulty from Section 1.3.

First, *dimensionality collapses*. The uncertainty ξ may live in ten or a hundred dimensions; the residual of an individual constraint is a scalar, always, and the residuals of a joint constraint form a vector of dimension R , the number of constraint components, which in the problems of this thesis is small. This is where the *curse of dimensionality*, named as the first cost of nonparametric estimation in Section 1.4.4, is addressed. The convergence rate of a kernel estimator deteriorates rapidly with the dimension of the estimated variable, so that holding the accuracy fixed requires a sample size

that grows roughly exponentially in that dimension (Silverman, 1986). The decisive observation is that the relevant dimension for the residual-space construction is the dimension of the estimated variable, not of the uncertainty: KDE is asked to perform one-dimensional density estimation from a few hundred samples, the task it is best suited to, and the curse in d is largely avoided by construction. The constraint function g itself performs the dimension reduction, and it does so without information loss for the question being asked, since Eq. (12) is an identity, not an approximation. The mitigation is not unconditional, however; it weakens when many constraints must hold jointly, since the residual vector then carries the dimension R , a point taken up in Section 1.5.3.

Second, *the integration region freezes*. The safe region in residual space is $(-\infty, 0]$ for every decision x ; all dependence on x has migrated into the samples $y_j = g(x, \xi_j)$ themselves. Integrating the kernel estimate over a fixed half-line is elementary: substituting Eq. (10) gives the estimated CDF

$$\hat{F}_{Y(x)}(0) = \frac{1}{N} \sum_{j=1}^N \mathcal{K}\left(\frac{-g(x, \xi_j)}{h}\right), \quad \mathcal{K}(t) = \int_{-\infty}^t K(s) ds, \quad (13)$$

a finite sum of evaluations of the kernel's CDF $\mathcal{K}$, available in closed form for every kernel used in this thesis.

Third, *smoothness is restored*. Comparing Eq. (13) with the sample-average form Eq. (9) reveals the substitution at the heart of the method: the discontinuous indicator $\mathbf{1}\{g(x, \xi_j) \leq 0\}$ has been replaced by the smooth sigmoid $\mathcal{K}(-g(x, \xi_j)/h)$, exactly the curves drawn in Fig. 5(b). If g is differentiable in x , then so is $\hat{F}_{Y(x)}(0)$, with gradients obtained by the chain rule through $\mathcal{K}$. The chance constraint Eq. (5) thereby becomes the deterministic, smooth, nonlinear constraint

$$\hat{F}_{Y(x)}(0) \geq 1 - \varepsilon, \quad (14)$$

which any sequential quadratic programming, interior-point, or direct-collocation solver consumes natively. Figure 8 shows the whole construction in one frame: residual samples, the kernel estimate, the frozen safe region, and the resulting probability estimate.

1.5.3 Joint constraints, bandwidth, and the safety guarantee

The construction extends to joint chance constraints with one further idea that previews the extension studied in this thesis. For Eq. (6), the natural scalar summary of the constraint system is the worst violation, $Y(x) = \max_r g_r(x, \xi)$, since all requirements hold exactly when the maximum residual is nonpositive. The maximum preserves the one-dimensionality of the estimation problem but reintroduces a kink; smoothing it with a log-sum-exp surrogate restores differentiability at the cost of a second, controllable approximation layer. This scalar violation statistic, its interaction with the KDE layer, and the alternative of low-dimensional multivariate KDE over the residual vector form the substance of research question RQ4 and are developed in Part 3.

The reformulation Eq. (14) is where the thesis begins rather than ends, because the two costs flagged in Section 1.4 are still standing and they define the methodological arc of Part 3. The bandwidth h now controls the geometry of an optimization problem, not just the fidelity of a density plot: $h \rightarrow 0$ recovers the non-smooth indicator and its numerical pathologies, while large h oversmooths and distorts the feasible region, and no classical bandwidth rule was derived with either effect in mind. And because an ordinary KDE can overestimate $F_{Y(x)}(0)$, the constraint Eq. (14) can certify a reliability the true system does not possess. The constructive answer is the biased KDE of Keil u. a. (2020, 2021): kernels engineered so that their integral function bounds the indicator from the safe side, making Eq. (14) a guaranteed conservative approximation of the true chance constraint while remaining smooth and bounded. The theoretical answer is the convergence framework of Schuster (2025), which establishes conditions under which solutions of the KDE-approximated problem converge to solutions of the true problem as $N \rightarrow \infty$. Engineering construction and convergence theory, developed independently, meet in residual space, and their meeting is the subject of this thesis.

1.6 Thesis Scope and Positioning

This closing section delimits the scope of the thesis. It positions the work between its two anchor papers, states which questions lie outside the scope, and formulates the four research questions together with a roadmap of the remaining chapters.

1.6.1 Disciplinary positioning and anchor literature

A thesis at the intersection of three fields must say plainly what it is and what it is not. This thesis is not a contribution to nonparametric statistics: it proves no new properties of kernel density estimators, and it takes the consistency theory of Parzen (1962); Silverman (1986); Devroye u. Györfi (1985) as given. It is not a contribution to control theory: it establishes no stability or recursive feasibility results for closed-loop predictive control. It is a data science and applied optimization contribution. Its object of study is the KDE-based reformulation pipeline itself, from samples through residuals through Eq. (14) into a numerical solver, and its method is to implement that pipeline rigorously, analyze its behavior, compare it against the established alternatives, and validate it across problem settings chosen to stress its assumptions. One further boundary deserves explicit mention. Every guarantee developed in this thesis is a statement about the distribution from which the samples were drawn. Whether such a guarantee transfers to the distribution encountered in deployment, when samples come from a simulator or from historical records that represent the target system imperfectly, is a separate question that this thesis does not address; distributionally robust formulations (Chen u. a., 2024) are a natural direction for that question.

Two anchor papers bound the territory. Keil u. a. (2021), with the earlier preprint Keil u. a. (2020) and the computational follow-up Keil u. Rao (2021), supply the constructive, engineering side: biased kernels whose integral functions bound the indicator, a safety-preserving deterministic reformulation, and a demonstration on chance-constrained soft lunar landing solved by direct collocation. Schuster (2025) supplies the analytical side: convergence theorems guaranteeing that, under verifiable conditions on the kernel, the bandwidth, and the problem, solving the estimated problem is asymptotically equivalent to solving the true one. The first anchor gives the method its practical credibility, the second its mathematical license; the thesis builds the bridge between them and load-tests it experimentally. The surrounding evidence base spans aerospace (Caillaud u. a., 2018), energy systems (Ciftci u. a., 2019; Wu u. a., 2021; Wu u. Kargarian, 2023; Qi u. a., 2026), process systems (Calfa u. a., 2015), and finance (Liu u. a., 2022), and the comparative baselines are the classical ones: scenario optimization (Calafiore u. Campi, 2006; Campi u. Garatti, 2008), sample average approximation (Luedtke u. Ahmed, 2008; Pagnoncelli u. a., 2009), parametric reformulation (Charnes u. Cooper, 1963), and convex risk surrogates (Rockafellar u. Uryasev, 2000; Nemirovski u. Shapiro, 2006). Further smooth alternatives exist for structured settings, notably polynomial chaos expansions (Xiu u. Karniadakis, 2002)

and spheric-radial gradient formulae for Gaussian-like laws (van Ackooij u. Henrion, 2014); a systematic benchmark against these lies outside the scope of this thesis and is noted as future work.

1.6.2 Research questions and roadmap

Four research questions organize everything that follows.

- **RQ1: formulation** how is a chance-constrained problem reformulated through KDE over constraint residuals, for individual and joint constraints, and under what conditions is the reformulation mathematically justified?
- **RQ2: numerical behavior** how does the KDE reformulation compare with deterministic, scenario-based, and parametric approaches in feasibility, objective value, conservatism, runtime, and sensitivity to sample size?
- **RQ3: bandwidth and smoothing** how do bandwidth and smoothing choices shape feasibility, conservatism, and solver performance, given that h acts on the optimization geometry and not only on estimation error?
- **RQ4: extension** can the framework be strengthened through one focused extension, specifically scalar violation statistics for joint chance constraints, optionally with log-sum-exp smoothing?

The argument now proceeds in three movements. Part 2 supplies the technical vocabulary: the taxonomy of optimization under uncertainty, the theory of kernel density estimation, and the elements of numerical optimal control. Part 3 develops the mathematics in the order a practitioner would discover it: the blackbox sample-counting approach and its failure inside gradient-based solvers, the smooth unbiased KDE reformulation, the biased construction that restores the safety guarantee, and the convergence theory that certifies the whole. Part 4 validates the framework experimentally, from controlled benchmarks through the lunar landing problem, and distills the results into practical guidance. The aim is a defensible answer to a question that each of the systems in Section 1.1 raises implicitly: how safe is safe enough, and what is the cheapest decision that reliably delivers it?

2 Stochastic Optimization and Chance Constraints

Chapter 1 established two observations that this chapter takes as its starting point. The chance constraint is a natural mathematical language for reliability requirements, because its protection level is an explicit parameter under the decision maker's control. And it is hard to enforce for exactly three structural reasons: a generally non-convex feasible set, a probability with no closed form, and an indicator function with no usable derivative. What Chapter 1 motivated, this chapter equips. The argument proceeds in a deliberate order, because each piece is the precondition of the next. Section 2.1 fixes the taxonomy of optimization under uncertainty, since the claim that chance constraints occupy a unique position can only be defended once the alternatives are stated precisely. Section 2.2 develops the classical theory and its convexity islands, establishing both how much structure the field has secured and how little of it survives contact with realistic distributions. Section 2.3 then confronts the consequence: six approximation methods, surveyed in a fixed rhythm of core idea, main advantage, and primary limitation, each method responding to a limitation of the one before it. The survey includes the blackbox approach, the most assumption-free option of all, developed there in enough depth, with computed evidence, to motivate the methodological development of Part 3. With the landscape mapped and its gap identified, Section 2.4 introduces the instrument this thesis uses to fill it, kernel density estimation, built up from first principles because optimization readers will rarely have met it as an optimization tool. Section 2.5 closes the chapter by organizing the literature that has already combined KDE with chance constraints into a single narrative, whose one open end is the work of this thesis.

2.1 A Taxonomy of Optimization Under Uncertainty

This section organizes the main paradigms of optimization under uncertainty along two axes, the distributional information a model consumes and the protection its solution provides. It first characterizes the four classical answers and then places them on a single conceptual map, which locates the position that chance-constrained programming occupies.

2.1.1 Four paradigms

Every approach to optimization under uncertainty answers the same two questions: how much information about the random parameter ξ the model consumes, and how much protection against ξ the solution provides. These two axes are more than a convenience of exposition; they capture much of how the field divides in practice. Four canonical answers organize the classical landscape, and fixing them precisely now supports the position claimed for chance constraints in Section 2.1.2.

Deterministic optimization consumes the least information and provides the least protection. The random parameter is replaced by a single representative value, almost always its mean or a nominal forecast, and the problem is solved as if that value were certain. The formulation, its systematic failure mechanism, and the near 50% violation rates it produces were dissected at length in Chapter 1 and need no repetition; in this taxonomy it serves as the zero point against which every other paradigm is measured.

Robust optimization consumes set information: the modeler specifies an uncertainty set Ξ believed to contain every relevant realization, and feasibility is enforced for all of them simultaneously, $g(x, \xi) \leq 0$ for all $\xi \in \Xi$ (Soyster, 1973; Ben-Tal u. a., 2009). The resulting problems are minimax in character, and a mature duality theory converts many of them into tractable deterministic equivalents. Protection is maximal over Ξ , but it is an all-or-nothing property: every point of the set is defended at equal priority, including points of vanishing probability, and the objective pays for all of them.

Stochastic programming consumes the full distribution and optimizes in expectation. In its two-stage form, a here-and-now decision x is taken before ξ is revealed and a recourse decision $y(\xi)$ repairs the outcome afterwards, with the objective $f(x) + \mathbb{E}[Q(x, \xi)]$ where Q is the optimal recourse cost (Dantzig, 1955; Birge u. Louveaux, 2011). The paradigm is the natural choice when violations are not failures but costs: a shortfall that can be bought back at a price belongs in the objective, not in a constraint. Its protection statement is correspondingly soft. The solution is good on average, but nothing bounds the probability that any individual realization behaves badly.

Chance-constrained programming, the framework this thesis lives in, also consumes the full distribution (or samples from it) but spends that information on a hard probabilistic guarantee rather than an average: $\mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon$ (Charnes u. Cooper, 1959; Prékopa, 1995). Where stochastic programming prices violations, chance constraints ration them, and the ration ε is chosen by the decision maker. This is the

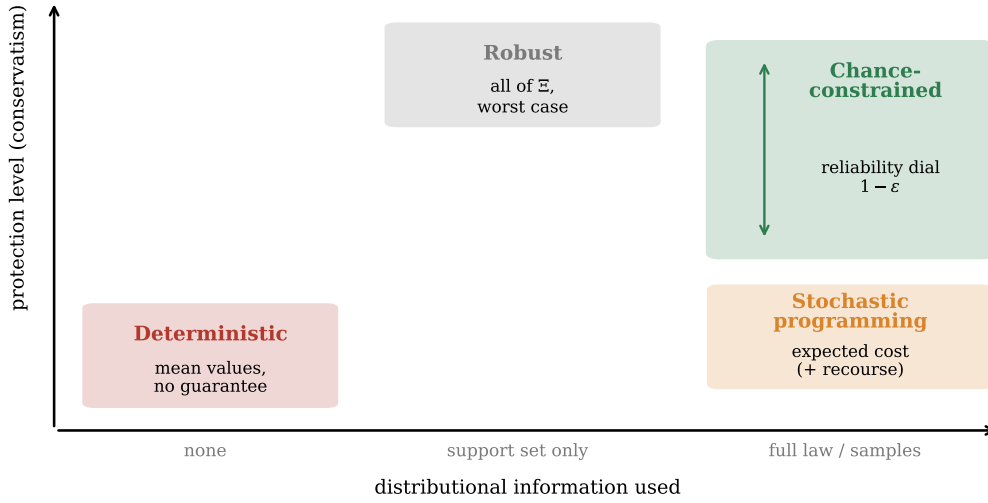


Figure 9: The four paradigms of optimization under uncertainty arranged by the distributional information they consume (horizontal) and the protection they provide (vertical). Deterministic optimization uses point estimates and guarantees nothing; robust optimization defends a whole set at maximal conservatism; stochastic programming optimizes expected cost, possibly with recourse, without bounding individual failure probabilities. Chance-constrained programming is the only paradigm with a continuously tunable protection level, the reliability dial $1 - \epsilon$.

correct paradigm when a violation is a discrete failure event whose frequency, not whose average cost, is regulated: a lander crash, a load-shedding event, a breach of a risk mandate.

2.1.2 A conceptual map and where this thesis sits

Figure 9 arranges the four paradigms on the two axes just described. Reading the map yields three observations that recur throughout the thesis. First, the paradigms are not competitors on a single scale of sophistication; they occupy different points of a two-dimensional trade-off, and each is the right tool somewhere, as Chapter 1 acknowledged for the two extremes. Second, chance-constrained programming is the only paradigm whose protection level is a continuous dial rather than a fixed point: sweeping ϵ from 0.5 toward 0 moves the solution continuously from mean-value behavior toward worst-case behavior over the support. Third, the dial comes at the price identified in Chapter 1: the probability operator destroys the convexity, the evaluability, and the smoothness that the other three paradigms retain, which is why the approximation methods of Section 2.3 exist at all.

Two boundary remarks complete the map. Hybrid formulations exist and are actively studied: distributionally robust chance constraints enforce the probabilistic guarantee for every distribution in an ambiguity ball around the empirical law, combining the chance-constrained dual with robust-style protection against estimation error (Chen *u. a.*, 2024; Liu *u. a.*, 2022). They reappear in Section 2.3.7 as a row of the comparison table. And the boundaries between paradigms are formulation choices, not laws of nature: the same engineering problem can often be written in several of them, and part of the modeler’s craft is recognizing which paradigm matches the actual reliability language of the application. The applications of this thesis, surveyed in Chapter 1, all speak in failure probabilities, which fixes the choice.

2.2 Classical Chance-Constrained Programming

The taxonomy assigned chance-constrained programming its position; this section establishes its content. The order of business is the one any mature theory imposes: first the canonical formulations and the equivalent forms the rest of the thesis computes with, then the strongest structural result the field possesses, and finally an honest accounting of how far that result reaches. The accounting matters, because the methods of Section 2.3 exist precisely in the territory the classical theory does not cover, and the size of that territory is the size of the problem this thesis addresses.

2.2.1 Origins and canonical formulations

Chance constraints emerged from a concrete scheduling problem. Planning heating-oil production under uncertain demand, Charnes *u. a.* (1958) faced requirements that no fixed plan could satisfy on every realization, and drew a conclusion the field has built on since: a planning constraint need not hold always, only sufficiently often, and the required frequency belongs in the model as a first-class quantity. Charnes *u. Cooper* (1959) formalized the resulting notion of chance-constrained programming, Charnes *u. Cooper* (1963) supplied early deterministic equivalents for such programs, and Miller *u. Wagner* (1965) extended the framework from individual to joint constraints, recognizing that a system of requirements demands a probability statement about the system, not about each row separately. The decisive consolidation came with Prékopa, whose log-concavity theory (Section 2.2.2) and monograph (Prékopa, 1995) elevated a modeling device into a branch of mathematical programming with its own structural theorems. The formulations fixed in Chapter 1 are restated here in the notation used

for the remainder of the thesis. With decision $x \in X \subseteq \mathbb{R}^n$, random parameter $\xi \in \mathbb{R}^d$ with law $\mathbb{P}$, constraint map g , and risk level $\varepsilon \in (0, 1)$, the individual and joint chance constraints read

$$\mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon, \quad \mathbb{P}(g_r(x, \xi) \leq 0, r = 1, \dots, m) \geq 1 - \varepsilon. \quad (15)$$

A third, equivalent form deserves its own display because the methodological chapters lean on it constantly. Writing the requirement in terms of the violation event,

$$\mathbb{P}(g(x, \xi) > 0) \leq \varepsilon, \quad (16)$$

makes the safety interpretation direct: the optimizer may move x anywhere it likes as long as the estimated probability of violation stays below the budget ε . The violation form is also the form in which sample-based methods naturally operate, since counting violations among samples estimates the left side of Eq. (16) directly. Throughout Part 3, the scalar random variable $Y(x) = g(x, \xi)$ introduced in Chapter 1 carries this event: $\{g(x, \xi) > 0\}$ is exactly $\{Y(x) > 0\}$, and every approximation method in this chapter is at bottom a different way of estimating the upper tail of $Y(x)$ at zero.

The pair in Eq. (15) conceals an asymmetry that practitioners learn quickly. Individual constraints control each requirement in isolation and say nothing about simultaneous satisfaction; the joint constraint controls the system but is strictly harder, both statistically (the event is an intersection) and computationally (the feasible set geometry worsens). The classical bridge is Boole's inequality: enforcing individual constraints at levels ε_r with $\sum_r \varepsilon_r \leq \varepsilon$ implies the joint constraint at level ε , at a cost in conservatism that grows with m . The sharper treatment of joint constraints through a scalar violation statistic is deferred to Part 3, where it forms research question RQ4.

2.2.2 Convexity islands: the log-concavity theory

The single most important positive structural result in the field is due to Prékopa. A measure is log-concave when the logarithm of its density is concave, a class that includes the Gaussian, the uniform on a convex set, the exponential, and many other standard laws. Prékopa's theorem states that if ξ has a log-concave distribution and the set $\{(x, \xi) : g(x, \xi) \leq 0\}$ is convex, then the probability function $p(x) = \mathbb{P}(g(x, \xi) \leq 0)$ is log-concave in x , and consequently every upper level set $\{x : p(x) \geq 1 - \varepsilon\}$ is convex (Prékopa, 1971, 1995, 2003). On this island, chance-constrained programming

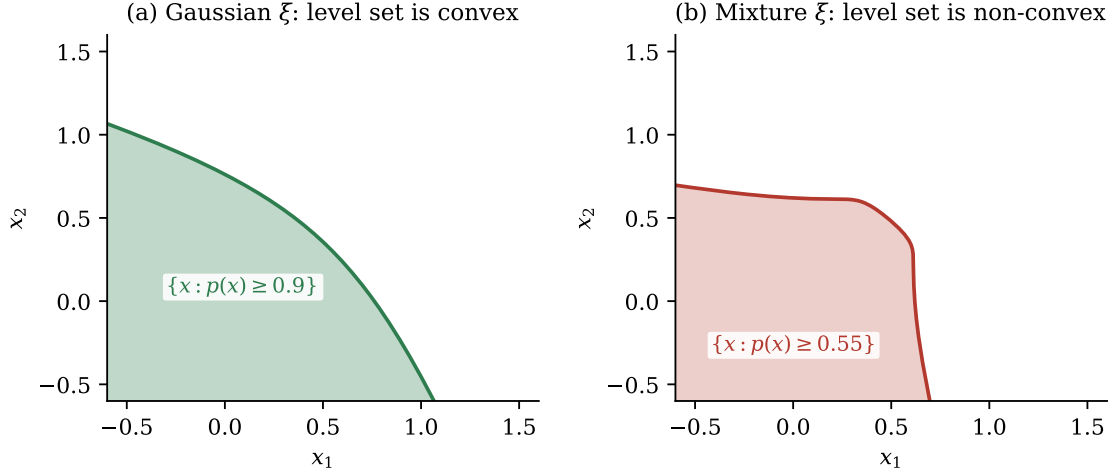


Figure 10: Prékopa’s theorem and its boundary, computed for the bilinear constraint $\xi^\top x \leq 1$ in $n = 2$. (a) For Gaussian ξ , a log-concave law, the probabilistic feasible set $\{x : p(x) \geq 0.9\}$ is convex, as the theory guarantees. (b) For a two-component Gaussian mixture, which is not log-concave, the level set $\{x : p(x) \geq 0.55\}$ develops the non-convex geometry anticipated by the discrete example of Chapter 1: the region bulges along two directions, one per mixture mode, and contains pairs of points whose midpoints are infeasible.

is convex programming: local solutions are global, duality holds, and mature solvers apply.

Figure 10 makes the theorem and its boundary visible in a computed example. For a Gaussian parameter and a bilinear constraint, the 90% level set of $p(x)$ is convex, panel (a). Replacing the Gaussian by a two-component mixture, a minimal departure from log-concavity, produces the non-convex level set of panel (b), the continuous analogue of the union-of-half-spaces construction in Chapter 1. The figure clarifies a question of attribution that matters for everything downstream: the non-convexity here is not caused by the constraint function, which is identical in both panels, but by the shape of the distribution. The geometry of the feasible set is inherited from the geometry of the law of ξ , and modifications on the constraint side alone cannot restore what the distributional side has removed.

This inheritance is exactly why the log-concavity island, elegant as it is, carries so little of the practical load. The distributions documented in the application domains of Chapter 1, skewed and heavy-tailed forecast errors, multimodal disturbance mixtures, heavy-tailed returns, are precisely the laws that leave the log-concave class. The conclusion is worth stating plainly: the positive theory applies where the uncertainty is well behaved, and the uncertainty that motivates chance constraints in the first place often is not. For the problems considered in this thesis, the general case is therefore

the typical case rather than an exception.

2.2.3 The general case

Outside the islands, the three difficulties of Chapter 1 stand in full force, and restating them in the violation-form language of Eq. (16) is more than housekeeping: the methods of the next section are organized by which difficulty each one attacks, and the restatement is the index of that organization. The feasible set $\{x : \mathbb{P}(Y(x) > 0) \leq \varepsilon\}$ is in general non-convex, so global optimality is out of reach and the realistic target is a high-quality local solution from a nonlinear programming method. The probability $\mathbb{P}(Y(x) > 0)$ has no closed form: it is an integral of an often unknown density over a region that moves with x , so it must be estimated, almost always from samples. And the natural sample estimate is an average of indicator functions, a piecewise-constant object whose gradient is zero almost everywhere, useless to exactly the solvers that handle nonlinearity well.

The landscape of Section 2.3 is best previewed as a matrix of concessions. Each classical method buys back tractability by conceding one of three things: *generality* (the parametric route assumes the distribution), *exactness* (the convex restrictions enforce a stronger constraint than required), or *smoothness* (the sample-enforcement routes hand the solver a non-differentiable or combinatorial problem). To the best of our knowledge, no classical method preserves all three at once, which reflects where decades of research have located the trade-offs. The case this thesis builds, beginning in Section 2.4, is that estimating the density of $Y(x)$ with a kernel estimator concedes none of the three outright, paying instead with a bias that can be controlled and, in the biased-KDE construction, even directed to the safe side.

2.3 Standard Approximation Methods and Their Trade-offs

This section positions the thesis relative to the landscape of existing approaches, and it is built as a progression rather than a catalogue. Each method is presented in the same fixed rhythm, core idea, main advantage, primary limitation, and each method enters the survey as the field’s answer to the limitation of the one before it: the scenario approach removes the distributional assumption the Gaussian route cannot survive without, sample average approximation recovers the slack the scenario approach forfeits, the convex restrictions restore the tractability that sample averaging destroys, quantile smoothing recovers the exactness the restrictions surrender, and the blackbox

approach strips away the last remaining assumption and pays the full price of doing so. The section closes with the comparison this ordering has been building toward, first as a table and then as a computed experiment on a problem where the truth is known exactly.

2.3.1 Parametric and Gaussian reformulations

The core idea of the parametric route is to assume the law of ξ and exploit it analytically. The Gaussian case with a constraint linear in ξ is the showpiece. For $g(x, \xi) = \xi^\top x - b$ with $\xi \sim \mathcal{N}(\mu, \Sigma)$, the residual $Y(x) = \xi^\top x - b$ is itself Gaussian with mean $\mu^\top x - b$ and variance $x^\top \Sigma x$, and the chance constraint collapses to an exact deterministic statement,

$$\mathbb{P}(\xi^\top x \leq b) \geq 1 - \varepsilon \iff \mu^\top x + \Phi^{-1}(1 - \varepsilon) \|\Sigma^{1/2} x\|_2 \leq b, \quad (17)$$

where Φ^{-1} is the standard normal quantile (Charnes u. Cooper, 1963; Shapiro u. a., 2014). For $\varepsilon \leq 1/2$ the right side of Eq. (17) is a second-order cone constraint: convex, smooth, and solvable at scale by every modern conic solver.

The advantages within its domain are substantial: exactness, convexity, smoothness, and speed, simultaneously. This is why the Gaussian reformulation dominates practice in fields such as chance-constrained optimal power flow (Bienstock u. a., 2014) and why it serves as the reference baseline of the experimental chapters of this thesis. When the Gaussian assumption holds, the problem identified in Chapter 1 simply does not exist.

The limitation is that the construction rests entirely on the distributional assumption. Each step of Eq. (17) follows from the normality of $Y(x)$, and Chapter 1 quantified what can happen when that normality does not hold: a skewed law whose tail mass is underestimated by a factor of seven turns a certified 0.5% failure rate into an actual rate above 1.5%. The reformulation offers no internal warning in such cases; it reports the same certificate whether or not the underlying assumption is met. Extensions to other named families and to Gaussian-like laws exist, including analytic gradient formulae for the probability function via spheric-radial decomposition (van Ackooij u. Henrion, 2014), but they relocate the assumption rather than remove it.

2.3.2 The scenario approach

If the Gaussian route fails because of its assumption, the natural response is to make none, and the scenario approach makes none with full commitment: it abandons the distribution entirely and enforces the constraint on data. Given independent samples $\xi_1, \dots, \xi_N$, the chance constraint is replaced by N hard deterministic constraints,

$$g(x, \xi_i) \leq 0, \quad i = 1, \dots, N. \quad (18)$$

The randomness now lives in the optimization problem itself, whose solution x_N^* is a random variable, and the theory asks how likely that solution is to satisfy the original chance constraint.

The advantage is the strength of the answer. For convex problems, Calafiore u. Campi (2006) showed that $N \geq \frac{2}{\varepsilon}(\ln \frac{1}{\beta} + n)$ samples suffice for x_N^* to satisfy the chance constraint at level ε with confidence $1 - \beta$, and Campi u. Garatti (2008) sharpened this to an exact, distribution-free feasibility certificate. These are finite-sample, a priori guarantees of a kind almost no other method offers, obtained without assuming anything about the law of ξ . The implementation is also disarmingly simple: sample, then solve a deterministic program.

The limitations are structural. The program grows with N , one hard constraint per sample, so precisely the regime where the statistics are comfortable is the regime where the optimization is large. The approach cannot exploit the slack the chance constraint permits: it defends every sampled scenario, including the $\lfloor \varepsilon N \rfloor$ worst ones it is entitled to ignore, so it is conservative by construction and grows more conservative as N increases, a behavior the experiment of Section 2.3.7 displays vividly. And the guarantees are proven for convex problems; in the nonlinear optimal control settings of this thesis, they do not transfer.

2.3.3 Sample average approximation

The scenario approach is conservative because it defends samples the chance constraint entitles it to ignore. Sample average approximation is the correction: it keeps the samples but restores the slack, allowing exactly the violation budget the constraint

grants. The violation probability in Eq. (16) is replaced by its empirical counterpart,

$$\hat{v}_N(x) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{g(x, \xi_i) > 0\} \leq \hat{\varepsilon}, \quad (19)$$

where the surrogate level $\hat{\varepsilon}$ may be set below ε to buy statistical confidence. The estimator is unbiased and the approach is statistically consistent: as $N \rightarrow \infty$, solutions of the approximated problem converge to solutions of the true one under mild conditions (Luedtke u. Ahmed, 2008; Pagnoncelli u. a., 2009). Unlike the scenario approach, SAA is allowed to discard its worst samples, so it tracks the true constraint rather than restricting it.

The advantage is exactly this statistical fidelity: Eq. (19) is the right quantity to compute, the empirical frequency of the event the constraint speaks about, with tunable conservatism through $\hat{\varepsilon}$ and well-developed asymptotic theory.

The limitation is that Eq. (19) is computationally challenging. The indicator makes $\hat{v}_N$ piecewise constant in x : its gradient vanishes almost everywhere and fails to exist on the boundary, so derivative-based nonlinear solvers receive no signal. The standard remedy is to encode the counting with binary variables, $g(x, \xi_i) \leq Mz_i$, $\sum_i z_i \leq \lfloor \hat{\varepsilon}N \rfloor$, $z_i \in \{0, 1\}$, which is exact but converts the problem into a mixed-integer program whose difficulty grows combinatorially with N (Luedtke u. Ahmed, 2008). SAA thus poses a choice between a landscape with no gradients and a program with 2^N corners, and neither option fits the smooth optimal control pipelines this thesis targets.

2.3.4 CVaR and convex restrictions

SAA's failure is computational, not statistical, and the convex-restriction route responds on exactly that front: it trades the exact constraint for a stronger one whose enforcement is tractable, accepting conservatism as the price of convexity. Its central instrument is the conditional value-at-risk. For the residual $Y(x)$ and level ε , the Rockafellar-Uryasev representation

$$\text{CVaR}_{1-\varepsilon}(Y(x)) = \min_{t \in \mathbb{R}} \left\{ t + \frac{1}{\varepsilon} \mathbb{E}[(Y(x) - t)_+] \right\}, \quad (20)$$

defines the average of the worst ε fraction of outcomes (Rockafellar u. Uryasev, 2000). Since the conditional value-at-risk dominates the value-at-risk, enforcing $\text{CVaR}_{1-\varepsilon}(Y(x)) \leq 0$ implies $\mathbb{P}(Y(x) > 0) \leq \varepsilon$, so the CVaR constraint is an inner approximation of the chance constraint, and Nemirovski u. Shapiro (2006) established it as the tightest

convex one in a natural family.

The advantages are convexity and safety in the same package. If $Y(x)$ is convex in x , then so is the CVaR constraint, the representation Eq. (20) is smooth after the standard reformulation of the positive part, and every feasible point of the restriction is provably feasible for the original chance constraint. Nothing in the construction requires knowing the distribution; the expectation in Eq. (20) is routinely replaced by a sample average, which preserves convexity.

The limitation is the direction of the inequality. CVaR controls the magnitude of violations, not merely their frequency, so it enforces more than the chance constraint asks: a decision with rare but large exceedances is rejected even when its violation probability is well inside the budget. The conservatism is largest exactly where chance constraints are most natural, under heavy-tailed residuals whose tail mean is far from the tail quantile. The experiment of Section 2.3.7 quantifies this gap on a skewed law: the CVaR solution overshoots the target reliability by a visible, structural margin that no amount of data removes.

2.3.5 Quantile smoothing

The survey has so far traded exactness for smoothness or smoothness for exactness; the question it forces is whether a method can refuse the trade. Quantile smoothing is the first demonstration that it can, and it is the closest methodological competitor to the approach of this thesis. Its starting point is the quantile equivalence. The chance constraint holds exactly when the $(1 - \varepsilon)$ -quantile of the residual is nonpositive,

$$\mathbb{P}(Y(x) \leq 0) \geq 1 - \varepsilon \iff Q_{1-\varepsilon}(Y(x)) \leq 0, \quad (21)$$

and the empirical quantile, an order statistic of the residual samples, estimates the left side of Eq. (21) consistently. The empirical quantile is itself non-smooth in x , so Peña-Ordieres u. a. (2020) smooth it, constructing a differentiable sample-based approximation of $Q_{1-\varepsilon}(Y(x))$ and embedding it in a sequential quadratic programming framework with trust regions tailored to the resulting nonlinear program.

The advantages mirror those claimed for KDE in Chapter 1: distribution-free operation, smoothness, gradients that mainstream nonlinear solvers consume, and statistical guarantees for the smoothed estimator. The method demonstrates convincingly that the smooth-sample-approximation paradigm works at scale, and its existence is part of the evidence that this paradigm, rather than enforcement or restriction, is the

right one for nonlinear problems.

The difference, and the limitation relative to this thesis, is the object being smoothed. Quantile smoothing approximates one functional of the residual law, the $(1 - \varepsilon)$ -quantile, and its smoothing is tied to that functional and that level. The KDE route estimates the residual density itself, from which the CDF at zero, its gradient, and quantiles at every level follow simultaneously in closed form; the same estimate also supports the biased construction of Part 3, which tilts the entire density surrogate to the safe side, a manipulation with no direct analogue at the level of a single quantile. The two methods are siblings, and the experimental chapters treat quantile smoothing as the fairest competitor.

2.3.6 The blackbox approach

Every method so far has exploited some structure: a distributional form, convexity, an analytic constraint map. The final method exploits none, and it closes the survey as the limiting case in which nothing is presumed about the problem. It is also the approach a practitioner with a simulator and no further structure would try first, so its mode of failure motivates much of what this thesis builds; for that reason this subsection develops it in more depth than the survey rhythm of the preceding methods, with computed evidence in place of a summary verdict. The core idea is to treat the violation probability as a blackbox. The estimator $\hat{v}_N(x)$ of Eq. (19) can be evaluated at any x by running the simulator N times and counting failures; nothing about g , the law of ξ , or the shape of the feasible set needs to be known. Optimization then proceeds with methods that consume only function values: penalty formulations driven by derivative-free solvers such as Nelder-Mead or pattern search (Nelder u. Mead, 1965; Audet u. Hare, 2017), surrogate-assisted search, or grid-and-refine heuristics over low-dimensional decision spaces.

The advantage is universality. The blackbox approach applies to discontinuous simulators, legacy codes, and constraint maps with no analytic form, settings where every other method in this section is inapplicable in principle. It is also honest: $\hat{v}_N(x)$ is an unbiased estimate of the true violation probability, so the quantity being thresholded is statistically the right one, with none of the conservatism of scenarios or CVaR. Statistically, Eq. (19) is sound; the sample average approximation theory cited in Section 2.3.3 establishes that solutions of the problem constrained by $\hat{v}_N(x) \leq \varepsilon$ converge to solutions of the true problem. If statistical correctness were the only requirement, little more would be needed.

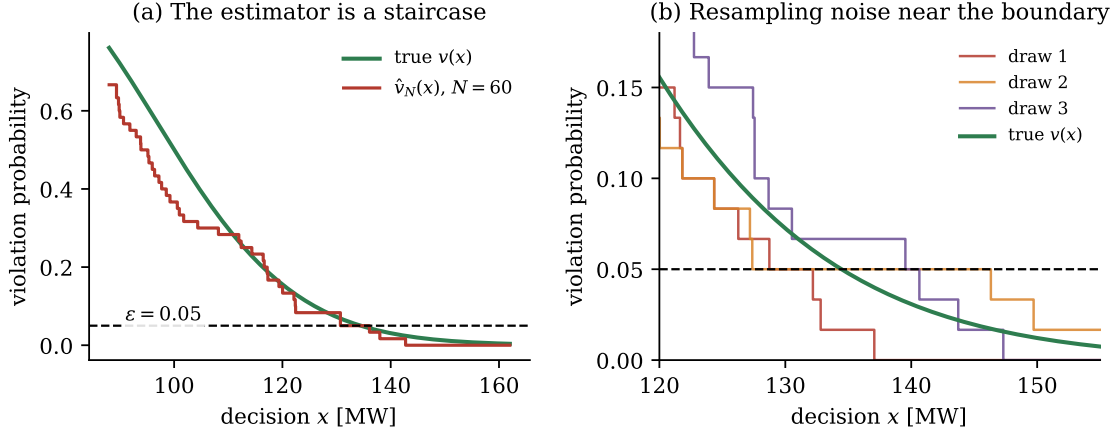


Figure 11: The empirical indicator estimator on the dispatch problem of Chapter 1 ($g(x, \xi) = \xi - x$, lognormal demand). (a) With $N = 60$ samples, $\hat{v}_N$ is a staircase: piecewise constant with jumps of $1/N$ at the samples, flat everywhere else. Its gradient is zero almost everywhere and undefined at the jumps. (b) Three independent sample draws, magnified near the feasibility boundary $\varepsilon = 0.05$: the staircases disagree with each other and with the true curve by amounts comparable to ε itself, so the estimated boundary wanders across tens of megawatts. The solver’s problem is therefore twofold: no usable gradient anywhere, and a noisy constraint where precision matters most.

It is not, and Fig. 11 shows why, computed on the dispatch problem of Chapter 1. As a function of the decision, $\hat{v}_N$ is a staircase: piecewise constant, jumping by exactly $1/N$ each time the moving decision crosses a sample, and flat everywhere else, so its gradient is zero almost everywhere and undefined at the jumps. A sequential quadratic programming or interior-point method, whose central mechanism is the exploitation of first-order information, receives essentially no signal; the solver is not merely slowed down but left without usable derivative information. Panel (b) exposes the statistical half of the difficulty, which compounds the first exactly where it matters most. Near the feasibility boundary, the region the optimizer must resolve precisely, the staircase is built from the extreme order statistics of the sample, among the noisiest quantities the sample provides, and independent draws of the same size produce estimated boundaries that wander across tens of megawatts, far beyond the optimality tolerances of typical nonlinear programming methods.

Two repair strategies exist within the paradigm, and both carry substantial costs. The counting in Eq. (19) can be encoded with binary variables, at the price of the mixed-integer growth already noted in Section 2.3.3, or derivatives can be abandoned altogether in favor of derivative-free optimization (Audet u. Hare, 2017). Derivative-free solvers tolerate the staircase but pay in evaluations, and their cost grows rapidly with the dimension of x ; in the optimal control problems of this thesis, where x is a dis-

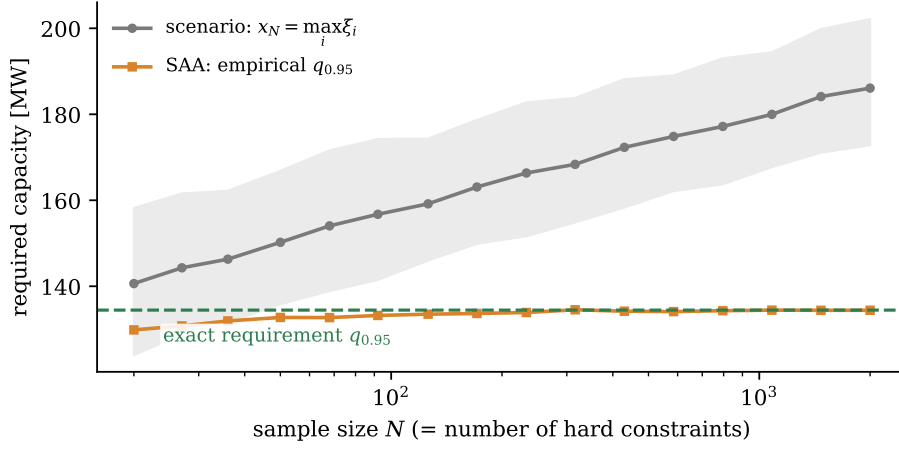


Figure 12: The price of structure, computed on the dispatch problem ($\varepsilon = 0.05$, 600 trials per sample size, bands are 10th-to-90th percentiles). The scenario decision must cover every sample, so it tracks the sample maximum and drifts away from the exact requirement $q_{0.95}$ (dashed) as N grows; the empirical quantile of the indicator paradigm converges to the exact answer. The horizontal axis is simultaneously the sample size and the number of hard constraints in the resulting program.

cretized state and control trajectory with hundreds of components, the route becomes impractical.

The scenario approach of Section 2.3.2 can be read as a *structured* member of the same family: it stops estimating the probability and starts enforcing the samples, thereby buying back smoothness, since each constraint in Eq. (18) is as smooth as g itself. What this concession costs is quantified in Fig. 12 on the problem where the exact answer is known. The scenario decision must cover every sample, so it tracks the sample maximum and drifts away from the exact requirement as N grows, exceeding it by nearly 40% at $N = 2000$, while the empirical quantile of the indicator paradigm converges to the exact answer throughout; the horizontal axis is simultaneously the sample size and the number of hard constraints in the resulting program, so statistical comfort and computational burden grow together. The two members of the family thus exhibit complementary limitations. The indicator approach estimates the appropriate quantity but presents the solver with a function that carries no usable derivative information; the scenario approach presents well-behaved functions but enforces a stronger, worst-case requirement that the chance constraint does not impose.

The shared limitation can now be stated precisely, and it is the observation that motivates the remainder of this thesis. Neither member of the family gives a gradient-based solver directional information about the violation probability itself. Between samples,

$\hat{v}_N$ reports that nothing changes, so a decision that violates the constraint with probability 0.50 and one just inside the budget at $\varepsilon = 0.05$ produce the same local signal, although the two are operationally very different; the scenario constraints carry perfectly good gradients, but of individual scenario residuals rather than of $v(x)$. The specification for a repair follows naturally: a single deterministic constraint, not N of them, whose value approximates the violation probability, whose dependence on x is as smooth as the constraint map itself, and whose gradient is available analytically. One further distinction is worth recording for the analysis ahead: the indicator estimator carries sampling error only, while a smooth reformulation introduces a second, controllable smoothing error on top of it, and keeping these two error sources separate is important for interpreting the developments of Chapters 3 and 4. Constructing an estimator that meets this specification, from the residual-space transformation through an analytic gradient, is the subject of Chapter 3.

2.3.7 Trade-offs in one view

The progression is complete, and its lesson can now be stated as a single claim the rest of the subsection substantiates: the classical methods partition the desirable properties among themselves, and none holds all of them at once. Table 1 condenses the survey into the four properties that matter for this thesis: whether the method is distribution-free, whether it hands a smooth problem to the solver, whether it is conservative, and what kind of feasibility guarantee it offers. The two KDE rows preview Part 3, and the distributionally robust row records the hybrid paradigm noted in Section 2.1.2, which wraps a Wasserstein ambiguity ball around the empirical law and obtains exact reformulations with finite-sample guarantees at the price of additional conservatism (Chen u. a., 2024).

Tables flatten behavior, so Fig. 13 runs the comparison as an experiment on the one-dimensional dispatch problem of Chapter 1, where the lognormal demand law is known exactly and the true reliability of any decision can be computed in closed form. Five methods produce a capacity decision from N samples at target reliability 0.95, and the figure reports the reliability actually achieved, averaged over 400 independent sample draws per N . The behaviors anticipated by the survey appear clearly. The Gaussian reformulation converges to a value near 0.937, below the target, so that an unsafe plan is certified as safe, and additional data does not correct this because the error lies in the assumption rather than in the estimate. The scenario approach climbs toward reliability one, drifting further from the target as N grows, the conser-

Table 1: Approximation methods for chance constraints, compared on the four properties most relevant to nonlinear optimization and optimal control. “Smooth” means the reformulated problem presents differentiable constraints to the solver. Guarantees are classified as finite-sample (holding at fixed N with stated confidence), statistical (confidence statements for the estimator), or asymptotic (consistency as $N \rightarrow \infty$).

Method	Distribution-free	Smooth	Conservative	Guarantee
Gaussian reformulation	No	Yes	No	None if law wrong
Scenario approach	Yes	No	Moderate, grows in N	Finite-sample*
SAA (indicator)	Yes	No	Tunable via $\hat{\epsilon}$	Asymptotic
CVaR restriction	Partial	Yes	Yes, structural	Deterministic bound
Quantile smoothing	Yes	Yes	No	Statistical
Blackbox (DFO)	Yes	No	No	Asymptotic
KDE, unbiased	Yes	Yes	No	Asymptotic
KDE, biased	Yes	Yes	By construction	Asymptotic + safe side
Wasserstein DRO	Yes	Partial	Yes	Finite-sample

*For convex problems (Campi u. Garatti, 2008); does not transfer to the nonlinear setting.

vatism that defends samples it is allowed to ignore. CVaR stabilizes near 0.978, safe but structurally above target, paying for tail magnitudes the constraint never asked it to control. SAA and KDE both converge to the target, the defining behavior of methods that estimate the constraint rather than assume or restrict it; their curves are nearly indistinguishable here because the quantity is a scalar quantile, but only the KDE decision was produced by a smooth, differentiable construction, which is the property that matters once the decision variable stops being a scalar.

Two further smooth routes deserve mention for completeness, although they lie outside the comparison of Table 1 because they exploit structure that the data-driven setting of this thesis does not assume. Polynomial chaos expansions (Xiu u. Karniadakis, 2002) propagate uncertainty through smooth dynamics with high accuracy when the law of ξ is known and sufficiently regular, and spheric-radial decomposition (van Ackooij u. Henrion, 2014) yields analytic gradients of the probability function for Gaussian and Gaussian-like laws. Both are strong alternatives within their domains of validity; a systematic benchmark against them lies outside the scope of this thesis.

The experiment makes the gap this thesis addresses concrete. What appears to be missing from the classical landscape is a method that is simultaneously distribution-free, smooth, non-conservative by default but conservative on demand, and equipped with convergence guarantees. The two KDE rows of Table 1 claim this combination. The remainder of the chapter introduces the estimator behind the claim and the literature that has already tested it.

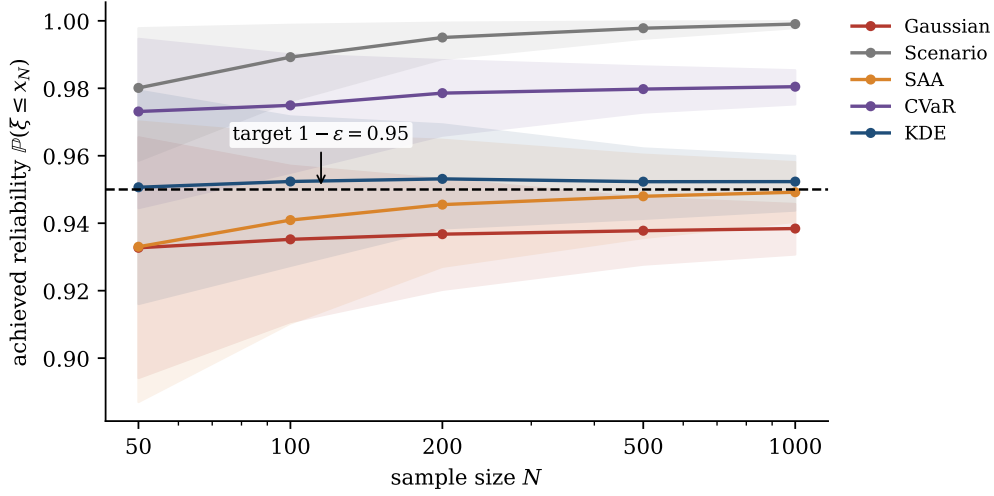


Figure 13: The survey of Section 2.3 as a computed experiment. Five methods choose a capacity from N samples of the lognormal demand of Chapter 1 at target reliability $1 - \varepsilon = 0.95$ (dashed); curves show the true reliability of the resulting decisions, averaged over 400 trials, with 10th-to-90th percentile bands. The Gaussian reformulation converges to an unsafe 0.937; the scenario approach and CVaR are persistently conservative; SAA and the KDE-smoothed quantile converge to the target. Only the KDE construction among the convergent pair is smooth in the decision variable.

2.4 Kernel Density Estimation

The survey ended on a specification: a method that is distribution-free, smooth, non-conservative by default and conservative on demand, with convergence guarantees. An estimator with many of the required properties has been available for decades. Kernel density estimation is standard equipment in statistics and largely unknown equipment in optimization, so this section introduces it deliberately, from the definition through the choices a user must make, with one organizing emphasis throughout: the property optimization cares about and statistics rarely examines, the behavior of the estimator as a function to be differentiated and embedded in a solver. Read with that emphasis, the classical theory of this section is not background; it is an inventory of exactly the assets and exactly the liabilities the reformulation of Part 3 inherits.

2.4.1 The Parzen-Rosenblatt estimator

Given scalar samples $y_1, \dots, y_N$ drawn from an unknown density ρ , the kernel density estimator of Rosenblatt (1956) and Parzen (1962) is

$$\hat{\rho}_h(y) = \frac{1}{Nh} \sum_{j=1}^N K\left(\frac{y - y_j}{h}\right), \quad (22)$$

where the kernel K is a nonnegative function integrating to one and the bandwidth $h > 0$ sets the smoothing scale. The mechanism is transparent: each sample is replaced by a small probability bump of width proportional to h , and the bumps are averaged. The estimator is itself a density, it inherits the smoothness of K , and its cumulative distribution function is the average of integrated kernels, $\hat{F}_h(y) = \frac{1}{N} \sum_j \mathcal{K}((y - y_j)/h)$ with $\mathcal{K}(t) = \int_{-\infty}^t K(s) ds$, available in closed form whenever $\mathcal{K}$ is. These are precisely the objects assembled in the residual-space construction of Chapter 1.

The estimator is consistent under remarkably weak conditions: if $h \rightarrow 0$ and $Nh \rightarrow \infty$ as $N \rightarrow \infty$, then $\hat{\rho}_h$ converges to ρ in mean integrated squared error for essentially any density of interest (Parzen, 1962; Devroye u. Györfi, 1985). No shape is assumed and none is imposed; given enough data the estimator recovers skewness, heavy tails, and multimodality alike. This distribution-free consistency is the property that earns KDE its place in this thesis, since the distributions of Chapter 1 are exactly the ones that defeat parametric fits.

For the optimization context, one more elementary observation carries disproportionate weight. The estimator Eq. (22) is a finite sum of shifted, scaled copies of a single fixed function. When the samples $y_j = g(x, \xi_j)$ depend on a decision variable, every analytic property of the estimate, its value, its integral, its derivative with respect to x , reduces by the chain rule to evaluations of K , $\mathcal{K}$, and the derivatives of g . There is no inner optimization, no numerical integration, and no nondifferentiable operation anywhere in the pipeline. The statistical literature prizes Eq. (22) for its flexibility; this thesis prizes it equally for being, computationally, a closed-form object.

2.4.2 Kernel choices

The kernel determines the local shape of the bumps, and three choices matter for this thesis. The Gaussian kernel $K(u) = (2\pi)^{-1/2} e^{-u^2/2}$ is smooth everywhere, has the standard normal CDF as its integrated kernel, and yields infinitely differentiable estimates

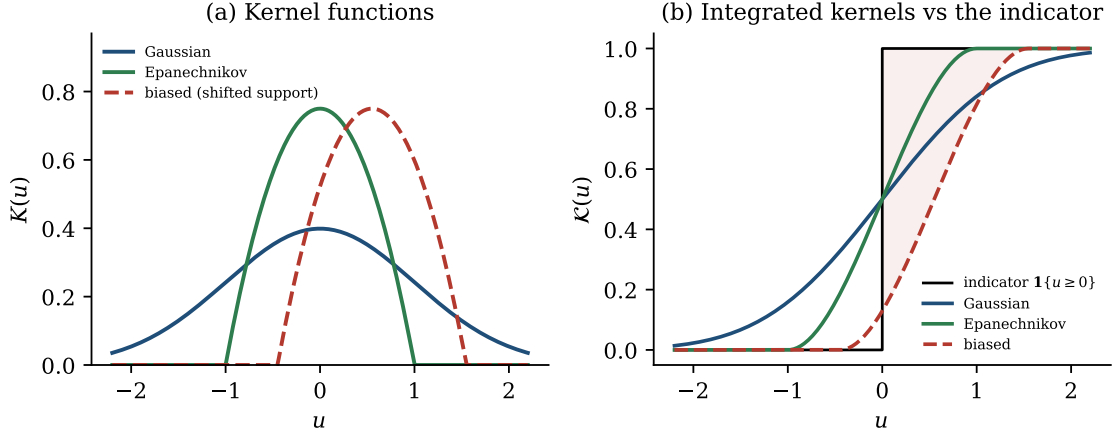


Figure 14: Kernels and their integrated forms. (a) The Gaussian and Epanechnikov kernels, both symmetric, and a biased kernel obtained by relocating the Epanechnikov support, the simplest illustration of the biasing idea formalized by the split-Bernstein construction of Keil u. a. (2021). (b) The corresponding integrated kernels against the indicator they replace. Symmetric kernels smooth the step from both sides and can overestimate the safe probability; the biased kernel’s integrated form sits on one side of the step (shaded gap), so the probability estimate it produces errs toward safety by construction.

with convenient gradients; it is the default choice throughout the unbiased developments of Part 3. The Epanechnikov kernel $K(u) = \frac{3}{4}(1 - u^2)_+$ is optimal for density estimation in the mean integrated squared error sense (Epanechnikov, 1969); its compact support also makes it the natural raw material for kernel modifications, since the entire kernel can be relocated. The split-Bernstein kernels of Keil u. a. (2021) are piecewise polynomial constructions engineered for a property neither classical kernel has: their integrated kernels bound the indicator function from the safe side, which is the seed of the biased, conservative reformulation developed in Part 3.

Figure 14 displays all three kernels and, in panel (b), the property that organizes Part 3. A symmetric kernel’s integrated form crosses the indicator at its jump: it underestimates the contribution of safely negative residuals and overestimates that of violating ones, and the net effect on the probability estimate can fall on either side. The biased kernel’s integrated form sits entirely on one side of the indicator, so every sample’s contribution errs in the same, safe direction. Anticipating the formal treatment, this single picture is the biased-KDE method: choose the kernel so that smoothing and conservatism are the same act.

Beyond these three, classical theory offers a reassurance that simplifies the rest of the thesis: the choice among standard symmetric kernels affects estimation efficiency only marginally, with the Gaussian sacrificing a few percent of efficiency relative to

the Epanechnikov optimum (Silverman, 1986). The choice of kernel is therefore of secondary importance for estimation quality; the bandwidth is the considerably more consequential choice.

2.4.3 Bandwidth and the bias-variance trade-off

The bandwidth is the decisive choice. Small h produces an estimate that follows the samples closely: low bias, high variance, and a ragged density whose spurious modes are artifacts of the particular draw. Large h produces a stable but blurred estimate: low variance, high bias, with genuine features such as modes and sharp tails smeared away. The mean integrated squared error makes the trade-off exact. For a symmetric kernel and a twice-differentiable density, the asymptotic expansion

$$\text{AMISE}(h) = \underbrace{\frac{R(K)}{Nh}}_{\text{variance}} + \underbrace{\frac{h^4 \mu_2(K)^2 R(\rho'')}{4}}_{\text{squared bias}}, \quad R(\phi) = \int \phi(s)^2 ds, \quad \mu_2(K) = \int s^2 K(s) ds, \quad (23)$$

is minimized at a bandwidth of order $N^{-1/5}$, whose constant depends on $R(K)$, $\mu_2(K)$, and the curvature term $R(\rho'')$, giving the optimal rate $\text{AMISE}(h^*) = O(N^{-4/5})$ (Silverman, 1986). Figure 15 shows both faces of the trade-off computed on the bimodal residual law used throughout this thesis: panel (a) the visual pathology of under- and oversmoothing on a single sample, panel (b) the empirical bias-variance decomposition over 120 replications, with the V-shaped total error and its interior minimizer.

Since h^* depends on the unknown curvature $R(\rho'')$, practice relies on data-driven rules. Silverman’s rule of thumb, $h = 0.9 \min(\hat{\sigma}, \text{IQR}/1.34) N^{-1/5}$, calibrates the constant to a Gaussian reference and is robust enough to serve as the default in most software (Silverman, 1986; Wand u. Jones, 1995). Scott’s rule, $h = \hat{\sigma} N^{-1/(d+4)}$ per dimension, generalizes the same idea to multivariate estimation (Scott, 1992). Cross-validation and plug-in methods estimate the curvature term from the data and adapt to non-Gaussian shapes at additional computational cost. For density estimation as an end in itself, any of these rules is adequate, and the near-optimal bandwidth of Fig. 15(a) is the value such rules produce on this sample.

A caveat must be placed on record here, because it grows into a research question later. Every rule above minimizes density estimation error, the AMISE of Eq. (23). Nothing in their derivation knows that the estimate will be thresholded at zero, differentiated with respect to a decision variable, and handed to a nonlinear solver as a constraint.

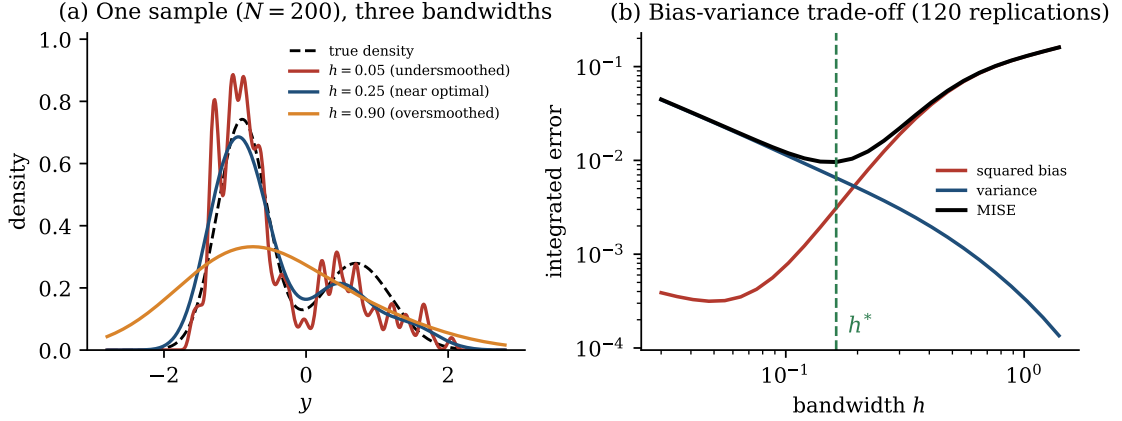


Figure 15: The bandwidth trade-off, computed on the bimodal law used in the residual-space illustration of Chapter 1. (a) One sample of size $N = 200$ estimated at three bandwidths: $h = 0.05$ invents spurious structure, $h = 0.90$ erases the two genuine modes, and $h = 0.25$ recovers the truth (dashed). (b) Empirical decomposition of the integrated error over 120 replications: variance falls and squared bias rises in h , and their sum is minimized at an interior h^* , here close to the $O(N^{-1/5})$ prediction of Eq. (23).

In the optimization context the bandwidth shapes the feasible region and the gradient field, and the h that best reconstructs the density need not be the h that best locates the optimizer. This distinction, AMISE optimality versus optimization optimality, is the substance of research question RQ3 and is studied experimentally in Part 4. A related caveat concerns the tails: kernel estimates are least accurate in sparsely sampled low-density regions, so risk levels stringent enough to push the binding region into the tails demand larger samples, while moderate levels keep it in the well-sampled bulk. This tail sensitivity is problem dependent, and the experimental chapters report results across a range of risk levels for that reason.

2.4.4 Multivariate KDE and the curse of dimensionality

The estimator extends to d -dimensional samples by replacing the scalar bandwidth with a bandwidth matrix H , $\hat{\rho}_H(y) = \frac{1}{N} \sum_j |H|^{-1/2} K(H^{-1/2}(y - y_j))$, and the AMISE analysis extends with it. The conclusion, however, reverses sign. The optimal bandwidth scales as $O(N^{-1/(d+4)})$ and the optimal error as $O(N^{-4/(d+4)})$ (Silverman, 1986; Scott, 1992): the convergence rate decays with every added dimension, and the sample size required to hold accuracy constant grows exponentially in d . Figure 16 translates the rate into numbers. Holding the accuracy achievable with $N = 100$ samples in one dimension fixed, five dimensions already demand on the order of 10^4 samples and ten dimensions on the order of 10^6 . This is the curse of dimensionality in a particularly

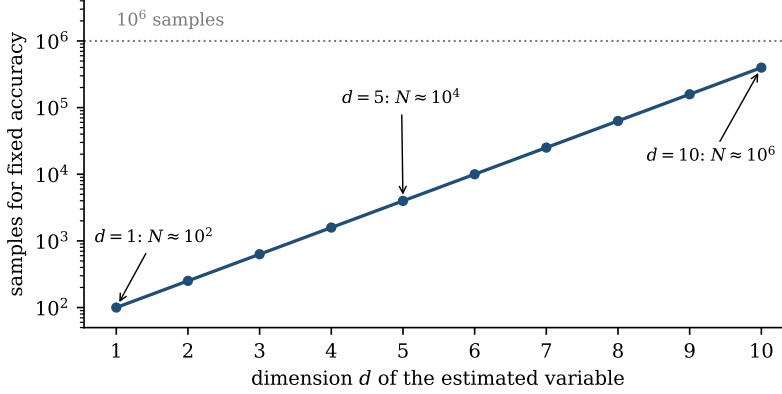


Figure 16: The curse of dimensionality for kernel density estimation: sample size required to maintain the accuracy achievable with $N = 100$ in one dimension, under the AMISE rate $O(N^{-4/(d+4)})$. The required sample size grows exponentially in d , reaching roughly 10^6 at $d = 10$. This figure is the quantitative case for estimating the scalar constraint residual rather than the uncertainty vector itself.

visible form, and it is a fundamental limitation of multivariate kernel estimation. Even for the multivariate normal distribution, evaluating orthant probabilities requires dedicated numerical methods (Genz u. Bretz, 2009), which indicates how demanding the general multivariate problem is.

The reader has already seen the answer this thesis gives. The residual-space construction of Chapter 1 never asks KDE to work in the dimension of ξ : an individual chance constraint produces a scalar residual whatever the dimension of the uncertainty, and the estimation problem is one-dimensional by design, where the $O(N^{-4/5})$ rate is available and a few hundred samples suffice. The curse retreats to one clearly marked location, joint chance constraints, whose residual vector has dimension m . Keeping that dimension out of the estimator, by collapsing the residual vector through a scalar violation statistic, is precisely the extension pursued as research question RQ4, and Fig. 16 is the reason it is worth pursuing.

2.5 KDE in the Chance-Constrained Literature: A Literature Map

The combination of kernel density estimation and chance constraints is no longer a proposal awaiting validation; it is a literature with a decade of results across four application domains and, since 2025, a convergence theory. What that literature has so far lacked is an organizing narrative, and supplying one is a substantive step of the argument rather than a convenience: when the papers are read as a single develop-

ment, it becomes visible that the engineering line and the theoretical line have not yet been connected systematically, which is the gap this thesis addresses. The narrative has three movements, displayed as a timeline in Fig. 17: independent invention, the safety turn, and the arrival of theory.

2.5.1 First constructions: process systems and aerospace

The data-driven construction enters the literature with Calfa u. a. (2015) in process systems engineering. Facing production planning problems whose right-hand-side uncertainty was known only through historical data, the authors replaced the unknown distribution with a kernel-smoothed estimate and derived deterministic reformulations for both individual and joint chance constraints, demonstrating them on industrial planning and scheduling cases. The paper established the template every later work follows, samples to smoothed estimate to deterministic constraint, and demonstrated that the resulting problems are solvable with standard nonlinear programming machinery.

Caillau u. a. (2018) carried the construction into optimal control, the setting of this thesis. Treating launcher ascent and the Goddard problem with uncertain parameters, the authors reformulated the chance constraints through kernel estimates and solved the resulting smooth optimal control problems with indirect and direct methods, establishing feasibility of the approach in exactly the trajectory-optimization context where the scenario and SAA routes are least comfortable. The aerospace line and the process systems line were developed independently, and their convergence on the same construction from different needs is early evidence that the construction is natural rather than contrived.

In parallel, the energy systems community arrived by a third route. Ciftci (2017) and Ciftci u. a. (2019) built adaptive-KDE chance constraints for microgrid energy management under renewable uncertainty, and the dispatch line that followed (Wu u. a., 2021; Wu u. Kargarian, 2023) pushed the construction to multivariate kernels for joint constraints in power-system scheduling. Three communities, three motivations, one estimator: by the end of this first movement the empirical case was broad, but every guarantee in it was inherited from generic KDE consistency, with nothing said about the optimization problems built on top.

2.5.2 The safety line: biased kernel density estimation

The first movement shares an unexamined weakness: a symmetric kernel smooths in both directions, so the estimated safe probability can exceed the true one, and a solution certified at level $1 - \varepsilon$ can be infeasible for the original constraint. The contribution of Keil u. a. (2020) and Keil u. a. (2021) is to make the smoothing itself safe. Their split-Bernstein kernels are constructed so that the integrated kernel bounds the indicator from the conservative side, as illustrated in Fig. 14(b); the resulting probability estimate is biased low by design, and feasibility for the approximated constraint implies feasibility for the true one up to the estimator’s statistical error. The method is demonstrated on chance-constrained soft lunar landing solved by direct collocation, the benchmark this thesis adopts.

The follow-up work of Keil u. Rao (2021) addresses the computational cost that conservatism introduces, proposing a warm-start strategy that solves a sequence of problems with decreasing bandwidth and switching kernels, using each solution to initialize the next. The pair of papers constitutes the primary algorithmic reference of this thesis: between them they supply the safe reformulation, the kernel constructions, and the practical solution strategy that Part 3 develops, dissects, and extends.

2.5.3 Networks, breadth, and the arrival of theory

The construction’s breadth test came from infrastructure. Probabilistic constraints on gas networks were studied by Gugat u. Schuster (2018), who treated stationary networks with compressor control under random loads. Building on this line, Schuster u. a. (2022) applied KDE-reformulated probabilistic constraints to stationary gas networks and dynamic water contamination networks, derived the gradients of the kernel-estimated probability analytically, and used them in necessary optimality conditions, comparing the KDE route against spheric-radial decomposition, the analytic alternative available under Gaussian-type laws (van Ackooij u. Henrion, 2014). Adjacent developments multiplied the evidence: nonparametric prediction intervals for wind power (Wan u. a., 2020), Bayesian nonparametric gas-power flow (Wang u. a., 2021), KDE-based tail-risk portfolio optimization (Liu u. a., 2022), KDE-driven distributionally robust load shedding (Qi u. a., 2026), and Hilbert-space embeddings of nonparametric chance constraints in robotics (Gopalakrishnan u. a., 2018).

What every entry above still lacked was a statement about the optimization problem itself: KDE consistency guarantees that the estimated probability converges, not

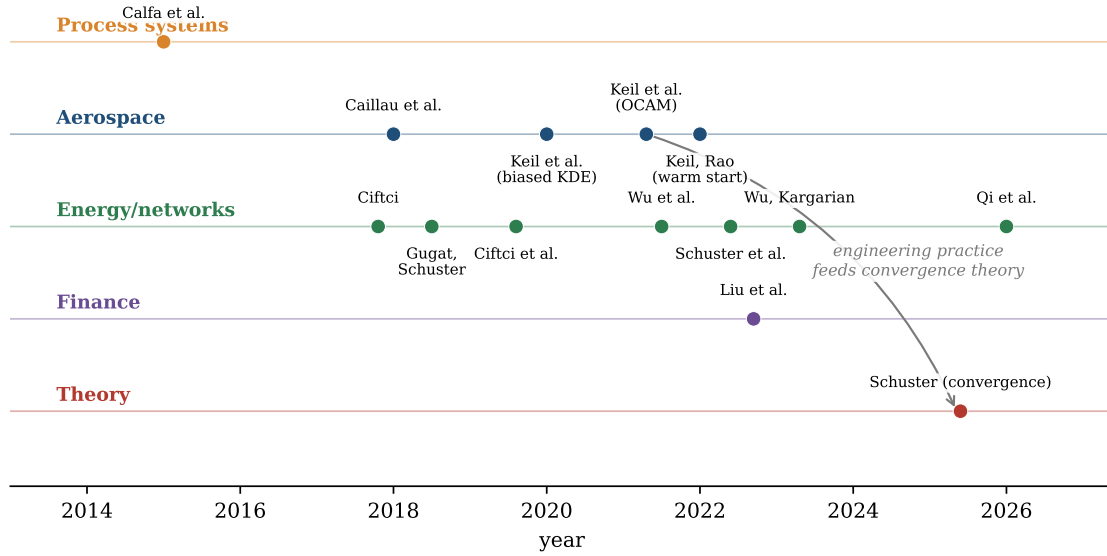


Figure 17: The KDE chance-constraint literature as a timeline, organized by application stream. The construction appears independently in process systems (Calfa u. a., 2015), aerospace (Caillaud u. a., 2018), and energy systems (Ciftci, 2017; Ciftci u. a., 2019); the biased-KDE safety line (Keil u. a., 2020, 2021; Keil u. Rao, 2021) and the network applications (Gugat u. Schuster, 2018; Schuster u. a., 2022) deepen it; and the convergence theory of Schuster (2025) arrives a decade after the first engineering constructions, closing the loop the gray arrow traces.

that the minimizer of the estimated problem converges to the minimizer of the true one. Schuster (2025) supplies exactly this missing piece, proving convergence of optimal values and solution sets of the KDE-approximated chance-constrained problem to those of the original as $N \rightarrow \infty$, under verifiable conditions on the kernel, the bandwidth sequence, and the problem data. With this result the literature acquires the shape visible in Fig. 17: a decade of engineering constructions, and a convergence theory developed separately that certifies the class they belong to.

The gap that remains is the connection itself, and it is the gap this thesis occupies. The engineering line and the theoretical line have not been brought into systematic contact: no work verifies the convergence conditions for the biased constructions used in practice, no controlled study quantifies how bandwidth choices made by AMISE rules behave inside an optimizer, and no unified experimental comparison places the KDE family against the classical methods of Section 2.3 across sample sizes, risk levels, and problem types, including the optimal control setting. Building that bridge, formulation by formulation in Part 3 and experiment by experiment in Part 4, is the work the next chapter begins, starting where any practitioner would: with the blackbox.

3 Smooth Approximation with Standard (Unbiased) KDE

Chapter 2 closed its method survey on a specification written by the limitations of the blackbox (subsection 2.3.6): a single deterministic constraint that approximates the safe probability, depends smoothly on the decision, and carries an analytic gradient. This chapter constructs that object. The construction has five steps, and the order matters because each step is what makes the next one possible: the change of variables that moves estimation into residual space (Section 3.1), the residual samples that the change of variables produces (Section 3.2), the kernel density estimate built on them (Section 3.3), the integrated-kernel identity that collapses the probability into a finite smooth sum (Section 3.4), and the analytic gradient that the identity makes available (Section 3.5). With the object built, Section 3.6 assembles the full reformulated non-linear program, and Section 3.7 states its properties with the candor the thesis owes them, including the one property it lacks, whose repair is the business of Chapter 4.

3.1 The Residual-Space Transformation

The fundamental insight behind the KDE approach, previewed in Chapter 1 and made load-bearing here, is a change of the space in which probabilities are estimated. The question the chance constraint literally poses is geometric: what is the probability that ξ lands in the safe region $M(x) = \{\xi : g(x, \xi) \leq 0\}$, a subset of $\mathbb{R}^d$ whose shape deforms with every candidate decision? The question this thesis poses instead is scalar: what is the probability that the constraint residual

$$Y(x) = g(x, \xi), \tag{24}$$

is non-positive? The two questions have the same answer, identically and without approximation,

$$\mathbb{P}(Y(x) \leq 0) = \mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon, \tag{25}$$

because the event $\{Y(x) \leq 0\}$ is the event $\{\xi \in M(x)\}$, written in the variable the constraint actually speaks about.

What changes is not the answer but the tractability of computing it, and the change is substantial rather than cosmetic. The original question demands the integral of an unknown d -dimensional density over a moving region, the second of the three difficulties catalogued in Chapter 1, with the curse of dimensionality of Chapter 2 attached to any

attempt to estimate that density nonparametrically. The transformed question asks for a one-dimensional cumulative distribution function evaluated at a fixed point, zero. The dimension of ξ no longer enters the estimation problem: whether the uncertainty lives in two dimensions or two hundred, the residual of an individual constraint is a scalar, and the constraint map g itself has performed the dimension reduction, without information loss, because Eq. (25) is an identity. This is the construction Schuster (2025) places at the foundation of the convergence theory, and it is the reason the KDE machinery of Chapter 2 can be deployed in exactly the one-dimensional regime where it is strongest.

3.2 Residual Samples

For a fixed decision x and the samples $\xi_1, \dots, \xi_N$ of Chapter 2, define the residual samples

$$y_j(x) = g(x, \xi_j), \quad j = 1, \dots, N. \quad (26)$$

These are realizations of the random variable $Y(x)$, obtained by pushing each uncertainty sample through the constraint map at the current decision, and they are the raw material of everything that follows.

One property of Eq. (26) must be understood precisely, because it is the property that distinguishes this construction from ordinary density estimation and shapes the analysis of the entire thesis: the residual samples are functions of the decision variable. The uncertainty samples ξ_j are drawn once and frozen; the residual cloud $\{y_j(x)\}$ moves whenever the optimizer moves x . Estimation and optimization are therefore not sequential stages but a single coupled process. The estimator is rebuilt, implicitly, at every iterate, and the constraint the solver sees is not a fixed statistical object but a family of them indexed by x . This coupling is what makes the smoothness of the next two subsections consequential rather than decorative: a smooth estimator of a fixed density would be a statistical convenience, while a smooth estimator of a moving one is a differentiable constraint.

3.3 KDE of the Residual Distribution

The Parzen-Rosenblatt estimator of Chapter 2, applied to the residual samples, estimates the density of $Y(x)$ as

$$\hat{\rho}_h(z; x) = \frac{1}{Nh} \sum_{j=1}^N K\left(\frac{z - y_j(x)}{h}\right), \quad (27)$$

with kernel K and bandwidth $h > 0$ carrying the meanings and the selection theory established there. The estimated probability that the constraint is satisfied is the mass this density places on the safe half-line,

$$\hat{p}_h(x) = \int_{-\infty}^0 \hat{\rho}_h(z; x) dz. \quad (28)$$

Two features of Eq. (28) should be recognized as the direct payoff of the residual-space transformation. The integration region is fixed: it is $(-\infty, 0]$ for every decision, every problem, and every distribution, because all dependence on x has migrated into the samples. And the integrand is a known finite sum of kernel bumps, not an unknown density, so the integral is not a numerical task but an algebraic one, which the next subsection carries out.

3.4 The Integrated Kernel Reformulation

Substituting Eq. (27) into Eq. (28) and integrating term by term against the integrated kernel $F_K(t) = \int_{-\infty}^t K(s) ds$ yields the closed form

$$\boxed{\hat{p}_h(x) = \frac{1}{N} \sum_{j=1}^N F_K\left(\frac{-y_j(x)}{h}\right) = \frac{1}{N} \sum_{j=1}^N F_K\left(\frac{-g(x, \xi_j)}{h}\right)}, \quad (29)$$

and, for the Gaussian kernel, $F_K = \Phi$, the standard normal CDF, so

$$\hat{p}_h(x) = \frac{1}{N} \sum_{j=1}^N \Phi\left(\frac{-g(x, \xi_j)}{h}\right). \quad (30)$$

The deterministic approximation of the chance constraint is then the single smooth inequality

$$\hat{p}_h(x) \geq 1 - \varepsilon. \quad (31)$$

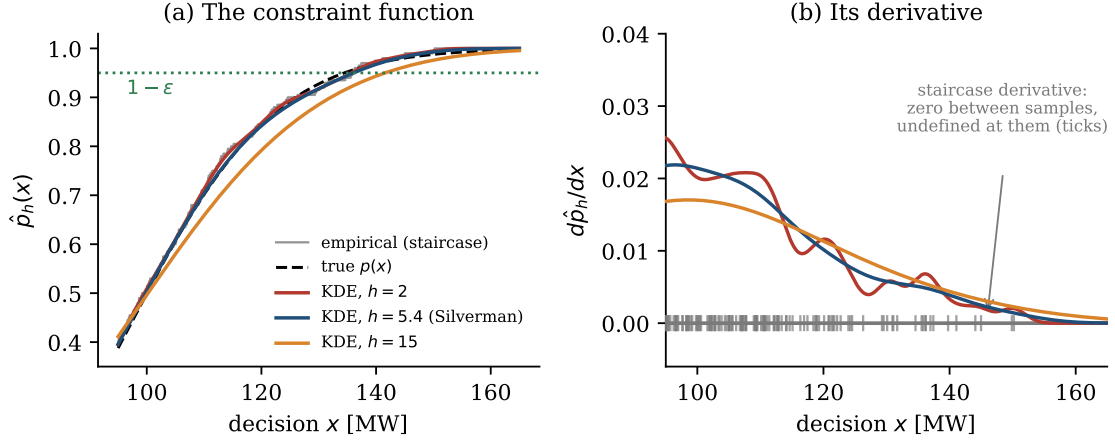


Figure 18: What the reformulation hands the solver, computed on the dispatch problem with $N = 200$ samples. (a) The empirical staircase, the true probability $p(x)$, and the KDE constraint $\hat{p}_h(x)$ at three bandwidths: small h tracks the staircase including its noise, Silverman’s $h = 5.4$ tracks the truth, and large h distorts the curve and shifts the feasibility boundary where it crosses $1 - \varepsilon$. (b) The derivatives. The staircase derivative is zero between samples and undefined at them (ticks on the axis); the KDE derivative is an honest, analytically available density estimate that gives the solver usable directional information at every x .

Equation (29) is among the most important equations in this thesis, and the claim deserves its justification. Set Eq. (29) beside the indicator estimator of Chapter 2: the two expressions are term-for-term identical except that the discontinuous indicator $\mathbf{1}\{g(x, \xi_j) \leq 0\}$ has been replaced by the smooth sigmoid $F_K(-g(x, \xi_j)/h)$, exactly the substitution drawn in Chapter 2’s kernel figure. Every sample still votes on feasibility, but the vote is no longer binary; a sample near the constraint boundary contributes a graded value that responds continuously as the decision moves. The staircase has been replaced by a ramp, and the entire discontinuous probabilistic feasibility requirement of Chapter 2 by a smooth, differentiable, deterministic nonlinear constraint. Figure 18(a) shows the replacement on the dispatch problem: the staircase, the true probability, and the KDE constraint at three bandwidths, with the bandwidth visibly controlling how tightly the ramp hugs the step.

3.5 Gradient Computation

Differentiating Eq. (29) by the chain rule, using $F'_K = K$, gives the gradient in closed form,

$$\nabla_x \hat{p}_h(x) = -\frac{1}{Nh} \sum_{j=1}^N K\left(\frac{-g(x, \xi_j)}{h}\right) \nabla_x g(x, \xi_j). \quad (32)$$

The formula repays a moment of reading, because its structure is central to the mechanism of the method. The gradient is a kernel-weighted average of the constraint gradients $\nabla_x g(x, \xi_j)$, and the weights $K(-g(x, \xi_j)/h)$ are largest for the samples whose residuals lie within a bandwidth of the boundary. The estimator automatically concentrates the solver's attention on the scenarios that are about to change status, which is exactly the directional information Chapter 2 showed the indicator withholds. For the Gaussian kernel every quantity in Eq. (32) is elementary, second derivatives follow by the same chain rule, and sequential quadratic programming and interior-point methods receive exact first- and second-order information rather than finite differences of a staircase (Wächter u. Biegler, 2006). Figure 18(b) displays the comparison directly: the staircase derivative is zero except at isolated undefined points, while the KDE derivative is a smooth, analytically available function at every decision. This differentiability is the core computational advantage of the KDE reformulation over both blackboxes of Chapter 2.

3.6 Stochastic Optimization Reformulated via Unbiased KDE

Assembling the pieces, the chance-constrained program of Chapters 1 through 3,

$$\underset{x \in X}{\text{minimize}} \ f(x) \quad \text{subject to} \quad \mathbb{P}(g(x, \xi) \leq 0) \geq 1 - \varepsilon, \quad (33)$$

becomes, after the KDE substitution,

$$\underset{x \in X}{\text{minimize}} \ f(x) \quad \text{subject to} \quad \frac{1}{N} \sum_{j=1}^N F_K\left(\frac{-g(x, \xi_j)}{h}\right) \geq 1 - \varepsilon. \quad (34)$$

Problem Eq. (34) is a deterministic nonlinear program, and its structure should be read off explicitly because it answers, point by point, the specification Chapter 2 issued. One smooth nonlinear inequality replaces one probabilistic constraint; the program does not grow with the sample size, in deliberate contrast to the scenario approach, because the samples ξ_j enter as fixed parameters inside a single sum. The decision variable enters only through the residuals $g(x, \xi_j)$, so the constraint is exactly as smooth as the constraint map itself. Standard solvers, IPOPT, SNOPT, and SQP implementations, consume Eq. (34) directly, with the analytic derivatives of Section 3.5 supplied through the modeling layer (Wächter u. Biegler, 2006). Figure 19 draws the complete loop: the frozen samples and the moving decision meet in the residuals, the KDE turns the residuals into a smooth constraint value and gradient, and the solver

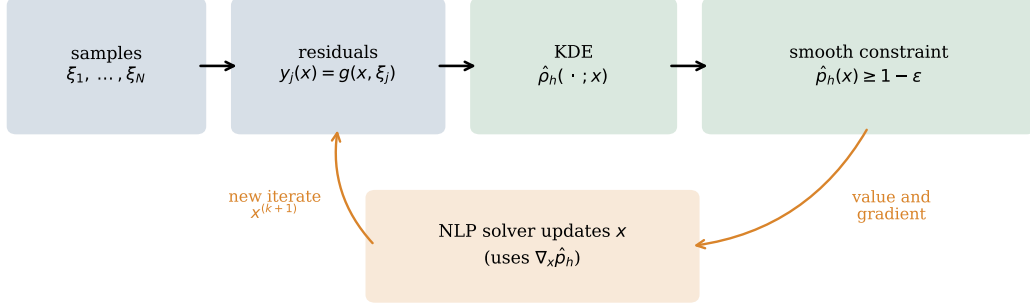


Figure 19: The reformulation as a computational loop. Uncertainty samples are drawn once and frozen; at each iterate the constraint map produces the residual cloud $y_j(x)$, the kernel estimator converts the cloud into the smooth constraint $\hat{p}_h(x) \geq 1 - \varepsilon$ with value and analytic gradient, and the nonlinear programming solver returns the next iterate to the residual map. One smooth inequality replaces one probabilistic constraint, and the program size is independent of N .

feeds the next iterate back into the residual map. This loop is the computational object the remainder of the thesis studies, extends, and stress-tests.

3.7 Properties of the Unbiased KDE Approximation

The reformulation Eq. (34) has four properties, and the thesis states all four with equal directness because the fourth is a genuine limitation. First, it is smooth: for kernels with differentiable F_K , the Gaussian above all, the constraint is twice continuously differentiable in x wherever g is, and the derivatives are the closed forms of Section 3.5. Second, it is consistent: as $N \rightarrow \infty$ with $h \rightarrow 0$ and $Nh \rightarrow \infty$, the estimate $\hat{p}_h(x)$ converges to the true probability $\mathbb{P}(g(x, \xi) \leq 0)$, inheriting the classical consistency of the kernel estimator (Parzen, 1962; Silverman, 1986), and under the conditions of Schuster (2025) the convergence lifts from the constraint to the optimization problem itself, optimal values and solution sets included. It is instructive to separate the two error sources at play, as anticipated in Chapter 2: a random sampling error of order $N^{-1/2}$, inherited from the finite sample, and a deterministic smoothing error controlled by the bandwidth, introduced by the kernel (Silverman, 1986; Wand u. Jones, 1995). The conditions $h \rightarrow 0$ and $Nh \rightarrow \infty$ balance exactly these two contributions, and the biased construction of Chapter 4 acts on the second source alone, giving it a sign rather than removing it. Third, the bandwidth controls a genuine trade-off: Fig. 18(a) shows small h reproducing the staircase together with its sampling noise and large h distorting the feasible boundary, so h is not a nuisance parameter but a design variable, the

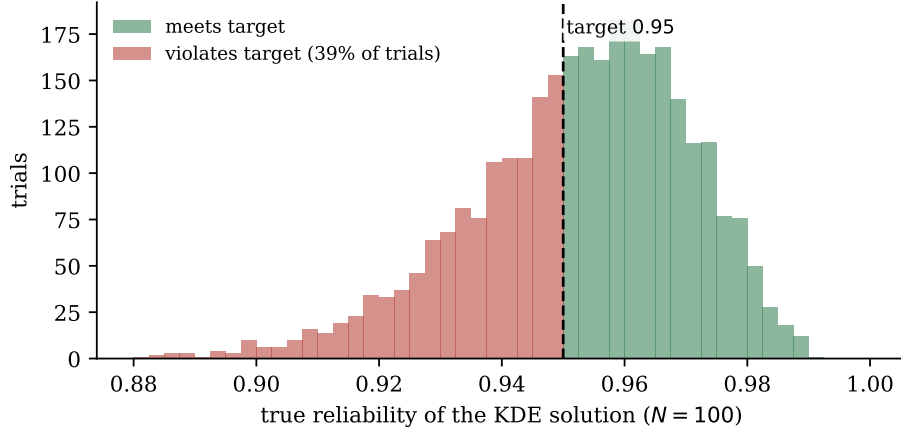


Figure 20: The unbiased reformulation is accurate but not safe. For 3000 independent trials of the dispatch problem at $N = 100$, the decision certified by the unbiased KDE constraint at $1 - \varepsilon = 0.95$ is evaluated against the true demand law. The achieved reliability concentrates near the target, the signature of a consistent estimator, but falls below it in 39% of trials: the certificate $\hat{p}_h(x) \geq 1 - \varepsilon$ is optimistic almost as often as not. This finite-sample optimism is the defect the biased construction of Chapter 4 removes.

optimization-versus-AMISE question flagged in Chapter 2 and studied experimentally in Part 4.

Fourth, and importantly for the structure of this thesis, the approximation is not conservative. A symmetric kernel smooths the indicator from both sides, so in finite samples the estimate $\hat{p}_h(x)$ can exceed the true probability, and the constraint $\hat{p}_h(x) \geq 1 - \varepsilon$ can certify a decision the true chance constraint rejects. Figure 20 measures the consequence on the dispatch problem: across three thousand independent trials at $N = 100$, the decision certified by the unbiased KDE violates the true reliability target in 39% of trials. The failures are individually small, the estimator is consistent and the misses concentrate near the target, but they are systematic, and a reformulation whose certificate fails in roughly four of ten trials functions as an accuracy method rather than a safety method. For applications where ε is a hard requirement, a lander, a grid code, a risk mandate, accuracy alone does not meet the requirement; a conservative guarantee is needed. Constructing a KDE reformulation that keeps every property of this chapter while erring, by design, only on the safe side is the contribution of the biased kernels of Keil u. a. (2021), and it is the business of the next chapter.

4 The Biased KDE Approach and Conservative Reformulation

Chapter 3 closed on a measured verdict: the unbiased reformulation is an accuracy method, not a safety method, and its certificate fails almost as often as not. This chapter converts the verdict into a construction. The repair is not a patch applied after estimation but a redesign of the estimator itself, due to Keil u. a. (2020, 2021): choose the kernel so that smoothing and conservatism become the same act, exactly the property previewed in Chapter 2’s kernel figure and now developed in full. The chapter states the requirement the unbiased estimator fails (Section 4.1), formalizes the domination condition that defines a biased kernel (Section 4.2), examines the three kernels that matter in practice and the precise sense in which each satisfies or approximates the condition (Section 4.3), assembles the conservative reformulation pipeline and prices its conservatism on a known-truth problem (Section 4.4), and then extends the framework to joint chance constraints through three competing routes (Section 4.5), of which the scalar violation statistic with log-sum-exp smoothing (Section 4.6) is this thesis’s recommended extension and one of its main implementation contributions. The chapter closes with the bandwidth-continuation strategy that makes the whole construction solvable in practice (Section 4.7).

4.1 Why Smoothing Alone Is Not Safe

The limitation must be located precisely before it can be addressed, because it does not lie where one might first look. The unbiased estimator of Chapter 3 is consistent, its bandwidth theory is classical, and its certificate concentrates near the target; none of its statistical properties is at fault. The limitation is directional. A symmetric kernel’s integrated form crosses the indicator at the constraint boundary, so each sample’s contribution to $\hat{p}_h(x)$ errs toward optimism for violating residuals and toward pessimism for safe ones, and nothing in the construction controls which effect dominates at the decision the optimizer actually returns. The optimizer, moreover, is not a neutral party: it systematically favors the cheapest certified decision, which tends to be a decision at which optimistic error contributed to the certificate being granted. Chapter 3 measured the result at 39% certificate failure.

For engineering and safety-critical applications this behavior is problematic, and the requirement that replaces it can be stated in one line. What is needed is an estimator

that never credits a violating sample: a surrogate $\hat{p}_h^b$ whose every per-sample contribution is bounded by the indicator it replaces, so that $\mathbb{E}[\hat{p}_h^b(x)] \leq \mathbb{P}(g(x, \xi) \leq 0)$ at every decision, and any solution certified by the surrogate satisfies the true chance constraint in expectation. Equivalently, in the violation form of Chapter 2, the surrogate violation estimate must dominate the true violation probability from above. The estimator may still err, but the construction ensures that its errors no longer point in the unsafe direction.

4.2 The Biasing Construction

The construction of Keil u. a. (2020) unifies two half-solutions, each inadequate alone. The first replaces the violation indicator $\mathbf{1}\{t > 0\}$ with any function $w(t) \geq \mathbf{1}\{t > 0\}$: averaging w over samples always upper-bounds the violation probability, but w need not be bounded by one, so the average is not a valid probability and distorts the optimization. The second is the standard KDE surrogate of Chapter 3, a valid CDF bounded in $[0, 1]$ that violates the domination requirement. The biased kernel is the object that holds both properties at once. A kernel K with integrated form F_K and bias term $b(h) \geq 0$ satisfies the *domination condition* when its safe-side surrogate is bounded by the indicator,

$$F_K\left(\frac{-t - b(h)}{h}\right) \leq \mathbf{1}\{t \leq 0\} \quad \text{for all } t \in \mathbb{R} \text{ and } h > 0, \quad (35)$$

equivalently, when the complementary violation surrogate dominates $\mathbf{1}\{t > 0\}$ from above for every residual. Condition Eq. (35) asks for exactly one thing beyond Chapter 3: the surrogate must vanish on the entire violating half-line $t > 0$, which the bias $b(h)$ accomplishes by shifting the kernel's mass to the safe side of the boundary.

The conservative estimator and constraint follow immediately. Averaging the dominated surrogate over the residual samples of Chapter 3 gives

$$\hat{p}_h^b(x) = \frac{1}{N} \sum_{j=1}^N F_K\left(\frac{-g(x, \xi_j) - b(h)}{h}\right), \quad \hat{p}_h^b(x) \geq 1 - \varepsilon, \quad (36)$$

and the safety statement is now a provable property rather than an empirical observation: since each term of Eq. (36) is bounded by the corresponding indicator, $\mathbb{E}[\hat{p}_h^b(x)] \leq \mathbb{P}(g(x, \xi) \leq 0)$ pointwise, so any decision satisfying the biased constraint satisfies the true chance constraint in expectation, with only sampling fluctuation, van-

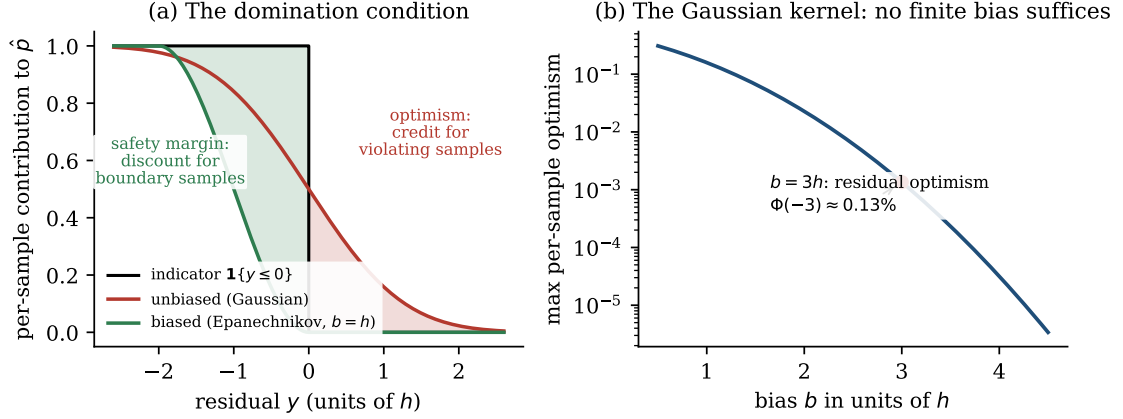


Figure 21: The biasing construction. (a) Per-sample contributions to the probability estimate, in units of the bandwidth. The unbiased Gaussian surrogate crosses the indicator and credits violating residuals (shaded right), the optimism measured in Chapter 3; the biased Epanechnikov surrogate with $b = h$ sits entirely beneath the indicator, so every per-sample error points to the safe side (shaded left). (b) Why the Gaussian kernel cannot be made exactly safe: its maximal per-sample optimism $\Phi(-b/h)$ is positive for every finite bias, decaying but never vanishing; $b = 3h$ leaves a residual optimism of 0.13% per sample.

ishing as N grows, standing between the certificate and the guarantee. Figure 21(a) draws the construction: the unbiased surrogate crosses the indicator and grants credit to violating samples (the optimism Chapter 3 measured), while the biased surrogate sits entirely beneath it, discounting boundary samples and crediting violations nothing at all. Smoothness has not been spent to buy safety; the biased surrogate is exactly as differentiable as the unbiased one, and every formula of Chapter 3, the closed-form constraint, the analytic gradient, the N -independent program size, survives with $-g(x, \xi_j)$ replaced by $-g(x, \xi_j) - b(h)$. In the language of the two error sources separated in Chapter 3, the bias acts on the smoothing error alone: the sampling error remains random and vanishes with N , while the smoothing error is now signed toward safety rather than merely made small.

4.3 Specific Biased Kernels

Condition Eq. (35) is a constraint on kernel tails, and the three kernels of Chapter 2 meet it in three instructively different ways. The *split-Bernstein kernel*, a piecewise-polynomial construction originating in approximation theory and adapted to chance constraints by Keil u. a. (2021), satisfies the domination condition exactly for appropriate parameter choices: its compact, one-sided structure is designed around Eq. (35) rather than retrofitted to it, and its integrated form is smooth and available in closed

form. It is the reference kernel whenever the strict guarantee is the priority. The *Epanechnikov kernel with bias* starts from the AMISE-optimal kernel of Chapter 2 and exploits its compact support: with $b(h) = h$, the shifted surrogate vanishes identically on the violating half-line, so Eq. (35) holds exactly, and in the lunar landing experiments of Keil u. a. (2021) this kernel achieves the lowest average optimal cost among the biased family, the statistical efficiency of Chapter 2 carrying over into the optimization.

The *Gaussian kernel* is the cautionary member, and its limitation here is structural rather than incidental: because its tails are strictly positive everywhere, no finite bias can make the surrogate vanish for $t > 0$, so the Gaussian cannot satisfy Eq. (35) exactly (Keil u. a., 2021). Figure 21(b) quantifies the gap. The maximal per-sample optimism after biasing is $\Phi(-b/h)$, positive for every finite b ; the practical compromise $b \approx 3h$ caps it at $\Phi(-3) \approx 0.13\%$, covering 99.87% of the tail and rendering the kernel *approximately* conservative. The Gaussian's compensating virtues are real, infinite smoothness and the fastest solver runtimes in the family, and the resulting division of labor is the one this thesis adopts: Gaussian kernels where speed and smoothness dominate, exact biased kernels where the guarantee does, and the continuation strategy of Section 4.7 to move between them within a single solve.

4.4 Stochastic Optimization Reformulated via Biased KDE

The full conservative pipeline assembles the chapters of Part 3 into five steps, stated once here as the procedure the experimental chapters implement. First, generate N samples $\xi_1, \dots, \xi_N$ by simulation, Markov chain Monte Carlo, or historical data, exactly as in Chapter 2. Second, for each candidate decision x , form the residuals $y_j(x) = g(x, \xi_j)$ of Chapter 3. Third, evaluate the biased constraint function $\hat{p}_h^b(x)$ of Eq. (36) with a kernel and bias from Section 4.3. Fourth, enforce $\hat{p}_h^b(x) \geq 1 - \varepsilon$. Fifth, solve the resulting deterministic nonlinear program with the analytic derivatives that the construction provides. The program retains every structural property of Chapter 3: one smooth inequality, size independent of N , gradients by the chain rule.

What the bias changes is the trade, and Fig. 22 prices it on the dispatch problem where the truth is computable. The comparison is the one promised in Chapter 3's closing: the unbiased reformulation is smooth and less conservative but carries no finite-sample safety, while the biased reformulation is slightly more conservative and carries it. The numbers make the asymmetry of the trade explicit. Across sample sizes, the unbiased

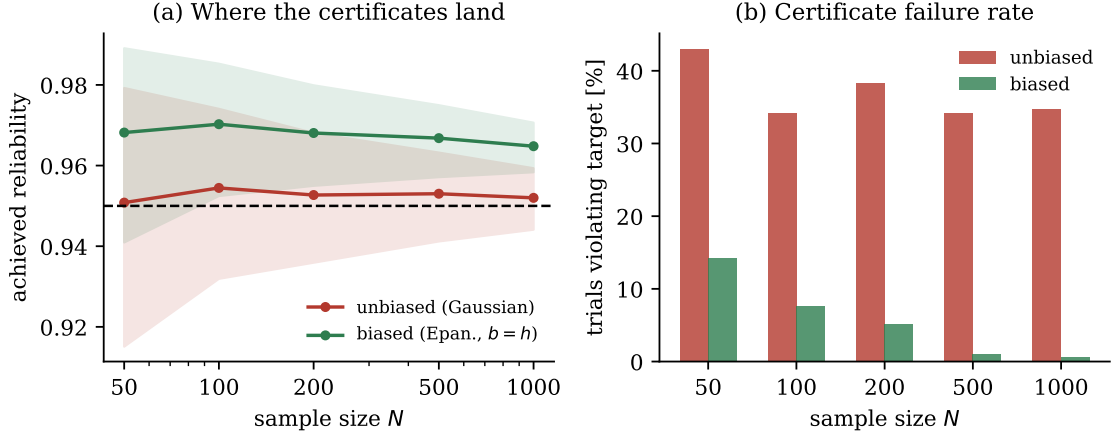


Figure 22: The price and the purchase of conservatism, on the dispatch problem ($\varepsilon = 0.05$, 800 trials per sample size, bands are 10th-to-90th percentiles). (a) The unbiased certificate straddles the target from below on a large fraction of trials at every N ; the biased certificate (Epanechnikov, $b = h$) lands above it by a margin that shrinks as the estimator sharpens. (b) Certificate failure rates: the unbiased rate is structural and does not improve with data; the biased rate is sampling fluctuation and collapses, falling below 1% from $N = 500$. The average objective premium paid for safety is 2.6 to 4%, decreasing in N .

certificate fails in roughly 34 to 43% of trials and does not improve with data, because its error is directional, not statistical; the biased certificate fails in 14% of trials at $N = 50$, falls to 5% by $N = 200$, and is below 1% from $N = 500$, the residual failures being pure sampling fluctuation around a safe-side mean. The premium paid for this is 2.6 to 4% of objective value, shrinking as N grows. This exchange is familiar from reliability engineering: a bounded and vanishing cost in return for removing a class of certificate failures. The unbiased method answers the question *what is the probability*; the biased method answers the question the application typically asks, *can this decision be trusted*.

4.5 Joint Chance Constraints: Three Approaches

Everything so far concerns a scalar residual. The joint constraint of Chapters 1 through 3, $\mathbb{P}(g_r(x, \xi) \leq 0, r = 1, \dots, m) \geq 1 - \varepsilon$, asks about m requirements at once, and the biased framework extends to it along three routes whose trade-offs this subsection fixes, because the choice among them is a design decision the experimental chapters revisit.

Approach A: Boole's inequality with risk allocation. The union bound $\mathbb{P}(\bigcup_r \{g_r > 0\}) \leq \sum_r \mathbb{P}(g_r > 0)$ yields a sufficient condition: enforce each component at level ε_r with $\sum_r \varepsilon_r \leq \varepsilon$, applying the one-dimensional biased KDE of Section 4.4 to each residual

separately (Prékopa, 1995; Keil u. a., 2021). The route is simple, fully tractable, and conservative twice over, once through the bias and once through the union bound; its conservatism grows with m and is most costly exactly when the component violations are strongly correlated, the case the union bound prices worst.

Approach B: multivariate KDE. Estimate the joint density of the residual vector $(g_1(x, \xi), \dots, g_m(x, \xi))$ with a multivariate kernel and bandwidth matrix H , and integrate over the safe orthant $(-\infty, 0]^m$ (Wu u. a., 2021; Wu u. Kargarian, 2023). The route is exact in the sense that no decomposition is imposed, but it reintroduces precisely the difficulty the residual-space construction was designed to avoid: the curse of dimensionality of Chapter 2 now acts in the constraint dimension m , and the sample sizes it demands grow exponentially.

Approach C: the scalar violation statistic. The recommended extension of this thesis collapses the joint constraint exactly. For positive weights w_r , define

$$s(x, \xi) = \max_{r=1, \dots, m} w_r g_r(x, \xi). \quad (37)$$

The reduction is an identity, not a bound: all components are nonpositive precisely when their weighted maximum is, so

$$\mathbb{P}(g_r(x, \xi) \leq 0, r = 1, \dots, m) = \mathbb{P}(s(x, \xi) \leq 0), \quad (38)$$

and the joint chance constraint is a scalar chance constraint on s . A one-dimensional biased KDE suffices; no Boole decomposition, no multivariate estimation, no curse of dimensionality. The constraint map itself has once again performed the dimension reduction, now in m as well as in d . The single catch is the one Chapter 2 anticipated: the max in Eq. (37) is nonsmooth, and a nonsmooth residual would re-break exactly the differentiability that Chapter 3 worked to establish. Removing that catch is the purpose of the next subsection.

4.6 Log-Sum-Exp Smoothing of the Scalar Violation Statistic

The max is smoothed by the log-sum-exp surrogate with temperature $\tau > 0$,

$$s_\tau(x, \xi) = \tau \log \left(\sum_{r=1}^m \exp \left(\frac{w_r g_r(x, \xi)}{\tau} \right) \right), \quad (39)$$

which satisfies the two-sided bound

$$\max_r w_r g_r(x, \xi) \leq s_\tau(x, \xi) \leq \max_r w_r g_r(x, \xi) + \tau \log m, \quad (40)$$

(Boyd u. Vandenberghe, 2004). The direction of the bound is the key property of the construction, and it should be read in the spirit of Eq. (35): the surrogate dominates the true statistic, so enforcing $\mathbb{P}(s_\tau \leq 0) \geq 1 - \varepsilon$ *implies* the true joint constraint, and the smoothing error, like the kernel bias, is conservative by construction. The temperature plays for the max exactly the role the bandwidth plays for the indicator: a smoothing parameter whose cost is a controlled, safe-side margin of at most $\tau \log m$. Figure 23(a) shows the geometry on a two-constraint slice, s_τ riding above the kink and collapsing onto it as τ shrinks.

The gradient completes the smoothness argument. Differentiating Eq. (39) gives

$$\nabla_x s_\tau(x, \xi) = \sum_{r=1}^m \pi_r(x, \xi) w_r \nabla_x g_r(x, \xi), \quad \pi_r = \frac{\exp(w_r g_r / \tau)}{\sum_q \exp(w_q g_q / \tau)}, \quad (41)$$

a softmax-weighted blend of the component gradients, smooth in x and analytically computable. Figure 23(b) shows the weights handing influence from one constraint to the other across the switching point, continuously, where the raw max would switch abruptly and nondifferentiably. Composing the pieces yields the full biased-KDE reformulation for joint constraints,

$$\frac{1}{N} \sum_{j=1}^N F_K \left(\frac{-s_\tau(x, \xi_j) - b(h)}{h} \right) \geq 1 - \varepsilon, \quad (42)$$

and Eq. (42) is one of this thesis's main implementation contributions, a claim supported by what the formulation absorbs: a joint, non-Gaussian, possibly high-dimensional chance constraint has been reduced to a single smooth one-dimensional deterministic inequality, with two stacked approximation layers, kernel bias and log-sum-exp temperature, both of which err on the safe side by construction. The interaction of those two layers, how conservatism compounds across h and τ and how the budget should be split between them, is a genuinely open question, and it is studied experimentally as research question RQ4 in Part 4.

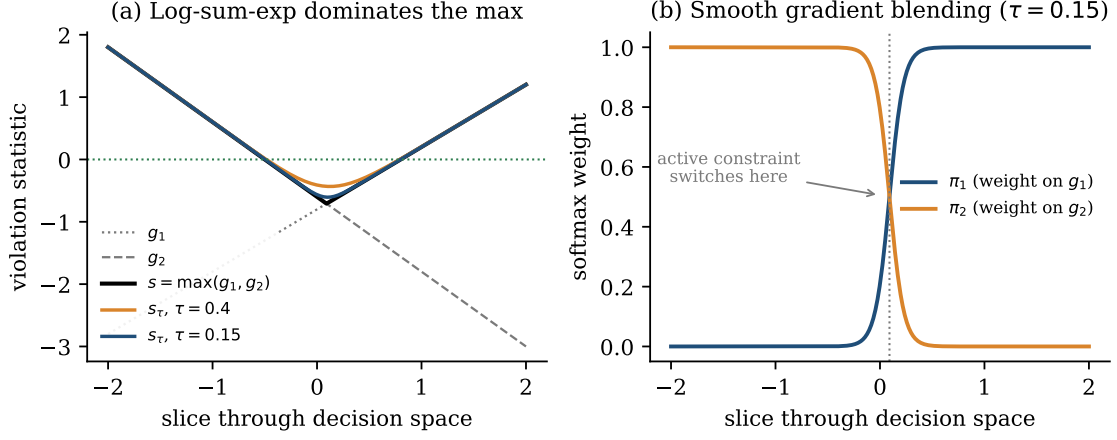


Figure 23: Log-sum-exp smoothing of the scalar violation statistic on a two-constraint slice. (a) The surrogate s_τ dominates $s = \max(g_1, g_2)$ everywhere, with gap at most $\tau \log m$ concentrated at the kink; as τ shrinks the surrogate collapses onto the max, so the smoothing error is a tunable, safe-side margin. (b) The softmax weights of the gradient formula hand influence from one constraint to the other continuously across the switching point, giving the solver a smooth blend where the raw max would give a nondifferentiable jump.

4.7 Bandwidth Continuation and Kernel Switching

The constructions of this chapter create a practical tension that one further idea resolves. Small bandwidths and exactly-safe kernels are where the guarantees live, but they are also where the optimization is hardest: a sharply smoothed constraint approaches the staircase of Chapter 2, and a solver started cold against it inherits a rugged, nearly nonsmooth landscape. The warm-start strategy of Keil u. Rao (2021) converts this tension into a schedule. Figure 24 draws it. Begin with a large bandwidth, where the constraint is heavily smoothed, the landscape benign, and the Gaussian kernel’s speed available; solve, and pass the solution forward as the initial point for a smaller bandwidth; repeat, increasing the sample size as the bandwidth shrinks so that statistical resolution keeps pace with geometric resolution; and near convergence, switch from the Gaussian to an exactly-safe kernel, the split-Bernstein, so that the final certificate carries the strict guarantee of Section 4.3 rather than the 99.87% approximation.

The schedule is a homotopy in the proper sense: a continuous deformation from an easy problem to the target problem along which the solution is tracked rather than rediscovered, and its effect in the experiments of Keil u. Rao (2021) is precisely the one homotopies exist to deliver, stabilized convergence and substantially reduced solve times relative to cold starts at the final bandwidth. With it, Part 3 is complete as a

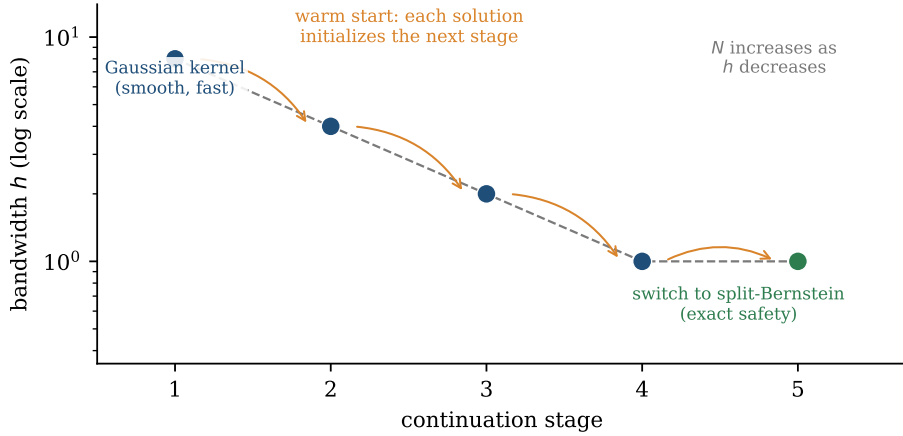


Figure 24: The bandwidth-continuation and kernel-switching schedule of Keil u. Rao (2021). Early stages use a large bandwidth and the fast Gaussian kernel, producing a benign landscape and a cheap solution; each stage’s solution warm-starts the next as the bandwidth shrinks and the sample size grows; the final stage switches to the split-Bernstein kernel so that the converged certificate carries the exact domination guarantee. The homotopy stabilizes the solver and avoids the numerical difficulty of attacking a small-bandwidth, nearly nonsmooth constraint from a cold start.

method: a conservative, smooth, one-dimensional, N -independent reformulation for individual and joint chance constraints, equipped with a practical solution strategy. What the method still lacks is a formal guarantee that its answers approximate *the true problem*, in the sense that as data accumulates the reformulated solutions converge to the true ones. Supplying that license is the convergence theory of Schuster (2025), and connecting it to the constructions of this chapter is the work the next chapter takes up.

5 Convergence Theory

Part 3 has so far been constructive. Chapter 4 built a smooth surrogate for the chance constraint, Chapter 5 redesigned it to err only on the safe side, and both chapters validated their constructions empirically on problems whose answers are known. What neither chapter could supply is the statement that elevates a construction to a method: the guarantee that solving the approximated problem is, in the limit of data, the same as solving the true one. That statement exists. The convergence theory of Schuster (2025) proves three theorems which together establish that the KDE-approximated chance-constrained problem is a well-posed approximation of the original: the approximation is exact where the constraint is inactive, the limits of approximate solutions are true solutions, and under a sharp-minimum condition such limits exist. This chapter is the theoretical center of the thesis. It states the gap the theory fills and why density-estimator consistency alone cannot fill it (Section 5.1), fixes the setting and assumptions (Section 5.2), develops each theorem with its statement, proof architecture, and practical meaning (Sections 5.3 to 5.5), draws the boundary of what the theory does and does not cover (Section 5.6), and closes on the structural gap between the two anchor papers of this thesis, which is arguably the most important open problem in the field (Section 5.7).

5.1 The Gap the Theory Fills

The biased construction of Chapter 5 provides a safety guarantee by construction: any feasible solution of the biased reformulation satisfies the original chance constraint in expectation. It is essential to recognize precisely what this guarantees and what it does not. It is a statement about the *feasibility of individual solutions*, made at finite N ; it says nothing about whether the optimizer, in certifying and comparing such solutions, is converging to the right one. A method could be safe at every N and still be solving, indefinitely, a problem whose minimizer differs substantially from the minimizer of the true problem. Safety and correctness are different properties, and Part 3 has so far delivered only the first.

The deeper question is therefore the one this chapter answers: as the sample size N grows, do solutions of the KDE-approximated problem converge to solutions of the true chance-constrained problem? Schuster (2025) provides the first rigorous answer, and the key insight that organizes the entire theory deserves to be stated before any theorem is: *density-estimator convergence does not automatically imply optimizer con-*

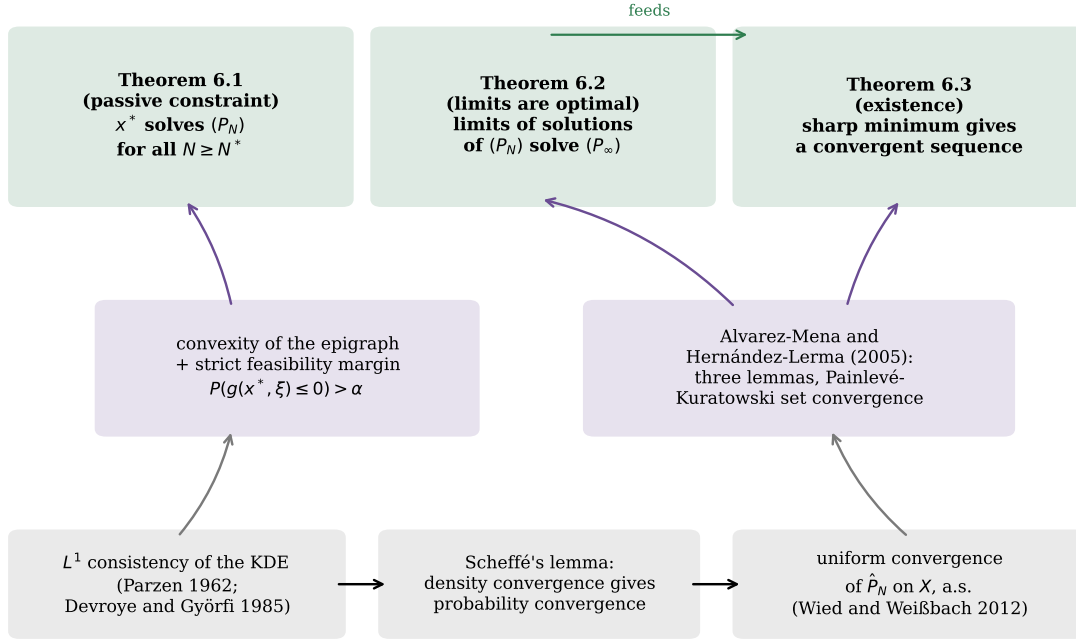


Figure 25: The architecture of the convergence theory of Schuster (2025). The statistical foundation (bottom row) is classical: L^1 consistency of the kernel estimator, upgraded to probability convergence by Scheffé’s lemma and to almost-sure uniform convergence on the feasible set under the conditions surveyed by Wied u. Weißbach (2012). Two analytical engines (middle row) convert statistical convergence into optimization statements: a convexity-of-epigraph argument with a strict feasibility margin, and the optimization-convergence theorem of Alvarez-Mena u. Hernández-Lerma (2005) verified through three lemmas. The three theorems (top row) divide the labor: exactness where the constraint is passive, optimality of limits, and existence of convergent solution sequences.

vergence. That the kernel estimate converges to the true density has been known since Parzen (1962), and Chapter 2 reviewed the classical theory in full; but an optimization problem is not a density. Its solution is determined by the interaction of an objective with the *boundary* of a feasible set, and a sequence of converging constraint functions can, in principle, produce feasible sets whose minimizers behave badly: solutions can fail to exist, can wander without converging, or can converge to limits that are infeasible or suboptimal for the true problem. A separate argument is needed to transfer convergence of the constraint approximation to convergence of the solution, and constructing that argument, in three theorems of increasing reach, is exactly what Schuster (2025) does. Figure 25 displays the architecture the rest of this chapter walks through: a statistical foundation inherited from classical KDE theory, two analytical engines, and the three theorems they power.

5.2 Setup and Assumptions

The theory is formulated at a level of generality worth pausing on, because it is part of the result. The decision space is not assumed finite-dimensional: Schuster (2025) works in a Banach space, which is precisely the generality required for the optimal control ambitions of this thesis, where decisions are trajectories before they are discretized. The standing assumptions are collected once.

Assumption 1 (Setting of Schuster, 2025). *V is a Banach space and $X \subseteq V$ is closed and nonempty. The objective f is convex and monotonically increasing. The random parameter ξ has an absolutely continuous distribution, so that for every $x \in X$ the pushforward density $\rho_{g(x)}$ of the residual vector $g(x, \xi) \in \mathbb{R}^m$ exists.*

Under Assumption 1 the chance constraint is the statement that the pushforward density places mass at least $\alpha = 1 - \varepsilon$ on the safe orthant,

$$\mathbb{P}(g(x, \xi) \leq 0) = \int_{(-\infty, 0]^m} \rho_{g(x)}(z) dz \geq \alpha, \quad (43)$$

which the reader will recognize as the residual-space identity of Chapter 4, here in its general m -dimensional form: the theory covers joint constraints through the multivariate residual, and the scalar case $m = 1$ of Chapters 4 and 5 is the special case used throughout this thesis. The approximation replaces $\rho_{g(x)}$ by its kernel estimate built from the residual samples $Y_i(x) = g(x, \xi_i)$, exactly the estimator of Chapter 4, and the pair of problems whose relationship the theory governs is

$$(P_\infty): \quad \underset{x \in X}{\text{minimize}} f(x) \quad \text{subject to} \quad \mathbb{P}(g(x, \xi) \leq 0) \geq \alpha, \quad (44)$$

$$(P_N): \quad \underset{x \in X}{\text{minimize}} f(x) \quad \text{subject to} \quad \hat{P}_N(g(x, \xi) \leq 0) \geq \alpha. \quad (45)$$

Every theorem below is a statement about how the solutions of (P_N) relate to those of (P_∞) , almost surely with respect to the sampling of $\xi_1, \xi_2, \dots$, and the three theorems are numbered here as Theorems 1 to 3, corresponding to Theorems 3.1, 3.2, and 3.3 of Schuster (2025) respectively.

5.3 The Passive Constraint Case

The first theorem settles the regime in which the chance constraint, at the optimum, is not doing any work.

Theorem 1 (Passive constraint; Theorem 3.1 of Schuster, 2025). *Let x^* solve (P_∞) with the chance constraint strictly inactive, $\mathbb{P}(g(x^*, \xi) \leq 0) > \alpha$. Then there exists N^* such that x^* also solves (P_N) for all $N \geq N^*$, almost surely.*

Proof architecture. The argument runs along the bottom-left of Fig. 25. The L^1 consistency of the kernel estimator (Parzen, 1962; Devroye u. Györfi, 1985) controls the integrated error of the density estimate; Scheffé’s lemma (Scheffé, 1947) converts L^1 convergence of densities into convergence of the probabilities they induce, so that $|\hat{P}_N - \mathbb{P}|$ at x^* falls, almost surely, below the strict feasibility margin $\gamma = \mathbb{P}(g(x^*, \xi) \leq 0) - \alpha > 0$ for all N beyond some N^* . From that point on, x^* remains feasible for every (P_N) , and a convexity-of-epigraph argument transfers the solution property: a point that is optimal among a larger feasible set and feasible for the smaller one is optimal for the smaller one as well.

Practical meaning. Where the chance constraint does not bind at the optimum, the KDE approximation introduces no error at all for large samples; the constraint is effectively transparent, and the problem behaves locally like the deterministic problem it would be without the constraint. Figure 26 draws the mechanism: the margin γ is a fixed buffer, the uniform estimation error is a tube around the true probability that shrinks with N , and once the tube fits inside the buffer the approximation can no longer disturb the solution. The theorem is the formal license for an instinct every practitioner has: approximation error only matters where the constraint matters.

5.4 Limits of Approximate Solutions Are Optimal

The second theorem addresses the regime the first one excludes, the one that matters most in practice: the constraint binds, the approximated solutions move with N , and the question is where they are allowed to go.

Theorem 2 (Limits are optimal; Theorem 3.2 of Schuster, 2025). *Suppose the kernel estimate converges almost surely uniformly for every $x \in X$. If (x_N) is a convergent sequence of solutions of (P_N) , then its limit solves (P_∞) , almost surely.*

Proof architecture. This is the deepest of the three results, and its proof is an exercise in verifying the hypotheses of a general optimization-convergence theorem due to Alvarez-Mena u. Hernández-Lerma (2005), which gives conditions under which minimizers of a sequence of problems converge to minimizers of a limit problem. Schuster

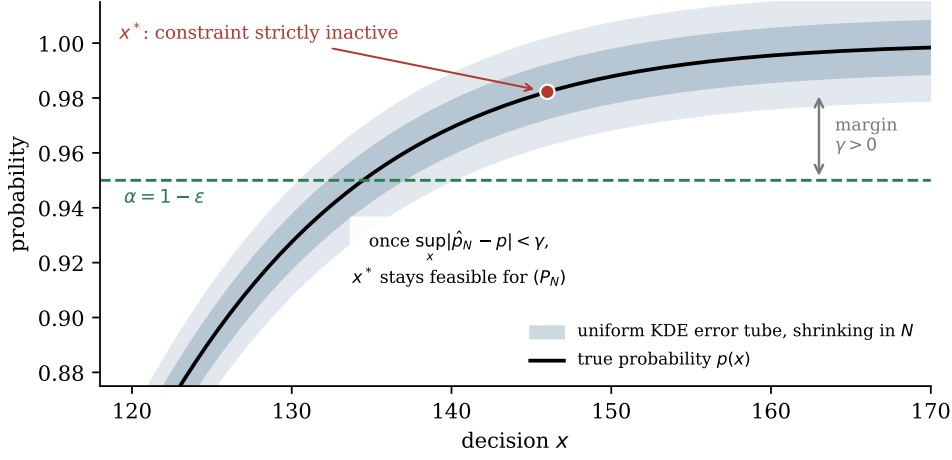


Figure 26: The mechanism of Theorem 1, drawn on the dispatch problem’s probability curve. At a solution x^* where the chance constraint is strictly inactive, the feasibility margin $\gamma = \mathbb{P}(g(x^*, \xi) \leq 0) - \alpha$ is a fixed positive buffer. The KDE error is a tube around the true probability that shrinks almost surely as N grows; once $\sup_x |\hat{p}_N - p| < \gamma$, the point x^* is feasible for every approximated problem and, by the epigraph argument, solves it.

(2025) establishes the three required conditions through three lemmas. The first upgrades the statistical foundation: uniform convergence of the kernel estimate yields uniform convergence of the constraint function $\hat{P}_N$ to $\mathbb{P}$ on X . The second translates constraint convergence into geometry: the admissible sets of (P_N) converge to the admissible set of (P_∞) in the Painlevé-Kuratowski sense, the notion of set convergence under which limits of feasible points are feasible. The third closes the loop from the other side: every feasible point of the limit problem can be approached by a sequence of points feasible for the approximating problems, so the approximations do not silently exclude parts of the true feasible set. With the three conditions in place, the theorem of Alvarez-Mena u. Hernández-Lerma (2005) delivers the conclusion.

Practical meaning. Whatever the approximated problems converge to, it is a true solution: the optimizer is approaching the right answer, not an artifact of the smoothing. The theorem converts the empirical convergence plots that this thesis produces in Part 4 from suggestive pictures into evidence about (P_∞) itself, because it guarantees that the plateau such plots approach is the true optimum and not an artifact of the approximation. What the theorem deliberately does not assert is that a convergent sequence exists; it is a statement about limits, conditional on their existence. Closing that gap is the third theorem’s job, and the division of labor between Theorems 2 and 3 is itself instructive: optimality of limits and existence of limits are separate mathematical events, requiring separate hypotheses.

5.5 Existence of a Convergent Solution Sequence

Theorem 3 (Existence; Theorem 3.3 of Schuster, 2025). *Suppose x^* is the unique solution of (P_∞) and satisfies a sharp-minimum condition: there exist $\bar{\varepsilon}, \delta > 0$ such that $f(x) > f(x^*) + \bar{\varepsilon}$ for every feasible x outside the ball $B_\delta(x^*)$. Then a convergent sequence of solutions of the approximated problems exists, almost surely.*

Proof architecture. The sharp-minimum condition is exactly what forbids the failure mode that Theorem 2 could not exclude. If near-optimal objective values were available far from x^* , the approximated solutions could achieve them and wander indefinitely; the growth condition makes every such excursion expensive, and combined with the set-convergence machinery of Section 5.4 it forces the solution sequence to be Cauchy, hence convergent. Figure 27 contrasts the two geometries: under a sharp minimum the approximate solutions are trapped in a shrinking neighborhood of x^* ; in a shallow valley, nothing prevents them from drifting between distant near-optimal points.

Practical meaning. The conditions under which the full theory operates are not exotic, and Schuster (2025) makes both halves of that claim concrete. On the geometry side, strict convexity of the objective already implies both the uniqueness of x^* and the growth condition, so the hypothesis of Theorem 3 is met by a large and recognizable problem class. On the statistical side, the uniform convergence that Theorems 2 and 3 require is not an abstract wish: a Gaussian product kernel with a diagonal bandwidth matrix, under the classical bandwidth conditions and uniform continuity of the true density, is identified as a concrete realization that guarantees it, drawing on the consistency conditions surveyed by Wied u. Weißbach (2012) and the L^1 theory of Devroye u. Györfi (1985). The theory’s assumptions can be discharged by objects this thesis already uses.

5.6 What the Theory Does and Does Not Cover

Taken together, the three theorems deliver the two assets this thesis claims from them. The first is mathematical credibility: the KDE approach is not a heuristic that happens to work but an asymptotically consistent method, with the consistency proven at the level that matters, the optimization problem, not merely the estimator inside it. The second is precision about conditions: the theory identifies exactly what convergence costs, namely uniform convergence of the estimate on the feasible set, plus either a

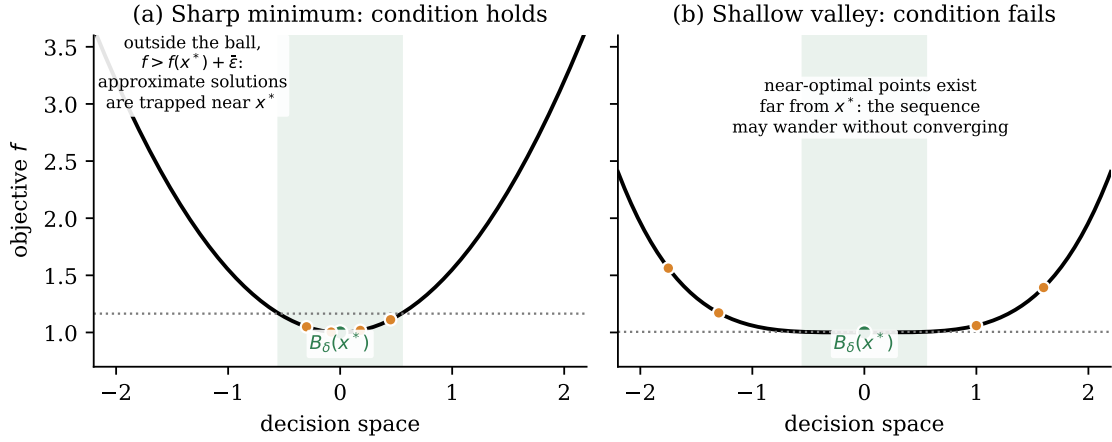


Figure 27: Why Theorem 3 needs the sharp-minimum condition. (a) When the objective grows by at least $\bar{\epsilon}$ outside the ball $B_\delta(x^*)$ (shaded), every approximate solution (orange) with near-optimal value is forced into the ball, and the set-convergence machinery makes the sequence Cauchy. (b) In a shallow valley the condition fails: near-optimal points exist far from x^* , and approximate solutions may alternate between them without converging, even though, by Theorem 2, any limit they did have would be optimal. Strict convexity of f implies the geometry of panel (a).

passive constraint or a sharp minimum, and it exhibits concrete, standard objects that pay the cost. A reader of Chapters 4 and 5 now knows not only how to build the reformulation but when to trust it.

Intellectual honesty about the boundary is equally part of the asset, and four items lie beyond it. There are, first, *no convergence rates*: the theory guarantees that the optimization error vanishes but not how fast, and no theorem links the kernel’s bias-variance behavior to the decay of $|f(x_N) - f(x^*)|$ or $\|x_N - x^*\|$; Schuster (2025) explicitly flags this as open. Figure 28 makes the gap tangible: on the dispatch problem the optimization error decays along a clean empirical power law of slope roughly -0.46 , a regular behavior that no available theorem currently predicts. Second, the theorems cover the *unbiased* estimator of Chapter 4; whether and how they extend to the biased construction of Chapter 5 is open, a point developed in Section 5.7. Third, the results lean on convexity of the objective, while the trajectory optimization problems of Part 4 are non-convex; comparable guarantees for the non-convex case do not exist. Fourth, the theory is asymptotic through and through: unlike the scenario approach of Chapters 2 and 3, it offers no N -dependent probability that a computed solution is feasible for the true problem. These four boundaries are not defects of the theory; they are its honest perimeter, and they define precisely where the experimental program of Part 4 carries the burden of evidence that mathematics does not yet carry.

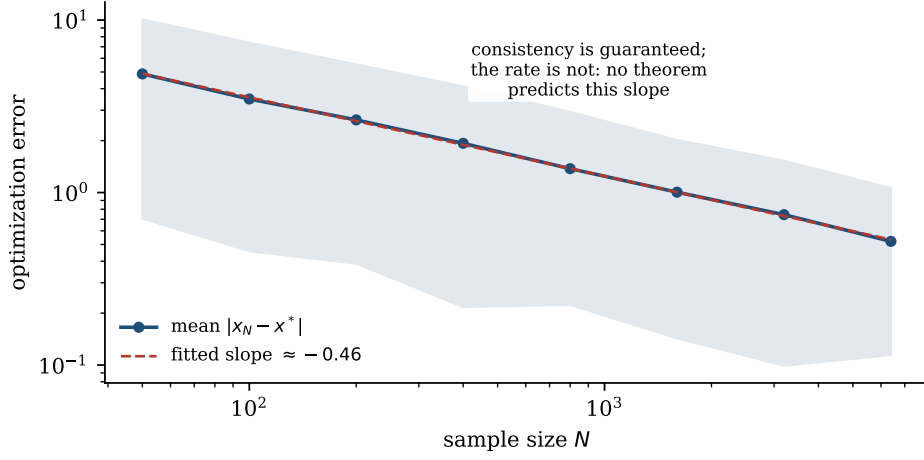


Figure 28: Consistency without rates. Mean optimization error $|x_N - x^*|$ of the unbiased KDE solution on the dispatch problem (500 trials per sample size, band is 10th-to-90th percentiles), on logarithmic axes. The error follows a clean empirical power law of slope ≈ -0.46 , but the convergence theory guarantees only that the error vanishes; no theorem predicts the slope. Quantifying this rate, and its dependence on the bandwidth rule, is the most important open problem the anchor papers leave

5.7 The Gap Between the Two Anchor Papers

One observation deserves to close Part 3’s theoretical development, stated as plainly as its importance warrants. The two anchor papers of this thesis work with the same high-level idea from opposite directions, and their results do not yet meet. Keil u. a. (2021) builds a *biased* integrated kernel that dominates the indicator, securing a finite-sample safety guarantee, but proves no convergence of the resulting optimizers. Schuster (2025) proves convergence of the optimizers, but for the *unbiased* kernel density estimate, with no safety guarantee at finite N . Neither paper’s theorems cover the other’s construction: the engineering line has safety without proven convergence, and the theoretical line has convergence without built-in safety.

The open problem this leaves is easy to state and, in the judgment of this thesis, arguably the most important one in the field: a unified convergence theory for the biased-KDE reformulation, establishing that the conservative construction of Chapter 5 inherits the asymptotic correctness of this chapter, ideally with rates. Its resolution would fuse the two guarantees that applications actually need, safety at every finite N and correctness in the limit, into a single certified method. This thesis does not claim to close that problem. What it does claim, and delivers in Part 4, is the systematic empirical groundwork on which a closure must rest: controlled experiments in which the biased and unbiased reformulations are run side by side, across sample

sizes, bandwidths, and risk levels, on problems whose true solutions are known, so that the conjectured convergence of the biased construction is measured even where it is not yet proven. The theory of this chapter tells us what such measurements mean; the gap between the anchors tells us why they are worth making.

6 Optimal Control Formulation

The optimal-control extension is the point at which statistical approximation, numerical transcription, and physical modelling become inseparable. A chance constraint is evaluated on a residual produced by a trajectory, but the trajectory itself is generated by a discretized differential equation. Consequently, a small test violation can coexist with a failed nonlinear program, and a reported objective can depend on the quadrature convention rather than on the physical control effort. This chapter states the formulation used by the local implementation, identifies the differences from the published lunar benchmark, and derives an audit identity that limits the interpretation of the local costs.

6.1 Stochastic optimal-control problem

Let $y(t) \in \mathbb{R}^{n_y}$ and $u(t) \in \mathbb{R}^{n_u}$ denote state and control trajectories, and let ξ denote uncertain parameters. A generic finite-horizon problem is

$$\begin{aligned} \min_{y,u} \quad & J(y,u) = \int_0^{t_f} L(y(t), u(t)) dt + \Phi(y(t_f)), \\ \text{s.t.} \quad & \dot{y}(t) = F(y(t), u(t), \xi), \\ & y(0) \in \mathcal{Y}_0, \quad u(t) \in \mathcal{U}, \\ & \mathbb{P}\{g_\ell(y, u, \xi) \leq 0\} \geq 1 - \varepsilon_\ell, \quad \ell = 1, \dots, m. \end{aligned} \tag{46}$$

The residual g_ℓ may be a terminal event, a path event evaluated at a time t , or a vector event reduced to a scalar statistic. The distinction matters: a terminal probability is not a pathwise probability over a continuum of times. In the implementation, path constraints are evaluated at mesh nodes. They are therefore nodewise surrogate constraints unless an additional union or supremum construction is imposed.

For training samples $\{\xi_i\}_{i=1}^n$ and a decision vector z containing the nodal states and controls, the residual observations are

$$r_{\ell i}(z) = g_\ell(y(z), u(z), \xi_i). \tag{47}$$

The empirical chance constraint used in Chapters 3–5 is replaced by an integrated-

kernel average,

$$\hat{p}_{\ell,h}(z) = \frac{1}{n} \sum_{i=1}^n \bar{K}\left(\frac{r_{\ell i}(z)}{h}\right), \quad \hat{p}_{\ell,h}(z) \leq \varepsilon_{\ell}, \quad (48)$$

where $\bar{K}$ is the integrated kernel and h is a bandwidth. Equation (48) is a deterministic NLP constraint after the samples and bandwidth have been fixed. It does not by itself provide a confidence level for the unknown probability. The biased construction discussed in Chapter 4 can provide an upper bound only when its bias and sampling assumptions are satisfied; the local shifted estimator is recorded as an implementation surrogate and is not identified with the Split–Bernstein construction of Keil u. a. (2020, 2021).

6.2 Direct transcription used in the experiments

The local fallback solver uses N uniformly spaced nodes $0 = t_0 < \dots < t_{N-1} = t_f$ and forward Euler defects. With $h_j = t_{j+1} - t_j$,

$$y_{j+1} - y_j - h_j F(y_j, u_j, \xi_0) = 0, \quad j = 0, \dots, N-2, \quad (49)$$

where ξ_0 is absent from the deterministic dynamics in the lunar proxy. The collocation vector includes all nodal controls, although the last control does not enter the forward Euler defects. Equality constraints enforce the initial state, the defects, and terminal velocity. Terminal position remains a chance-constrained quantity. Inequality constraints enforce control bounds and the KDE approximations of the terminal and path residuals.

The implementation separates four operations. The problem module defines dynamics and residuals; the collocation module maps a vector z to nodal states and controls; the KDE module evaluates integrated-kernel statistics; and the solver module supplies SLSQP with equality and inequality callables. This separation makes it possible to inspect a statistical estimator without silently changing the dynamics. It also exposes a numerical limitation: the same nodal control can contribute to a path residual and to the objective while not contributing to the state defects when it is the final control. That limitation is quantified below rather than hidden in a single objective number.

The continuation sequence is nominal, worst-case, scenario, unbiased KDE, and local shifted Epanechnikov. A stage is warm-started with the last successful vector. Warm starts are consistent with the continuation strategy described by Keil u. Rao (2021),

but the local implementation does not use LGR nodes, adaptive mesh refinement, IPOPT, or GPOPS-II. A failed SLSQP stage is retained in the result ledger with its status and constraint margin; it is not promoted to a feasible solution because its held-out violation happens to be below the target.

6.3 Local lunar landing model

The local benchmark has altitude y_1 , vertical velocity y_2 , and nonnegative thrust u :

$$\dot{y}_1 = y_2, \quad \dot{y}_2 = -g + u, \quad g = 1.622, \quad 0 \leq u \leq 3. \quad (50)$$

The local horizon is fixed at $t_f = 20$, the initial state is $(y_1(0), y_2(0)) = (100, 0)$, and only terminal velocity is an equality. Terminal altitude is subject to

$$\mathbb{P}(|y_1(t_f) - \xi_1| - \delta > 0) \leq \varepsilon_a, \quad \xi_1 \sim \mathcal{N}(0, 0.1^2), \quad \delta = 0.25, \quad \varepsilon_a = 0.1. \quad (51)$$

The nodewise path event is

$$\mathbb{P}(u(t_j) + \xi_2 - 3 > 0) \leq \varepsilon_b, \quad \varepsilon_b = 0.01, \quad (52)$$

with the locally documented mixture $\xi_2 \sim 0.6\mathcal{N}(-0.25, 0.10^2) + 0.4\mathcal{N}(0.35, 0.15^2)$. The mixture is a synthetic stress test for a bounded, asymmetric disturbance; it is not a claim that the local data reproduce the distribution used in the published study.

The comparison with Keil et al. must be made at the level of formulation. Their lunar study uses $y_1(0) = 10$, $y_2(0) = -2$, free final time, free terminal altitude, terminal velocity zero, and $J = \int u \, dt$; it solves the resulting problem with LGR/GPOPS-II/SNOPT and reports costs around 9.09 for the chance-constrained methods (Keil u. a., 2020, 2021). The local problem fixes a different initial altitude, initial velocity, horizon, mesh, and solver. The two objective values therefore cannot be compared numerically. The local experiment is a pipeline and audit experiment, not a replication.

6.4 A conservation audit of the local objective

The local transcription provides a stronger diagnostic than a solver status. Summing the velocity defects over the $N - 1$ intervals gives

$$y_{2,N-1} - y_{2,0} = \sum_{j=0}^{N-2} h_j(-g + u_j). \quad (53)$$

With uniform $h = 2$, $y_{2,0} = y_{2,N-1} = 0$, and $g = 1.622$, every feasible trajectory satisfies

$$\sum_{j=0}^{N-2} h u_j = g t_f = 32.44. \quad (54)$$

Thus, in this fixed-horizon model, the physically consistent piecewise-constant fuel integral is invariant over the feasible set. Trajectory shape still matters because terminal altitude imposes the weighted moment

$$\int_0^{t_f} (t_f - t)u(t) dt = \frac{g t_f^2}{2} - y_1(0) = 224.4, \quad (55)$$

when terminal altitude is zero. The stochastic terminal event changes the admissible timing of thrust, not the invariant integral implied by terminal velocity.

The code records a trapezoidal integral over all N controls, whereas the Euler defects use only $u_0, \dots, u_{N-2}$. On a uniform mesh,

$$J_{\text{trap}} = h \left(\frac{u_0}{2} + \sum_{j=1}^{N-2} u_j + \frac{u_{N-1}}{2} \right) = 32.44 + \frac{h}{2}(u_{N-1} - u_0). \quad (56)$$

The nominal record $J = 29.8052$ is explained by the endpoint term: the stored solution has approximately $u_0 = 2.6345$ and $u_{N-1} = 0$. The later records at 32.44 are not evidence that robust control requires more physical fuel. They reveal that the endpoint contribution becomes different when the optimizer is warm-started under another constraint. This audit is a substantive result of the local implementation and is the reason Chapter 8 reports the lunar objective only as a transcription diagnostic.

Table 2: Formulation differences between the published lunar study and the local audit benchmark.

Feature	Keil et al.	Local implementation
Initial state	$(10, -2)$	$(100, 0)$
Final time	free	fixed at 20
Terminal conditions	velocity equality; altitude free	velocity equality; altitude chance event
Transcription	LGR with adaptive machinery	11-node forward Euler
Solver	GPOPS-II/SNOPT	SciPy SLSQP
Uncertainty	terminal normal and path mixture in published model	synthetic normal and two-mode mixture
Interpretation of J	physical $\int u dt$	trapezoidal nodal diagnostic with endpoint mismatch

6.5 Stage logic and numerical interpretation

The local stage ledger contains five records. The nominal stage succeeds, with terminal test violation 0.51425. The worst-case, scenario, unbiased KDE, and local shifted stages all report SLSQP failure; their stored controls are useful for diagnosing continuation, but they are not feasible certified solutions. The robust-stage minimum chance margins are respectively approximately -0.738 , -0.280 , -0.0209 , and -0.0867 in the solver diagnostics. A terminal test frequency below 0.1 in a failed stage cannot repair a negative optimization margin, and no pathwise held-out validation was performed. The local experiment therefore supports a statement about solver difficulty and residual statistics, not a claim of successful robust lunar landing.

This distinction also clarifies what a warm start can and cannot do. Continuation reduces abrupt changes in the NLP, but it cannot create feasibility when the transcription, fixed horizon, and risk margins are incompatible. The warm-start paper of Keil u. Rao (2021) couples kernel changes to increasing sample sizes and mesh refinement. The local sequence changes the residual estimator while holding a coarse mesh and a fixed sample regime. It is consequently a partial implementation of the continuation idea, not a reproduction of its convergence behavior.

6.6 Scope of the formulation

The formulation is adequate for a controlled comparison of residual estimators in low dimension. It is not adequate for a claim about continuous-time path safety, because

nodewise constraints do not bound the inter-node maximum. It is not adequate for a physical fuel comparison until the quadrature is made consistent with the defect control parameterization. Finally, the finite-dimensional KDE constraint inherits the statistical qualifications developed earlier: convergence of a KDE sequence under assumptions, as in Schuster (2025), does not give a finite-sample confidence statement for the particular 250-sample local run. These limits determine the evidence hierarchy used in Chapters 8 and 9.

7 Implementation and Validation

The implementation follows a fixed data path:

training samples $\rightarrow$ residuals $\rightarrow$ KDE or baseline constraint $\rightarrow$ NLP/collocation $\rightarrow$
independent test Monte Carlo $\rightarrow$ saved result artifacts.

The modules are deliberately small. ‘uncertainty’ produces seeded distributions and disjoint train/test samples; ‘kde’ implements kernels, integrated kernels, and band-width rules; ‘constraints’ handles scalar, Boole, maximum, and log-sum-exp statistics; ‘problems’ contains the canonical static, dispatch, and lunar models; and ‘evaluation’ serializes metrics and provenance.

For a computed decision x^* and independent test samples ξ_r^{test} , the central validation statistic is

$$\hat{v}_{\text{test}}(x^*) = \frac{1}{M} \sum_{r=1}^M \mathbf{1}\{g(x^*, \xi_r^{\text{test}}) > 0\}. \quad (57)$$

The conservatism gap is

$$\text{gap} = \varepsilon - \hat{v}_{\text{test}}(x^*). \quad (58)$$

A result record also contains the training empirical violation, the KDE estimate used by the optimizer, objective value, runtime, solver status, bandwidth, sample sizes, seed, and benchmark metadata. Training samples never enter the test estimate, and the test seed is independent of the training seed.

The test sources cover kernel normalization and limits, analytic gradients by central differences, reproducibility under fixed seeds, residual dimensions, and agreement between saved metrics and independently recomputed violations. The gradient check uses relative tolerance 10^{-5} and absolute tolerance 10^{-7} ; the direct numerical smoke checks passed. These checks do not create a finite-sample certificate. The local run used a virtual environment with the installed NumPy and SciPy packages; Python compilation, direct numerical checks, the seeded runner, and LaTeX compilation were executed, while pytest and the optional CasADi/IPOPT path were unavailable and are not claimed as passed. The local shifted Epanechnikov surrogate dominates the empirical violation indicator, but out-of-sample feasibility remains an empirical question.

8 Experimental Study

This chapter reports the local evidence rather than treating every output field as a theorem. The experiments were designed to answer four questions. First, how do nominal, scenario, unbiased KDE, and locally shifted Epanechnikov constraints behave when the residual law is changed while the deterministic model is fixed? Second, does the same interface remain usable for a balance-constrained dispatch problem? Third, how sensitive are the estimated probabilities to bandwidth, sample size, and a joint statistic? Fourth, what can the lunar continuation experiment actually establish after its numerical audit? All experiments are synthetic, seeded, and run locally. The resulting measurements are empirical frequencies or estimator diagnostics; they are not out-of-sample certificates.

8.1 Design, estimands, and provenance

The static model has two decision variables and a nonlinear residual whose uncertainty enters through a scalar residual map. Four residual laws are used: Gaussian, bimodal, skewed, and heavy-tailed. For each law, 250 training observations define the optimization constraint and 4,000 independent test observations define the held-out frequency. The target is $\varepsilon = 0.1$. The method order is nominal, scenario, unbiased KDE, and local shifted Epanechnikov. The exact seed mapping is 20260906 for the Gaussian static run, 20260907 for dispatch and the bimodal row, 20260908 for the joint comparison and the skewed row, 20260909 for sensitivity and the heavy-tailed row, and 20260910 for the lunar continuation. The mapping is recorded explicitly because a statement such as “the five benchmark seeds” is otherwise not reproducible.

For a decision x , the principal held-out quantity is

$$\hat{v}_{\text{test}}(x) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbf{1}\{r(x, \xi_i) > 0\}. \quad (59)$$

The standard error of this frequency is sampling error in the test set, not uncertainty in the optimized decision. For example, the Gaussian unbiased KDE row has 248 violations out of 4,000, giving $\hat{v} = 0.0620$ and an approximate 95% binomial interval $[0.0547, 0.0699]$; the local row has 77 violations and interval $[0.0152, 0.0240]$. These intervals describe repeated test sampling conditional on the fitted decision. They do not account for selecting the decision after inspecting training data, and they do not

Table 3: Static Gaussian benchmark. The test frequency uses 4,000 independent observations and $\varepsilon = 0.1$.

Method	Objective	Train	Test	Est. diagnostic	Gap
Nominal	0.0234	0.5320	0.5180	0.5320	-0.4180
Scenario	0.6134	0.0040	0.0015	0.0040	0.0985
Unbiased KDE	0.1981	0.0520	0.0620	0.1000	0.0380
Local shifted Epanechnikov	0.3071	0.0320	0.0193	0.0531	0.0808

become scenario confidence bounds by relabelling them.

The result ledger contains objective, training violation, held-out violation, estimated violation, solver status, and gap fields. A source audit found that the field named `estimated_violation` is computed with a Gaussian KDE for both KDE methods. For the local shifted Epanechnikov rows it is therefore a Gaussian diagnostic, not the value of the biased estimator used in the constraint. The optimized local estimator is active at approximately 0.1 on successful rows. This distinction is retained throughout the chapter.

8.2 Static nonlinear benchmark

The Gaussian case provides the clearest baseline because its true radial probability can be evaluated independently. The local results are:

The nominal decision is not a marginally unsafe solution: its held-out frequency is 0.5180, five times the target. The scenario solution is strongly conservative on this split, while the two KDE solutions retain more objective value. The unbiased KDE is closer to the Gaussian reference than the local method in objective value, but the local method has the lower test frequency. This ordering is evidence about one fitted sample, not a universal ranking. The analytical Gaussian calculation gives an exact test probability near 0.5137 for the nominal decision and near 0.0572 for the unbiased KDE decision. It also gives a Gaussian chance-constrained reference objective near 0.1392, below the KDE objective 0.1981. The gap is consistent with smoothing and finite training data imposing extra slack.

Across the non-Gaussian laws, the pairs $(\hat{v}_{\text{test}}^{\text{unbiased}}, \hat{v}_{\text{test}}^{\text{local}})$ are (0.0150, 0.0003) for the bimodal law, (0.1053, 0.0700) for the skewed law, and (0.0613, 0.0405) for the heavy-tailed law. The skewed unbiased row exceeds the target. Its 95% test interval is approximately [0.0959, 0.1152], so the data do not establish a clear separation from

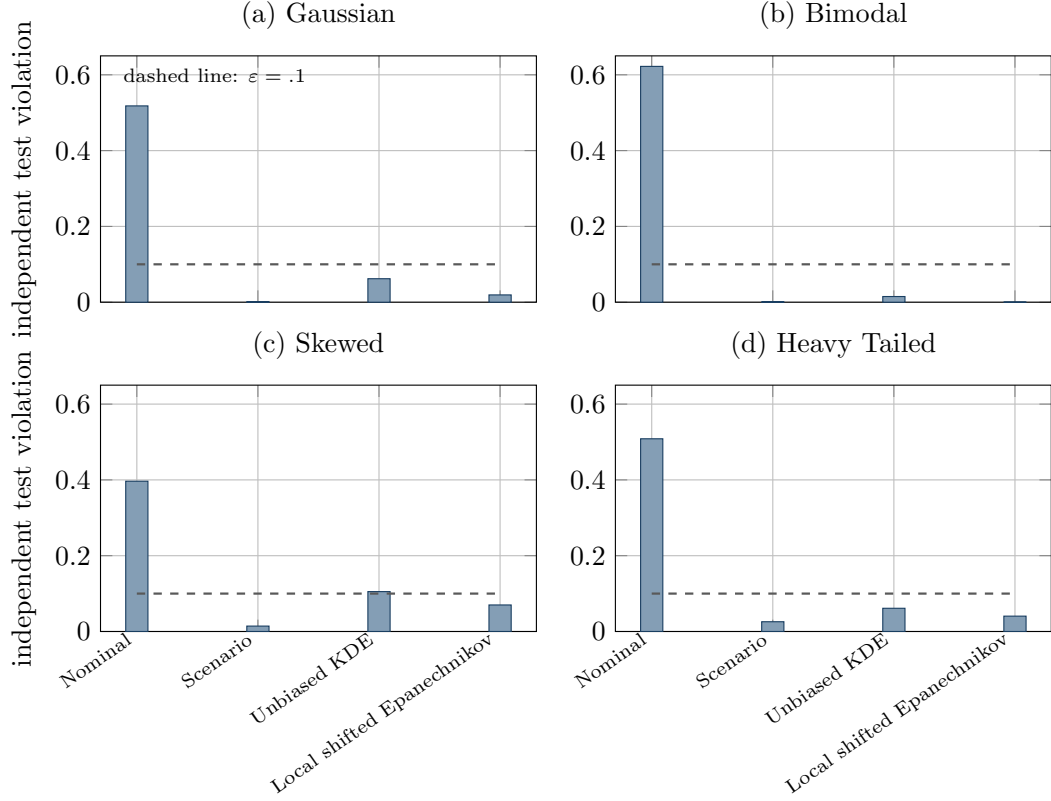


Figure 29: Independent test violation for the static nonlinear benchmark ($N_{\text{train}} = 250$, $N_{\text{test}} = 4000$, $\varepsilon = 0.1$). The dashed line is the design target.

0.1, but they also do not support calling the method safe. The scenario method fails its nonlinear program on the skewed and heavy-tailed rows despite low recorded test frequencies in the finite sample. This is a useful warning: a low test count from a failed run is not evidence of a feasible optimizer.

The four panels in Figure 29 display these frequencies. The dashed line is only the design target. Figure 30 shows the objective–violation trade-off. In both figures, successful and failed solver rows must be read together with the status field. The figures do not encode statistical confidence, repeated seeds, or a correction for trying four methods across four distributions.

The qualitative pattern is compatible with the motivation for KDE in process systems and data-driven chance constraints (Calfa u. a., 2015): a residual-space estimator can represent multimodality without committing to a single Gaussian family. It does not follow that KDE dominates scenario optimization in every regime. Scenario optimization has a different guarantee architecture, with explicit sample-complexity conditions for convex programs (Calafiore u. Campi, 2006; Campi u. Garatti, 2008). The local

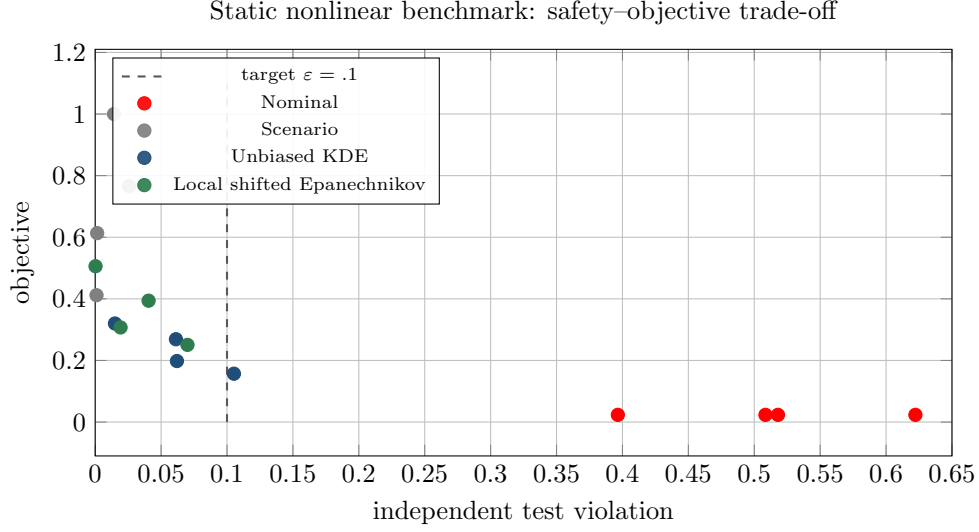


Figure 30: Objective versus independent test violation for the four static residual laws. Solver status and estimator identity are reported in the accompanying ledger.

Table 4: Two-generator dispatch benchmark. The uncertain balance residual is tested on an independent sample.

Method	Objective	Train	Test	Solver
Nominal	928.5686	0.5788	0.5788	success
Scenario	935.3871	0.0060	0.0060	success
Unbiased KDE	933.9233	0.0398	0.0398	success
Local shifted Epanechnikov	934.8865	0.0110	0.0110	success

nonlinear problem is outside the setting in which those results can simply be quoted.

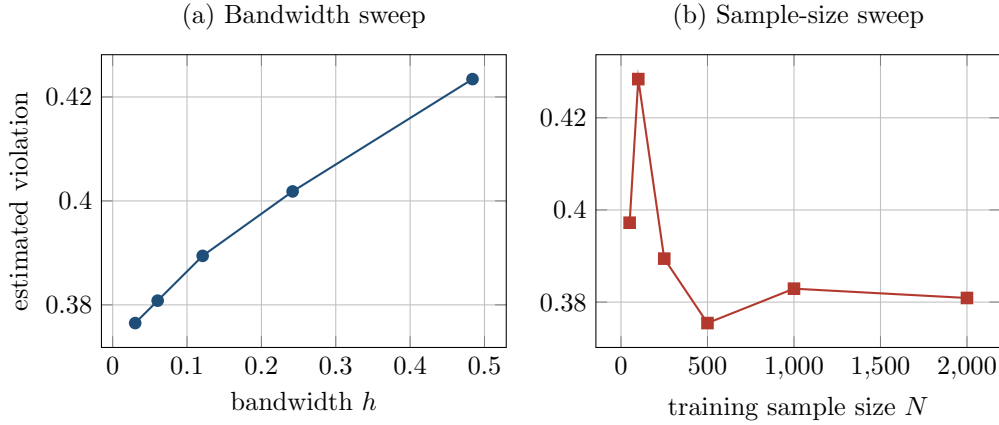
8.3 Energy dispatch and exact distributional check

The dispatch proxy has two generators and a balance residual. Capacity residuals are deterministic in this adapter, so the stochastic problem tests only renewable forecast error in the balance equation. The error is the mixture $0.6\mathcal{N}(-0.25, 0.10^2) + 0.4\mathcal{N}(0.35, 0.15^2)$. The objective values and held-out frequencies are:

Because the mixture is known, its CDF supplies an independent reference. The exact mixture probability at the nominal decision is approximately 0.5951, while the Monte Carlo test frequency is 0.5788. The corresponding probabilities for scenario, unbiased KDE, and local shifted decisions are approximately 0.0043, 0.0450, and 0.0107. The sample frequencies are therefore plausible finite-sample measurements, and the objective ranking is not an artefact of an unknown test distribution. The scenario solution

Table 5: Fixed-residual sensitivity values. Each row changes one input while the other is held fixed.

Bandwidth factor	0.25	0.5	1	2	4	Test
Estimated violation	0.3765	0.3808	0.3894	0.4018	0.4234	0.3950
Training size	50	100	250	500	1000	2000
Estimated violation	0.3972	0.4284	0.3894	0.3754	0.3829	0.3809

**Figure 31:** Sensitivity of a fixed-residual violation estimate to bandwidth and training sample size. The experiment does not reoptimize the decision.

is most conservative; the local estimator is closer to it than the unbiased KDE in violation while retaining slightly lower objective than scenario. This is a single synthetic balance experiment and says nothing about network congestion, unit commitment, temporal correlation, or capacity uncertainty.

8.4 Bandwidth, sample size, and joint statistics

The sensitivity experiment holds a scalar residual split fixed and changes only the estimator input. It is deliberately not an end-to-end reoptimization. The baseline bandwidth is $h_0 = 0.120976$ and the fixed test frequency is 0.3950. The curves in Figure 31 show a smooth bandwidth response but no monotone sample-size law. That is expected for a plug-in estimator evaluated at a fixed decision and a finite random sample; it is not evidence against consistency.

The joint experiment compares statistics rather than optimized risk allocations. Equal Boole allocation and the exact maximum statistic use the same Boolean event in the current runner, so their held-out rates coincide at 0.5530. The log-sum-exp statistic

Table 6: Local lunar continuation. The objective is the stored trapezoidal transcription value; robust stages have failed SLSQP status. Validation is terminal-position frequency only.

Stage	Stored J	Train terminal	Test terminal	Status
Nominal	29.8052	0.5080	0.5143	success
Worst-case	29.8052	0.5080	0.5143	failure
Scenario	31.3685	0.0400	0.0255	failure
Unbiased KDE	32.4400	0.1080	0.1128	failure
Local shifted Epanechnikov	32.4400	0.0520	0.0548	failure

with $\tau = 0.1$ gives 0.5745. These values are computed on the held-out half of a 4,000-sample pool; a separate ledger field averages over the full pool and must not be confused with the test rate. Component rates are approximately 0.3328 and 0.3255. The joint experiment therefore demonstrates how a smooth upper-envelope statistic changes the measured event, but it does not demonstrate a Boole allocation theorem or a joint chance-constrained optimizer. An actual allocation study would impose separate component constraints and report the union event on an independent test set.

8.5 Lunar continuation record

The lunar records are included to make the experimental chain auditable. The stage table is:

The table should be read with the formulation audit in Chapter 6. The nominal result is unsafe. The robust rows are failed NLP outputs, and their terminal frequencies do not validate the path event. The terminal-position test uses the Gaussian disturbance only; no saved control trajectory was passed through an independent all-time path test. The stored nominal value below 32.44 is explained by the mismatch in equation (56). These observations rule out a cost claim and a continuous-time safety claim, while preserving the value of the run as a reproducibility and failure-analysis record.

The difference from Keil et al. is expected. Their published costs around 9.09 arise from a free-time LGR formulation with different initial conditions and a different numerical stack (Keil u. a., 2020, 2021). The local rows cannot confirm or refute that result. They show instead that a locally assembled pipeline must expose transcription invariants and solver status before numerical comparisons are meaningful.

8.6 Evidence hierarchy

The strongest local evidence is the agreement of independent checks: static Gaussian analytical probabilities, the exact dispatch mixture CDF, and held-out test frequencies. The next level is the descriptive sensitivity experiment, which identifies bandwidth and sample-size dependence without pretending to optimize. The weakest evidence is a failed lunar stage with a low terminal frequency, because it lacks feasibility and pathwise validation. This hierarchy determines the claims made in Chapter 9. It also explains why a figure with four bars is useful only when its estimator, split, and solver status are stated beside it.

8.7 Reproducibility and reporting protocol

The experiment is reproducible only if the numerical context travels with the scalar result. Each record therefore needs the model name, residual law, method, seed, training size, test size, bandwidth, target risk, objective convention, solver status, and validation scope. A held-out violation of 0.0193 without those fields is not a portable observation: it could be a training frequency, a Gaussian diagnostic, a failed solve, or a value obtained after changing the test set. The source ledger and the machine-readable summary keep these meanings separate.

A second requirement is to preserve failed runs. Removing the skewed scenario row because SLSQP returned failure would make the table look cleaner while erasing information about the interaction between nonconvex residuals and scenario constraints. The same applies to the lunar robust stages. Their negative chance margins explain why a low terminal frequency cannot be interpreted as a successful controller. In an experimental paper, a failure record is part of the denominator for claims about robustness and computational reliability.

The third requirement is to distinguish diagnostic replication from confirmatory evaluation. The current run fixes one sample split per row and uses the test set after fitting. It is appropriate for checking that the implementation is deterministic and for exposing numerical inconsistencies. It is not a study of the distribution of objective or violation over repeated datasets. A confirmatory study would pre-register the seed set, fit on each training sample, reserve an untouched test sample, and report the median, dispersion, and failure rate over repetitions. If the target is a rare event, the test size must be selected from the desired interval width rather than inherited from

a convenient default.

The local findings consequently support a transparent hybrid reproducibility claim: the code, seeds, result schema, figures, and interpretation can be rerun locally, while the statistical statements remain conditional on the synthetic design. This is a stronger and more useful claim than calling a single successful SLSQP result a guarantee. It also creates a direct path for future extensions: corrected path validation, repeated seeds, consistent control quadrature, and an explicit finite-sample confidence construction can be added without changing the evidence vocabulary used here.

9 Discussion and Practical Guidance

The experiments support a qualified conclusion. Residual-space KDE constraints can preserve useful objective value on low-dimensional synthetic problems with non-Gaussian uncertainty, but the local implementation does not establish a finite-sample safety guarantee and does not yet provide a valid cost comparison for the lunar transcription. The most informative result is therefore methodological: statistical approximation, solver feasibility, and physical discretization must be audited separately. A single test frequency cannot stand in for all three.

9.1 What the static evidence says

The static Gaussian experiment gives a clean ordering of failure modes. The nominal decision has test violation 0.5180, so replacing a chance constraint by a mean constraint is materially unsafe in this nonlinear residual. The scenario decision has violation 0.0015 and objective 0.6134, which is conservative on the observed split. The unbiased KDE and local shifted Epanechnikov decisions have violations 0.0620 and 0.0193, respectively. Their objectives, 0.1981 and 0.3071, lie between the nominal and scenario solutions. This is the expected qualitative role of a smooth empirical probability constraint: it can use slack in the residual distribution instead of enforcing every training observation.

The analytical Gaussian check strengthens but does not generalize that observation. The exact Gaussian probability of the nominal decision is about 0.5137, close to the test frequency. The unbiased KDE decision has exact probability about 0.0572, also close to its test frequency. The analytical reference objective, approximately 0.1392, is below the KDE objective because smoothing and finite training samples alter the feasible boundary. This does not show that KDE is inefficient; it shows that the estimator defines a different finite-sample optimization problem. For the bimodal and heavy-tailed laws, KDE remains below the target on the recorded split. For the skewed law, the unbiased row is 0.1053, and the scenario run fails. The evidence therefore favours distributional flexibility while rejecting a blanket claim of safety across residual shapes.

The local shifted method has lower test frequency than the unbiased method in all four static rows, but that ordering should not be attributed to a theorem. The runner stores a Gaussian KDE diagnostic under `estimated_violation` for the local row, whereas

the actual shifted Epanechnikov constraint is active at approximately 0.1. The local method is not the biased integrated-kernel estimator in Keil et al. Their construction uses a bandwidth-dependent shift that dominates the indicator under a stated condition and relies on a law-of-large-numbers premise for the sample sequence (Keil u. a., 2020, 2021). A local 250-sample Silverman bandwidth and an empirical shift are not enough to transfer that guarantee. The correct interpretation is that the local method is a useful conservative surrogate in these runs, with its conservatism measured by test data rather than certified by the cited theorem.

9.2 Scenario optimization and KDE answer different questions

The scenario rows should not be treated as a baseline that KDE must beat. Scenario optimization is designed around a confidence statement for sampled constraints under structural assumptions such as convexity and an appropriate support or dimension measure (Calafiore u. Campi, 2006; Campi u. Garatti, 2008). The local static residual is nonlinear, and the solver can fail while still returning a low finite-sample violation. KDE instead approximates a probability functional and supplies a differentiable constraint. It trades a direct finite-sample scenario argument for assumptions about sampling, bandwidth, residual regularity, and optimization convergence. These are different contracts.

The observed dispatch ordering illustrates the distinction. The exact mixture CDF gives probabilities approximately 0.5951, 0.0043, 0.0450, and 0.0107 for nominal, scenario, unbiased KDE, and local decisions. Scenario is most conservative, the unbiased KDE retains objective value, and the local shifted method is close to scenario in violation. The stochastic constraint concerns only balance; capacities are deterministic, and there is no temporal coupling. The result supports portability of the software interface, not a claim about realistic unit commitment or renewable operation.

9.3 Bandwidth and data requirements

Bandwidth is a modelling parameter, not a cosmetic plotting choice. On the fixed residual split, multiplying the baseline by 0.25 to 4 moves the estimate from 0.3765 to 0.4234. At a fixed decision, the independent test frequency is 0.3950. The sample-size sweep ranges from 0.3754 to 0.4284 without monotone improvement. This behaviour is consistent with finite-sample bias and variance: a small bandwidth follows

individual residuals, while a large bandwidth spreads mass across the threshold. Neither sweep reoptimizes the decision, so it cannot quantify the full optimizer–estimator interaction.

A defensible workflow should therefore select bandwidth on a training or validation split, retain an untouched test set, and repeat the complete optimization over multiple seeds. Reporting one seed is acceptable for a reproducibility smoke test; it is inadequate for estimating a method ranking. For rare events, $n = 4,000$ is also too small to resolve probabilities much below 10^{-3} without a correspondingly wide confidence interval. Tail-focused sampling, importance sampling, or distributionally robust bounds would be needed for that regime. The convergence results of Schuster (2025) concern sequences of approximate optimization problems under their assumptions. They do not turn a single bandwidth and a single sample into a finite-sample certificate.

9.4 Joint constraints require a real allocation experiment

The joint comparison is deliberately modest. Equal Boole allocation and the exact maximum statistic coincide in the current runner because both evaluate the same Boolean event. The log-sum-exp statistic produces a higher observed rate, 0.5745 versus 0.5530, with $\tau = 0.1$. This is evidence that smoothing changes the residual statistic; it is not evidence that log-sum-exp enforces a union chance constraint. The recorded risk allocation is metadata rather than an optimized set of component constraints, and one ledger field averages over the full sample pool rather than the held-out half.

A stronger experiment would define component residuals, allocate $\varepsilon_1 + \varepsilon_2 = \varepsilon$, optimize both component constraints, and report the union event on a third sample. It should also compare the smooth statistic with a nonsmooth maximum under the same decision and use repeated seeds. That design would connect the practical construction to the Boole and maximum-statistic theory reviewed earlier. The current result is still useful because it identifies precisely what the implementation does and what remains to be implemented.

9.5 The lunar result is a numerical audit

The local lunar experiment cannot support the statement that a KDE controller uses 29.8052 units of fuel. The forward Euler defects and terminal velocity equality imply

$\sum hu_j = 32.44$ for every feasible fixed-horizon trajectory. The stored trapezoidal objective includes the final nodal control, which does not enter the defects, and satisfies

$$J_{\text{trap}} = 32.44 + \frac{h}{2}(u_{N-1} - u_0). \quad (60)$$

The nominal value 29.8052 is an endpoint quadrature effect. The robust values at 32.44 expose the same mismatch under a different continuation stage. A physically interpretable comparison would use the interval controls consistently in both dynamics and objective, or would adopt a higher-order collocation scheme with a matching quadrature rule.

The local robust stages also fail SLSQP. Their terminal test frequencies (0.0255, 0.1128, and 0.0548 for scenario, unbiased KDE, and local shifted stages) cannot override negative solver margins, and the validation does not test the path event over time. The nominal stage has terminal test violation 0.5143. The defensible conclusion is that this coarse fixed-horizon proxy is a useful stress test for infeasibility and implementation diagnostics. It is not a successful replication of Keil et al., whose free-time LGR formulation and different initial conditions yield costs near 9.09 (Keil u. a., 2020, 2021). The discrepancy is explained by non-equivalent problem definitions and a local quadrature defect, not by a contradiction in the published result.

9.6 Threats to validity

Five limitations bound the claims. First, every distribution is synthetic; there is no external dataset against which the residual laws or dispatch mixture can be calibrated. Second, the static and dispatch studies use one primary seed per benchmark row, so the figures show conditional outcomes rather than a sampling distribution over fitted decisions. Third, the test frequencies are Monte Carlo estimates and are not corrected for method selection, repeated comparisons, or solver failures. Fourth, the joint and lunar adapters do not yet implement independent component or continuous-time path validation. Fifth, the local objective audit shows that a correct statistical constraint can still be attached to an inconsistent discretization.

These limitations are actionable. A publishable follow-up should repeat each static and dispatch method over at least 30 seeds, report binomial or concentration intervals for held-out rates, record KKT and chance margins, and separate successful from failed solves before ranking objectives. The lunar model should use a consistent piecewise-constant or collocation quadrature, refine the mesh, validate path maxima on a denser

grid, and compare against the published formulation only after matching initial conditions, free-time treatment, disturbance laws, and solver tolerances. A finite-sample KDE claim would additionally require an explicit confidence construction or a distributionally robust envelope.

9.7 Practical decision rule

For a low-dimensional nonlinear problem with data-driven, multimodal uncertainty, the local evidence supports using KDE as a smooth exploratory constraint. The method should be accompanied by an untouched test set, a bandwidth sensitivity plot, solver-margin checks, and an explicit statement that the result is empirical. If a convex scenario formulation and an a priori confidence level are available, the scenario method has the stronger certification story even when its objective is more conservative. If the residual is known parametrically and the resulting chance constraint is tractable, the parametric reformulation remains the most economical choice.

The appropriate reporting unit is therefore a tuple: formulation, estimator, bandwidth, training sample, solver status, held-out frequency, and validation scope. Omitting any member makes the numerical claim ambiguous. The experiments show why. A four-bar plot can suggest that a shifted KDE is best, yet its estimator field can be mislabelled, its optimizer can fail, or its objective can be a quadrature artefact. Once those distinctions are made explicit, the local contribution is credible: a reproducible implementation, an evidence hierarchy for KDE-based chance constraints, and a concrete set of numerical conditions that must be repaired before stronger claims are justified.

References

- [van Ackooij u. Henrion 2014] ACKOOIJ, Wim van ; HENRION, René: Gradient formulae for nonlinear probabilistic constraints with Gaussian and Gaussian-like distributions. In: *SIAM Journal on Optimization* 24 (2014), Nr. 4, S. 1864–1889. <http://dx.doi.org/10.1137/130922689>. – DOI 10.1137/130922689
- [Alvarez-Mena u. Hernández-Lerma 2005] ALVAREZ-MENA, Jorge ; HERNÁNDEZ-LERMA, Onésimo: Convergence and approximation of optimization problems. In: *SIAM Journal on Optimization* 15 (2005), Nr. 2, S. 527–539
- [Audet u. Hare 2017] AUDET, Charles ; HARE, Warren: *Derivative-Free and Blackbox Optimization*. Cham : Springer, 2017. <http://dx.doi.org/10.1007/978-3-319-68913-5>. <http://dx.doi.org/10.1007/978-3-319-68913-5>
- [Babić u. a. 2023] BABIĆ, Luka u. a.: Data-driven non-parametric chance-constrained model predictive control for microgrids energy management using small data batches. In: *Frontiers in Control Engineering* 4 (2023), S. 1237759. <http://dx.doi.org/10.3389/fcteg.2023.1237759>. – DOI 10.3389/fcteg.2023.1237759
- [Ben-Tal u. a. 2009] BEN-TAL, Aharon ; EL GHAOU, Laurent ; NEMIROVSKI, Arkadi: *Robust Optimization*. Princeton, NJ : Princeton University Press, 2009 (Princeton Series in Applied Mathematics)
- [Bienstock u. a. 2014] BIENSTOCK, Daniel ; CHERTKOV, Michael ; HARNETT, Sean: Chance-constrained optimal power flow: risk-aware network control under uncertainty. In: *SIAM Review* 56 (2014), Nr. 3, S. 461–495. <http://dx.doi.org/10.1137/130910312>. – DOI 10.1137/130910312
- [Birge u. Louveaux 2011] BIRGE, John R. ; LOUVEAUX, François: *Introduction to Stochastic Programming*. 2. New York : Springer, 2011. <http://dx.doi.org/10.1007/978-1-4614-0237-4>. <http://dx.doi.org/10.1007/978-1-4614-0237-4>
- [Blackmore u. a. 2010] BLACKMORE, Lars ; ONO, Masahiro ; BEKTASSOV, Askar ; WILLIAMS, Brian C.: A probabilistic particle-control approximation of chance-constrained stochastic predictive control. In: *IEEE Transactions on Robotics* 26 (2010), Nr. 3, S. 502–517. <http://dx.doi.org/10.1109/TRO.2010.2044948>. – DOI 10.1109/TRO.2010.2044948

- [Boyd u. Vandenberghe 2004] BOYD, Stephen ; VANDENBERGHE, Lieven: *Convex Optimization*. Cambridge : Cambridge University Press, 2004. <http://dx.doi.org/10.1017/CBO9780511804441>. <http://dx.doi.org/10.1017/CBO9780511804441>
- [Caillaud u. a. 2018] CAILLAUD, Jean-Baptiste ; CERF, Max ; SASSI, Achille ; TRÉLAT, Emmanuel ; ZIDANI, Hasnaa: Solving chance constrained optimal control problems in aerospace via kernel density estimation. In: *Optimal Control Applications and Methods* 39 (2018), Nr. 5, S. 1833–1858. <http://dx.doi.org/10.1002/oca.2445>. – DOI 10.1002/oca.2445
- [Calafiore u. Campi 2006] CALAFIORE, Giuseppe C. ; CAMPI, Marco C.: The scenario approach to robust control design. In: *IEEE Transactions on Automatic Control* 51 (2006), Nr. 5, S. 742–753. <http://dx.doi.org/10.1109/TAC.2006.875041>. – DOI 10.1109/TAC.2006.875041
- [Calfa u. a. 2015] CALFA, Bruno A. ; GROSSMANN, Ignacio E. ; AGARWAL, Anshul ; BURY, Scott J. ; WASSICK, John M.: Data-driven individual and joint chance-constrained optimization via kernel smoothing. In: *Computers & Chemical Engineering* 78 (2015), S. 51–69. <http://dx.doi.org/10.1016/j.compchemeng.2015.04.012>. – DOI 10.1016/j.compchemeng.2015.04.012
- [Campi u. Garatti 2008] CAMPI, Marco C. ; GARATTI, Simone: The exact feasibility of randomized solutions of uncertain convex programs. In: *SIAM Journal on Optimization* 19 (2008), Nr. 3, S. 1211–1230. <http://dx.doi.org/10.1137/07069821X>. – DOI 10.1137/07069821X
- [Charnes u. Cooper 1959] CHARNES, Abraham ; COOPER, William W.: Chance-constrained programming. In: *Management Science* 6 (1959), Nr. 1, S. 73–79. <http://dx.doi.org/10.1287/mnsc.6.1.73>. – DOI 10.1287/mnsc.6.1.73
- [Charnes u. Cooper 1963] CHARNES, Abraham ; COOPER, William W.: Deterministic equivalents for optimizing and satisficing under chance constraints. In: *Operations Research* 11 (1963), Nr. 1, S. 18–39. <http://dx.doi.org/10.1287/opre.11.1.18>. – DOI 10.1287/opre.11.1.18
- [Charnes u. a. 1958] CHARNES, Abraham ; COOPER, William W. ; SYMONDS, Gifford H.: Cost horizons and certainty equivalents: an approach to stochastic programming of heating oil. In: *Management Science* 4 (1958), Nr. 3, S. 235–263. <http://dx.doi.org/10.1287/mnsc.4.3.235>. – DOI 10.1287/mnsc.4.3.235

- [Chen u. a. 2024] CHEN, Zhi ; KUHN, Daniel ; WIESEMANN, Wolfram: Data-driven chance constrained programs over Wasserstein balls. In: *Operations Research* 72 (2024), Nr. 1, S. 410–424. <http://dx.doi.org/10.1287/opre.2022.2330>. – DOI 10.1287/opre.2022.2330
- [Ciftci 2017] CIFTCI, Okan: *Microgrid Energy Management with Flexibility Constraints: A Data-Driven Solution Method*, Louisiana State University, Diplomarbeit, 2017. http://dx.doi.org/10.31390/gradschool_theses.4351. – DOI 10.31390/gradschool_theses.4351
- [Ciftci u. a. 2019] CIFTCI, Okan ; MEHRTASH, Mahdi ; KARGARIAN, Amin: Data-driven nonparametric chance-constrained optimization for microgrid energy management. In: *IEEE Transactions on Industrial Informatics* 16 (2019), Nr. 4, S. 2447–2457. <http://dx.doi.org/10.1109/TII.2019.2932078>. – DOI 10.1109/TII.2019.2932078
- [Cont 2001] CONT, Rama: Empirical properties of asset returns: stylized facts and statistical issues. In: *Quantitative Finance* 1 (2001), Nr. 2, S. 223–236. <http://dx.doi.org/10.1080/713665670>. – DOI 10.1080/713665670
- [Dantzig 1955] DANTZIG, George B.: Linear programming under uncertainty. In: *Management Science* 1 (1955), Nr. 3-4, S. 197–206. <http://dx.doi.org/10.1287/mnsc.1.3-4.197>. – DOI 10.1287/mnsc.1.3-4.197
- [Devroye u. Györfi 1985] DEVROYE, Luc ; GYÖRFI, László: *Nonparametric Density Estimation: The L_1 View*. New York : Wiley, 1985
- [Epanechnikov 1969] EPANECHNIKOV, V. A.: Non-parametric estimation of a multivariate probability density. In: *Theory of Probability and its Applications* 14 (1969), Nr. 1, S. 153–158. <http://dx.doi.org/10.1137/1114019>. – DOI 10.1137/1114019
- [Genz u. Bretz 2009] GENZ, Alan ; BRETZ, Frank: *Lecture Notes in Statistics*. Bd. 195: *Computation of Multivariate Normal and t Probabilities*. Berlin : Springer, 2009. <http://dx.doi.org/10.1007/978-3-642-01689-9>. <http://dx.doi.org/10.1007/978-3-642-01689-9>
- [Gopalakrishnan u. a. 2018] GOPALAKRISHNAN, Bharath ; SINGH, Arun K. ; KRISHNA, K. M. ; MANOCHA, Dinesh: Solving chance constrained optimization under nonparametric uncertainty through Hilbert space embedding. In: *arXiv preprint* (2018). – arXiv:1811.09311

- [Gugat u. Schuster 2018] GUGAT, Martin ; SCHUSTER, Michael: Stationary gas networks with compressor control and random loads: optimization with probabilistic constraints. In: *Mathematical Problems in Engineering* 2018 (2018), S. 7984079. <http://dx.doi.org/10.1155/2018/7984079>. – DOI 10.1155/2018/7984079
- [Hodge u. Milligan 2011] HODGE, Bri-Mathias ; MILLIGAN, Michael: Wind power forecasting error distributions over multiple timescales. In: *Proceedings of the IEEE Power and Energy Society General Meeting*. Detroit, MI, 2011
- [Keil u. a. 2020] KEIL, Rachel E. ; MILLER, Alexander T. ; KUMAR, Mrinal ; RAO, Anil V.: *Method for chance constrained optimal control using biased kernel density estimators*. arXiv preprint. – arXiv:2003.08010
- [Keil u. a. 2021] KEIL, Rachel E. ; MILLER, Alexander T. ; KUMAR, Mrinal ; RAO, Anil V.: Method for solving chance constrained optimal control problems using biased kernel density estimators. In: *Optimal Control Applications and Methods* 42 (2021), Nr. 1, S. 330–354. <http://dx.doi.org/10.1002/oca.2675>. – DOI 10.1002/oca.2675
- [Keil u. Rao 2021] KEIL, Rachel E. ; RAO, Anil V.: Warm start method for solving chance constrained optimal control problems using biased kernel density estimators. In: *Journal of Dynamic Systems, Measurement, and Control* 143 (2021), Nr. 12, S. 124502. <http://dx.doi.org/10.1115/1.4052173>. – DOI 10.1115/1.4052173
- [Liu u. a. 2022] LIU, Wei ; YANG, Li ; YU, Bo: Kernel density estimation based distributionally robust mean-CVaR portfolio optimization. In: *Journal of Global Optimization* 84 (2022), S. 1053–1077. <http://dx.doi.org/10.1007/s10898-022-01177-5>. – DOI 10.1007/s10898-022-01177-5
- [Luedtke u. Ahmed 2008] LUEDTKE, James ; AHMED, Shabbir: A sample approximation approach for optimization with probabilistic constraints. In: *SIAM Journal on Optimization* 19 (2008), Nr. 2, S. 674–699. <http://dx.doi.org/10.1137/070702928>. – DOI 10.1137/070702928
- [Miller u. Wagner 1965] MILLER, Bruce L. ; WAGNER, Harvey M.: Chance constrained programming with joint constraints. In: *Operations Research* 13 (1965), Nr. 6, S. 930–945. <http://dx.doi.org/10.1287/opre.13.6.930>. – DOI 10.1287/opre.13.6.930
- [Nelder u. Mead 1965] NELDER, John A. ; MEAD, Roger: A simplex method for function minimization. In: *The Computer Journal* 7 (1965), Nr. 4, S. 308–313. <http://dx.doi.org/10.1093/comjnl/7.4.308>. – DOI 10.1093/comjnl/7.4.308

- [Nemirovski u. Shapiro 2006] NEMIROVSKI, Arkadi ; SHAPIRO, Alexander: Convex approximations of chance constrained programs. In: *SIAM Journal on Optimization* 17 (2006), Nr. 4, S. 969–996. <http://dx.doi.org/10.1137/050622328>. – DOI 10.1137/050622328
- [Pagnoncelli u. a. 2009] PAGNONCELLI, Bernardo K. ; AHMED, Shabbir ; SHAPIRO, Alexander: Sample average approximation method for chance constrained programming: theory and applications. In: *Journal of Optimization Theory and Applications* 142 (2009), Nr. 2, S. 399–416. <http://dx.doi.org/10.1007/s10957-009-9523-6>. – DOI 10.1007/s10957-009-9523-6
- [Parzen 1962] PARZEN, Emanuel: On estimation of a probability density function and mode. In: *Annals of Mathematical Statistics* 33 (1962), Nr. 3, S. 1065–1076. <http://dx.doi.org/10.1214/aoms/1177704472>. – DOI 10.1214/aoms/1177704472
- [Peña-Ordieres u. a. 2020] PEÑA-ORDIERES, Alejandra ; LUEDTKE, James R. ; WÄCHTER, Andreas: Solving chance-constrained problems via a smooth sample-based nonlinear approximation. In: *SIAM Journal on Optimization* 30 (2020), Nr. 3, S. 2221–2250. <http://dx.doi.org/10.1137/19M1261985>. – DOI 10.1137/19M1261985
- [Prékopa 1971] PRÉKOPA, András: Logarithmic concave measures with application to stochastic programming. In: *Acta Scientiarum Mathematicarum* 32 (1971), S. 301–316
- [Prékopa 1995] PRÉKOPA, András: *Stochastic Programming*. Dordrecht : Kluwer Academic Publishers, 1995
- [Prékopa 2003] PRÉKOPA, András: Probabilistic programming. Version: 2003. [http://dx.doi.org/10.1016/S0927-0507\(03\)10005-9](http://dx.doi.org/10.1016/S0927-0507(03)10005-9). In: RUSZCZYŃSKI, Andrzej (Hrsg.) ; SHAPIRO, Alexander (Hrsg.): *Stochastic Programming* Bd. 10. Elsevier, 2003. – DOI 10.1016/S0927-0507(03)10005-9, S. 267–351
- [Qi u. a. 2026] QI, Xiaotong ; LIU, Wei ; YU, Bo ; YANG, Li: *Distributionally robust chance-constrained optimal load shedding model for active distribution networks based on KDE*. Optimization Online preprint, 2026. – Optimization Online ID 34324
- [Rockafellar u. Uryasev 2000] ROCKAFELLAR, R. T. ; URYASEV, Stanislav: Optimization of conditional value-at-risk. In: *Journal of Risk* 2 (2000), Nr. 3, S. 21–41. <http://dx.doi.org/10.21314/JOR.2000.038>. – DOI 10.21314/JOR.2000.038

- [Rosenblatt 1956] ROSENBLATT, Murray: Remarks on some nonparametric estimates of a density function. In: *Annals of Mathematical Statistics* 27 (1956), Nr. 3, S. 832–837. <http://dx.doi.org/10.1214/aoms/1177728190>. – DOI 10.1214/aoms/1177728190
- [Savage 2009] SAVAGE, Sam L.: *The Flaw of Averages: Why We Underestimate Risk in the Face of Uncertainty*. Hoboken, NJ : Wiley, 2009
- [Scheffé 1947] SCHEFFÉ, Henry: A useful convergence theorem for probability distributions. In: *Annals of Mathematical Statistics* 18 (1947), Nr. 3, S. 434–438. <http://dx.doi.org/10.1214/aoms/1177730390>. – DOI 10.1214/aoms/1177730390
- [Schuster 2025] SCHUSTER, Michael: On the convergence of optimization problems with kernel density estimated probabilistic constraints. In: *Applied Set-Valued Analysis and Optimization* 7 (2025), Nr. 2, S. 209–221
- [Schuster u. a. 2022] SCHUSTER, Michael ; STRAUCH, Elisa ; GUGAT, Martin ; LANG, Jens: Probabilistic constrained optimization on flow networks. In: *Optimization and Engineering* 23 (2022), S. 1–50. <http://dx.doi.org/10.1007/s11081-021-09619-x>. – DOI 10.1007/s11081-021-09619-x
- [Scott 1992] SCOTT, David W.: *Multivariate Density Estimation: Theory, Practice, and Visualization*. New York : Wiley, 1992. <http://dx.doi.org/10.1002/9780470316849>. <http://dx.doi.org/10.1002/9780470316849>
- [Shapiro u. a. 2014] SHAPIRO, Alexander ; DENTCHEVA, Darinka ; RUSZCZYŃSKI, Andrzej: *Lectures on Stochastic Programming: Modeling and Theory*. 2. Philadelphia, PA : SIAM, 2014. <http://dx.doi.org/10.1137/1.9781611973433>. <http://dx.doi.org/10.1137/1.9781611973433>
- [Silverman 1986] SILVERMAN, Bernard W.: *Density Estimation for Statistics and Data Analysis*. London : Chapman & Hall, 1986
- [Soyster 1973] SOYSTER, Allen L.: Convex programming with set-inclusive constraints and applications to inexact linear programming. In: *Operations Research* 21 (1973), Nr. 5, S. 1154–1157. <http://dx.doi.org/10.1287/opre.21.5.1154>. – DOI 10.1287/opre.21.5.1154
- [Wächter u. Biegler 2006] WÄCHTER, Andreas ; BIEGLER, Lorenz T.: On the implementation of an interior-point filter line-search algorithm for large-scale non-

- linear programming. In: *Mathematical Programming* 106 (2006), Nr. 1, S. 25–57. <http://dx.doi.org/10.1007/s10107-004-0559-y>. – DOI 10.1007/s10107-004-0559-y
- [Wan u. a. 2020] WAN, Can ; ZHAO, Changfei ; SONG, Yonghua: Chance constrained extreme learning machine for nonparametric prediction intervals of wind power generation. In: *IEEE Transactions on Power Systems* 35 (2020), Nr. 5, S. 3869–3884. <http://dx.doi.org/10.1109/TPWRS.2020.2986282>. – DOI 10.1109/TPWRS.2020.2986282
- [Wand u. Jones 1995] WAND, Matt P ; JONES, M. C.: *Kernel Smoothing*. London : Chapman & Hall, 1995. <http://dx.doi.org/10.1007/978-1-4899-4493-1>. <http://dx.doi.org/10.1007/978-1-4899-4493-1>
- [Wang u. a. 2021] WANG, Jingyao ; WANG, Cheng ; LIANG, Yile ; BI, Tao ; SHAFIE-KHAH, Miadreza ; CATALÃO, Joao P. S.: Data-driven chance-constrained optimal gas-power flow calculation: a Bayesian nonparametric approach. In: *IEEE Transactions on Power Systems* 36 (2021), Nr. 5, S. 4683–4698. <http://dx.doi.org/10.1109/TPWRS.2021.3065465>. – DOI 10.1109/TPWRS.2021.3065465
- [Wied u. Weißbach 2012] WIED, Dominik ; WEISSBACH, Rafael: Consistency of the kernel density estimator: a survey. In: *Statistical Papers* 52 (2012), Nr. 1, S. 1–21. <http://dx.doi.org/10.1007/s00362-010-0338-1>. – DOI 10.1007/s00362-010-0338-1
- [Wu u. Kargarian 2023] WU, Chutian ; KARGARIAN, Amin: Computationally efficient data-driven joint chance constraints for power systems scheduling. In: *IEEE Transactions on Power Systems* 38 (2023), Nr. 3, S. 2858–2869. <http://dx.doi.org/10.1109/TPWRS.2022.3195127>. – DOI 10.1109/TPWRS.2022.3195127
- [Wu u. a. 2021] WU, Chutian ; KARGARIAN, Amin ; JEON, Hyun W.: Data-driven nonparametric joint chance constraints for economic dispatch with renewable generation. In: *IEEE Transactions on Industry Applications* 57 (2021), Nr. 6, S. 6537–6546. <http://dx.doi.org/10.1109/TIA.2021.3105364>. – DOI 10.1109/TIA.2021.3105364
- [Xiu u. Karniadakis 2002] XIU, Dongbin ; KARNIADAKIS, George E.: The Wiener-Askey polynomial chaos for stochastic differential equations. In: *SIAM Journal on Scientific Computing* 24 (2002), Nr. 2, S. 619–644. <http://dx.doi.org/10.1137/S1064827501387826>. – DOI 10.1137/S1064827501387826

Erklärung

Hiermit versichere ich, dass ich die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe, dass alle Stellen der Arbeit, die wörtlich oder sinngemäß aus anderen Quellen übernommen wurden, als solche kenntlich gemacht sind und dass die Arbeit in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegt wurde.

Erlangen, den September 6, 2026

Hierher die Unterschrift

LEBENS LAUF

VORNAME NACHNAME

ZUR PERSON

Name Vorname Nachname
Geboren
Adresse Straße Hausnummer
 PLZ Nürnberg

SCHULE

09/1977 - 08/1981 1. Schulzeit
09/1981 - 06/1990 2. Schulzeit

STUDIUM

09/1991 - 08/1993 1. Universit"at
09/1993 - 06/1996 2. Universit"at

SONSTIGE T"ATIGKEITEN

09/1990 - 08/1991 Anlauf und Sprung
